SMT-Tutorial-for-AI-Class-2

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1 Statistical Machine Translation Tutorial (without Normalization)

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For AI Fundamental Class students
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1.1 Data

[1]:
```
%pwd
```

[1]:
```
'/home/ye/exp/SMT-NMT_tutorial'
```

[3]:
```
%cd clean-data/big/
```

```
/home/ye/exp/SMT-NMT_tutorial/clean-data/big
```

[4]:
```
!ls *.my
```

```
dev.my  test.my  train.my
```

[5]:
```
!ls *.th
```

```
dev.th  test.th  train.th
```

[8]:
```
!wc *.my
```

```
  10000  117871 1126953 dev.my
   8000   93908  896405 test.my
  62625  734667 7029308 train.my
  80625  946446 9052666 total
```

[9]:
```
!wc *.th
```

```
  10000   71013  825957 dev.th
   8000   56113  651909 test.th
  62625  439436 5110712 train.th
  80625  566562 6588578 total
```

ဒေတာ format ကတော့ စာကြောင်းတစ်ကြောင်းစီကို လိုင်းတစ်လိုင်းစီ ထားတဲ့ ပုံစံပါ။ အရေးကြီးတာက မြန်မာစာ နဲ့ ထိုင်းစာ ဘက်အခြမ်းက parallel corpus ဖြစ်နေမှ ရပါမယ်။

[10]:
```
!head train.th
```

[11]:
```
!head train.my
```

အောက် ပါ ရော ဂါ လက္ခ ဏာ တွေ များ ရှိ ရင်
စိတ် ကို အေး အေး ဆေး ဆေး ထား ပါ
နာ တာ ရှည် ရော ဂါ ရယ် လို့ မ ရှိ ပါ ဘူး ဗျ
လက္ခ ဏာ တွေ နဲ့
ပြီး တဲ့ အ ခါ နား ကြပ် ကို တံ တော င့် ဆစ် နား မှာ ထား မှာ မို့ ပါ ရှင်
ပု စွန် တော့ ကျွန် တော် စား တယ်
သို့ ပေ မဲ့ လည်း
ကျွန် တော် ခေါင်း ပေါ် ရေ ခဲ တင် ထား ရ မ လား
ကွန် ပြူ တာ ရောင် ခြည် နဲ့ ထပ် စစ် ဆေး ပါ မယ် နော်
ဒီ ကြား မှာ ခ ဏ လေး ထိုင် စော င့် ပေး လို့ ရ မ လား

[13]:
```
!tail test.th
```

[12]:
```
!tail test.my
```

အမ်
ခန္ဓာ ကိုယ် က မာ တော င့် နေ တယ် ရှ င့်
ဆ ရာ ဝန် တွေ ရော အ ထူး ကု တွေ နဲ့ ပါ ကု သ တာ
ဘယ် နား က နာ တာ လဲ ရှင်
ကိုယ် ဝန် ကို အပ် ထား ပါ
ကျွန် တော် အ တော် ကြာ ကား မ မောင်း ခဲ့ ဘူး

```
ဘယ် သူ က မှ လည်း သ မီး ဘာ ဖြစ် နေ တာ လဲ လို့ မ မေး ကြ ဘူး
အ ခြား ဘာ လက္ခ ဏာ တွေ ဖြစ် သေး လဲ
ဒေါက် တာ ဒီ ကိစ္စ ကို ဟိုး အ ရင် က ပြော ပြ ဖူး ပါ တယ်
သွား နဲ့ ကိုက် ထား တဲ့ ဟာ လာ မ ထုတ် မ ချင်း ငြိမ် နေ ပေး ပါ လုံး ဝ မ လှုပ် ပါ
နဲ့ ရှင်
```

ဒီတစ်ခါတော့ အတန်းထဲမှာ run ပြခဲ့သလို ဒေတာ နည်းနည်းကို မသုံးပဲ အပြည့်သုံးပြီး run ပါမယ်။ Moses ရဲ့ experimental management framework ရဲ့ config ကို သုံးမှာမို့လို့ test ဒေတာကိုလည်း SGM format ပြောင်းထားဖို့ လိုအပ်ပါတယ်။ ပြောင်းထားပြီးပါပြီ။ အောက်ပါအတိုင်းပါ။

[15]:
```
!tree .
```

```
.
  dev.my
  dev.th
  test.my
  test-sgm
     generate_sgms.pl
     ref2sgm.pl
     src2sgm.pl
     test.my.ref.sgm
     test.my.src.sgm
     test.th.ref.sgm
     test.th.src.sgm
  test.th
  train.my
  train.th

2 directories, 13 files
```

SGM format က အောက်ပါအတိုင်းပါ။

[16]:
```
!head ./test-sgm/test.my.src.sgm
```

```
<srcset setid="Thai-Myanmar_data" srclang="any">
<doc docid="none" genre="8000" origlang="my">
<seg id="1">ခွဲ မှ ရ မှာ လား </seg>
<seg id="2">မျက် ခံ လှုပ် တဲ့ လက္ခ ဏာ ရှိ မယ် </seg>
<seg id="3">အ ရက် ကို စွ န့် လွှတ် ရ မည် ဖြစ် ပြီး ပ ရို တင်း နှ င့် ဗီ တာ မင် ဓာတ် ကြွယ် ဝ သော အာ ဟာ ရ ရှိ သော အ စား အ စာ များ ကို စား သ င့် သည် </seg>
<seg id="4">ဟုတ် ကဲ့ သူ ဖျား ပါ တယ် သူ သီး သန့် တော့ မ နေ ခဲ့ ဘူး </seg>
<seg id="5">မောင် လေး နောက် ဆုံး အ ကြိမ် ဆေး သောက် ခဲ့ တာ ဘယ် ချိန် လဲ </seg>
<seg id="6">ကိုယ် မှာ ကူး စက် တာ က နေ ဦး နှောက် ထိ ပါ ကူး စက် လာ တယ် </seg>
<seg id="7">မစ် စ တာ ပ ရာ မို့ ရဲ့ ဆွေ မျိုး ဝင် လာ ဖို့ တစ် ချက် ခေါ် ပေး ပါ
ရှင် </seg>
```

```
<seg id="8">အ ချို ကို လျော့ ချ ကြ ည့် လိုက် ပါ </seg>
```

[17]:
```
!head ./test-sgm/test.th.src.sgm
```

```
<srcset setid="Thai-Myanmar_data" srclang="any">
<doc docid="none" genre="8000" origlang="th">
<seg id="1">        </seg>
<seg id="2">        </seg>
<seg id="3">
             </seg>
<seg id="4">                        </seg>
<seg id="5">                     </seg>
<seg id="6">                           </seg>
<seg id="7">                        </seg>
<seg id="8">                 </seg>
```

Source အပိုင်းကို ကြည့်ပြီးသွားပြီ။ **Reference** အပိုင်းကိုလည်း လေ့လာကြည့်ရအောင်။

[18]:
```
!head ./test-sgm/test.my.ref.sgm
```

```
<refset trglang="my" setid="Thai-Myanmar_data" srclang="any">
<doc sysid="ref" docid="none" genre="8000" origlang="any">
<seg id="1">ခွဲ မှ ရ မှာ လား </seg>
<seg id="2">မျက် ခံ လှုပ် တဲ့ လက္ခ ဏာ ရှိ မယ် </seg>
<seg id="3">အ ရက် ကို စွ န့် လွှတ် ရ မည် ဖြစ် ပြီး ပ ရိ တင်း နှ င့် ဗီ တာ မင်
ဓာတ် ကြွယ် ဝ သော အာ ဟာ ရ ရှိ သော အ စား အ စာ များ ကို စား သ င့် သည် </
↪seg>
<seg id="4">ဟုတ် ကဲ့ သူ ဖျား ပါ တယ် သူ သီး သန့် တော့ မ နေ ခဲ့ ဘူး </seg>
<seg id="5">မောင် လေး နောက် ဆုံး အ ကြိမ် ဆေး သောက် ခဲ့ တာ ဘယ် ချိန် လဲ </seg>
<seg id="6">ကိုယ် မှာ ကူး စက် တာ က နေ ဦး နှောက် ထိ ပါ ကူး စက် လာ တယ် </
↪seg>
<seg id="7">မစ် စ တာ ပ ရာ မို့ ရဲ့ ဆွေ မျိုး ဝင် လာ ဖို့ တစ် ချက် ခေါ် ပေး ပါ
ရှင် </seg>
<seg id="8">အ ချို ကို လျော့ ချ ကြ ည့် လိုက် ပါ </seg>
```

[19]:
```
!head ./test-sgm/test.th.ref.sgm
```

```
<refset trglang="th" setid="Thai-Myanmar_data" srclang="any">
<doc sysid="ref" docid="none" genre="8000" origlang="any">
<seg id="1">        </seg>
<seg id="2">        </seg>
<seg id="3">
             </seg>
<seg id="4">                        </seg>
<seg id="5">                     </seg>
<seg id="6">                           </seg>
<seg id="7">                        </seg>
<seg id="8">                 </seg>
```

1.2 Config File Preparations

အတန်းထဲမှာ သင်ပြပေးခဲ့တဲ့အတိုင်းပဲ။ ကိုယ့်စက်ထဲမှာ run မယ့် path တွေကို config ဖိုင် မှန်မှန်ကန်ကန် generate လုပ်နိုင်ဖို့ ဝင်ပြင်ပေးရပါမယ်။

[27]:
```
%pwd
```

[27]:
```
'/home/ye/exp/SMT-NMT_tutorial/clean-data/big'
```

[28]:
```
%cd ../../
```

```
/home/ye/exp/SMT-NMT_tutorial
```

ဒီတစ်ခေါက် လိုင်း ခြောက်သောင်းကျော် မြန်မာ-ထိုင်း ကောပတ်စ်ကို pbsmt-big ဆိုတဲ့ folder အောက်မှာ experiment လုပ်မှာမို့ ဖိုလ်ဒါအသစ် ဆောက်ပါမယ်။

[29]:
```
!mkdir pbsmt-big
```

အတန်းထဲမှာတုန်းက SMT demo လုပ်ပြစဉ်က သုံးခဲ့တဲ့ perl script တွေ၊ config ဖိုင်တွေကို pbsmt-big အောက်ကို ကော်ပီကူးယူပါမယ်။

[30]:
```
!cp ./pbsmt/*.pl ./pbsmt-big
```

[31]:
```
!cp ./pbsmt/config.baseline ./pbsmt-big/
```

အရင်က ပေးထားတဲ့ path တွေမှာ အဓိက ဝင်ပြင်ရမှာကတော့ ဒေတာ path (i.e. clean-data/big/) နဲ့ experimental folder path (i.e. pbsmt-big/) တွေပါပဲ။ ကိုယ့်စက်ထဲမှာတော့ moses ရဲ့ binary ဖိုင်တွေကို ကြိုတင်ပြင်ဆင်ထားဖို့လည်း မမေ့ပါနဲ့။

ဆရာ စက်ထဲမှာတော့ အောက်ပါ path ထဲမှာ moses binary ဖိုင်တွေကို ပြင်ဆင်ထားပြီးသားပါ။

/home/ye/tool/mosesbin/ubuntu-17.04/moses/

Edit လုပ်ပြီးသား perl scirpt နှစ်ဖိုင်နဲ့ config.baseline ဖိုင်တွေက အောက်မှာ မြင်ရတဲ့အတိုင်းပါပဲ။

[33]:
```
!cat ./pbsmt-big/generate_configs.pl
```

```
#!/usr/bin/perl

# Preparing configuration files for PBSMT experiments
# Output: one configuration file for baseline or PBSMT
# *** Before you run this script, you should prepare training, development and
test parallel-data in advance!!!

# Written by Finch-san and Ye
# We used this script for running our NAACL paper:
# Ye Kyaw Thu, Andrew Finch and Eiichiro Sumita,
# "Interlocking Phrases in Phrase-based Statistical Machine Translation",
# In Proceedings of the NAACL-HLT 2016, June 12-17, San Diego, US, pp.
1076-1081.
```

```
# Last updated 9 Oct 2019, @Machine Translation Research Summer School, YTU,
Myanmar

use strict;

my @langs;

# you have to update the PBSMT experiment path!!!
#my $smtpath = "/home/lar/experiment/kachin-myanmar/demo-mykc-smt";
#my $smtpath = "/media/lar/Transcend/student/lecture/mtrss/pbsmt-demo/pbsmt";
my $smtpath = "/home/ye/exp/SMT-NMT_tutorial/pbsmt-big";

# you have to update the data path for running PBSMT experiment!!!

#foreach my $trainFile (
</panfs/panltg2/users/finch/expt/multibtec/corpus/train.[a-z][a-z]> )
#foreach my $trainFile (
</panfs/panmt/users/ye/cluster/corpus/train.[a-z][a-z]>)
#foreach my $trainFile ( </home/tha/experiment/my-rk/data/train.[a-z][a-z]>)
#foreach my $trainFile ( </home/lar/experiment/kachin-myanmar/demo-mykc-
smt/4demo/train.[a-z][a-z]>)
#foreach my $trainFile ( </media/lar/Transcend/student/lecture/mtrss/pbsmt-
demo/pbsmt/data/train.[a-z][a-z]>)
foreach my $trainFile ( </home/ye/exp/SMT-NMT_tutorial/clean-
data/big/train.[a-z][a-z]>)
{
        $trainFile =~ m/train.([a-z][a-z])/;

        push @langs, $1;
}

foreach my $expt (qw/baseline/)
{

die("$smtpath/$expt") if (-d "$smtpath/$expt");
`mkdir $smtpath/$expt`;
    foreach my $src (@langs)
    {
        foreach my $trg (@langs)
        {
                next if $src eq $trg;
            #next if $src ne "my" and $trg ne "my";

                my $pair = "${src}-${trg}";
                my $configFile = "$smtpath/$expt/$pair/config.${expt}.$pair";

                `mkdir $smtpath/$expt/$pair`;
```

```
                open FILE, ">$configFile" or die;

                print FILE "# Config file for $pair ($expt)\n";
                print FILE "\n[GENERAL]\n";
                print FILE "working-dir = $smtpath/$expt/$pair\n";
                print FILE "input-extension = ${src}\n";
                print FILE "output-extension = ${trg}\n";
                print FILE "pair-extension = ${pair}\n";
                print FILE "\n";
                close FILE;

            `cat $configFile config.$expt > tmp`;
            `mv tmp $configFile`;
            }
    }
}
```

[34]:

```
!cat ./pbsmt-big/run-baseline.pl
```

```
#!/usr/bin/perl
use strict;
use warnings;

require IPC::System::Simple;
use autodie qw(:all);

# Written by Finch-san and Ye
# We used this script for running our NAACL paper:
# Ye Kyaw Thu, Andrew Finch and Eiichiro Sumita,
# "Interlocking Phrases in Phrase-based Statistical Machine Translation",
# In Proceedings of the NAACL-HLT 2016, June 12-17, San Diego, US, pp.
1076-1081.

# Last updated 9 Oct 2019, @Machine Translation Research Summer School, YTU,
Myanmar

# You have to update following path!!!

#my @configs = `find /home/ros/experiment/langacq/exp1/smt/{baseline,osm,hiero}
-name "config.*" | sort`;
#my @configs = `find /home/ros/experiment/langacq/exp2a/smt/*/ -name
"config.*.*" | sort`;
#my @configs = `find /home/ros/experiment/sl-my/smt1/*/ -name
"config.baseline.*" | sort`;
#my @configs = `find /home/ros/experiment/my-rk/smt1/*/ -name
"config.baseline.*" | sort`;
#my @configs = `find /media/lar/Transcend/exp/kachin-myanmar/demo-mykc-smt/
-name "config.baseline"  `;
```

```
#my @configs = `find /home/lar/experiment/kachin-myanmar/demo-mykc-smt/*/ -name
"config.baseline*" | sort`;
#my @configs = `find /media/lar/Transcend/student/lecture/mtrss/pbsmt-
demo/pbsmt/*/ -name "config.baseline*" | sort`;

my @configs = `find /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/*/ -name
"config.baseline*" | sort`;

#for debugging
#print "config files: @configs\n";
#exit;

#my @configs = qw(/panfs/panltg2/users/finch/expt/moses2.1/baseline/zh-
nl/config.baseline.zh-nl);
#my @langs = qw/mini_en mini_ja/;

foreach my $i (0..$#configs)
{
        # sleep until there are less than 40 of my jobs on the queue
        #while(`qstat | grep ye | wc` =~ /^\s*([0-9]+)/ && $1 > 700 )
        #{
        #       sleep 20;
        #}

#       my $o_config = $o_configs[$i];
#       my $b_config = $b_configs[$i];
        my $config = $configs[$i];

#       chomp $o_config;
#       chomp $b_config;
        chomp $config;

        my @toks = split /\./, $config;
        my $jobName = "$toks[$#toks]-$toks[$#toks - 1]";

        print "$jobName $config\n";

        #       `/home/lar/tool/moses/scripts/ems/experiment.perl -config
$config -exec -no-graph 2>&1 | tee -a run1.log`;

#/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/ems
        `/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/ems/experiment.perl
-config $config -exec -no-graph 2>&1 | tee -a run1.log`;
        #`/home/ros/mosesdecoder/scripts/ems/experiment.perl -config $config
-exec -no-graph`;

#       my @toks = split /\./, $o_config;
#       my $jobName = "$toks[$#toks]-$toks[$#toks - 1]";
```

```
#
#       print "$jobName $o_config\n";
#
#       `qsub -q mt128 -N "$jobName" --
/panfs/panltg2/users/finch/local/mosesdecoder.v21/scripts/ems/experiment.perl
-config $o_config -exec`;
#
#       @toks = split /\./, $b_config;
#       $jobName = "$toks[$#toks]-$toks[$#toks - 1]";
#
#       print "$jobName $b_config\n\n";
#
#       `qsub -q mt128 -N "$jobName" --
/panfs/panltg2/users/finch/local/mosesdecoder.v21/scripts/ems/experiment.perl
-config $b_config -exec`;
}
```

Update လုပ်ထားတဲ့ **config.baseline** ဖိုင်က အောက်ပါအတိုင်းပါ
ဒီဖိုင်က template ဖိုင်လိုမျိုးပါပဲ။ ဒီဖိုင်ကို အခြေခံပြီး မြန်မာ-ထိုင်း၊ ထိုင်း-မြန်မာ နှစ်မျိုးလုံးအတွက် configuration ဖိုင်ကို ထုတ်ပေးသွားမှာပါ။

[35]:
```
!cat ./pbsmt-big/config.baseline
```

```
### directories that contain tools and data
#
# moses
#moses-src-dir = /home/ros/mosesdecoder
#moses-src-dir = /home/lar/tool/moses/
#moses-src-dir = /home/ye/tool/mosesbin/ubuntu-17.04/moses
moses-src-dir = /home/ye/tool/mosesbin/ubuntu-17.04/moses

# moses binaries
moses-bin-dir = $moses-src-dir/bin

# moses scripts
moses-script-dir = $moses-src-dir/scripts

# directory where GIZA++/MGIZA programs resides
#external-bin-dir = /home/lar/tool/giza-pp-master/mkcls-v2
#external-bin-dir = /home/lar/tool/giza-pp-master/GIZA++-v2
#external-bin-dir = /home/lar/tool/giza-pp-master/
#external-bin-dir = /home/ye/tool/mgiza/mgizapp/bin
external-bin-dir = /home/ye/tool/giza-pp/GIZA++-v2

# srilm
#srilm-dir = /data/lttools/training/srilm/srilm-1.6.0/bin/i686-m64
#I haven't installed SRILM yet on Deep Learning Box computer
```

```
#srilm-dir = /home/ros/tool/srilm-1.7.1/bin/i686-m64

# irstlm
#irstlm-dir = $moses-src-dir/irstlm/bin

# randlm
#randlm-dir = $moses-src-dir/randlm/bin

# kenlm
lmplz = $moses-bin-dir/lmplz

# data
#myrk-data = /home/lar/experiment/my-rk/smt1/t9
#mykc-data = /home/lar/experiment/kachin-myanmar/demo-mykc-smt/4demo/
#mykc-data = /home/lar/experiment/kachin-myanmar/demo-mykc-smt/4demo2/
#myrk-data = /media/lar/Transcend/student/lecture/mtrss/pbsmt-demo/pbsmt/data/
thmy-data = /home/ye/exp/SMT-NMT_tutorial/clean-data/big/

### basic tools
#
# moses decoder
decoder = $moses-bin-dir/moses

# conversion of phrase table into binary on-disk format
#ttable-binarizer = $moses-bin-dir/processPhraseTable
#ttable-binarizer = $moses-bin-dir/processPhraseTableMin

# conversion of rule table into binary on-disk format
#ttable-binarizer = "$moses-bin-dir/CreateOnDiskPt 1 1 4 100 2"

# tokenizers - comment out if all your data is already tokenized
#input-tokenizer = "$moses-script-dir/tokenizer/tokenizer.perl -a -l $input-
extension"
#output-tokenizer = "$moses-script-dir/tokenizer/tokenizer.perl -a -l $output-
extension"

# truecasers - comment out if you do not use the truecaser
#input-truecaser = $moses-script-dir/recaser/truecase.perl
#output-truecaser = $moses-script-dir/recaser/truecase.perl
#detruecaser = $moses-script-dir/recaser/detruecase.perl

### generic parallelizer for cluster and multi-core machines
# you may specify a script that allows the parallel execution
# parallizable steps (see meta file). you also need specify
# the number of jobs (cluster) or cores (multicore)
#
generic-parallelizer = $moses-script-dir/ems/support/generic-parallelizer.perl
#generic-parallelizer = $moses-script-dir/ems/support/generic-multicore-
```

```
parallelizer.perl

### cluster settings (if run on a cluster machine)
# number of jobs to be submitted in parallel
#
#jobs = 10

# arguments to qsub when scheduling a job
#qsub-settings = ""

# project for priviledges and usage accounting
#qsub-project = iccs_smt

# memory and time
#qsub-memory = 4
#qsub-hours = 48

### multi-core settings
# when the generic parallelizer is used, the number of cores
# specified here
#cores = 8

#################################################################
# PARALLEL CORPUS PREPARATION:
# create a tokenized, sentence-aligned corpus, ready for training

[CORPUS]

### long sentences are filtered out, since they slow down GIZA++
# and are a less reliable source of data. set here the maximum
# length of a sentence
#
max-sentence-length = 80

[CORPUS:thmy]

### command to run to get raw corpus files
#
# get-corpus-script =

### raw corpus files (untokenized, but sentence aligned)
#
raw-stem = $thmy-data/train

### tokenized corpus files (may contain long sentences)
#
#tokenized-stem = $myrk-data/train
```

```
### if sentence filtering should be skipped,
# point to the clean training data
#
#clean-stem =

### if corpus preparation should be skipped,
# point to the prepared training data
#
#lowercased-stem =

#################################################################
# LANGUAGE MODEL TRAINING

[LM]

### tool to be used for language model training
# srilm
#lm-training = /data/lttools/training/srilm/srilm-1.6.0/bin/i686-m64/ngram-count
#lm-training = /home/ros/tool/srilm-1.7.1/bin/i686-m64/ngram-count
#settings = "-interpolate -kndiscount -unk"

# irstlm training
# msb = modified kneser ney; p=0 no singleton pruning
#lm-training = "$moses-script-dir/generic/trainlm-irst2.perl -cores $cores
-irst-dir $irstlm-dir -temp-dir $working-dir/tmp"
#settings = "-s msb -p 0"

# kenlm
# additional parameters to lmplz (check lmplz help message)
#settings = "-T $working-dir/tmp -S 10G"
# this tells EMS to use lmplz and tells EMS where lmplz is located
#lm-training = "$moses-script-dir/generic/trainlm-lmplz.perl -lmplz $lmplz"
#lm-training = "$moses-script-dir/ems/support/lmplz-wrapper.perl -bin $moses-
bin-dir/lmplz"

# kenlm training
lm-training = "$moses-script-dir/ems/support/lmplz-wrapper.perl -bin $moses-bin-
dir/lmplz"
settings = "--prune '0 0 1' -T $working-dir/lm -S 20% --discount_fallback"


#lm-binarizer = $moses-bin-dir/build_binary

# order of the language model
order = 5

### tool to be used for training randomized language model from scratch
# (more commonly, a SRILM is trained)
```

```
#
#rlm-training = "$randlm-dir/buildlm -falsepos 8 -values 8"

### script to use for binary table format for irstlm or kenlm
# (default: no binarization)

# irstlm
#lm-binarizer = $irstlm-dir/compile-lm

# kenlm, also set type to 8
lm-binarizer = $moses-bin-dir/build_binary
type = 8

### script to create quantized language model format (irstlm)
# (default: no quantization)
#
#lm-quantizer = $irstlm-dir/quantize-lm

### script to use for converting into randomized table format
# (default: no randomization)
#
#lm-randomizer = "$randlm-dir/buildlm -falsepos 8 -values 8"

### each language model to be used has its own section here

[LM:thmy]

### command to run to get raw corpus files
#
#get-corpus-script = ""

### raw corpus (untokenized)
#
raw-corpus = $thmy-data/train.$output-extension

### tokenized corpus files (may contain long sentences)
#
#tokenized-corpus = $my-data/train.$output-extension

### if corpus preparation should be skipped,
# point to the prepared language model
#
#lm =

#################################################################
# INTERPOLATING LANGUAGE MODELS

[INTERPOLATED-LM]
```

```
# if multiple language models are used, these may be combined
# by optimizing perplexity on a tuning set
# see, for instance [Koehn and Schwenk, IJCNLP 2008]

### script to interpolate language models
# if commented out, no interpolation is performed
#
# script = $moses-script-dir/ems/support/interpolate-lm.perl

### tuning set
# you may use the same set that is used for mert tuning (reference set)
#
#tuning-sgm =
#raw-tuning =
#tokenized-tuning =
#factored-tuning =
#lowercased-tuning =
#split-tuning =

### group language models for hierarchical interpolation
# (flat interpolation is limited to 10 language models)
#group = "first,second fourth,fifth"

### script to use for binary table format for irstlm or kenlm
# (default: no binarization)

# irstlm
#lm-binarizer = $irstlm-dir/compile-lm

# kenlm, also set type to 8
lm-binarizer = $moses-bin-dir/build_binary
type = 8

### script to create quantized language model format (irstlm)
# (default: no quantization)
#
#lm-quantizer = $irstlm-dir/quantize-lm

### script to use for converting into randomized table format
# (default: no randomization)
#
#lm-randomizer = "$randlm-dir/buildlm -falsepos 8 -values 8"

#################################################################
# MODIFIED MOORE LEWIS FILTERING

[MML] IGNORE
```

```
### specifications for language models to be trained
#
#lm-training = $srilm-dir/ngram-count
#lm-settings = "-interpolate -kndiscount -unk"
#lm-binarizer = $moses-src-dir/bin/build_binary
#lm-query = $moses-src-dir/bin/query
#order = 5

### in-/out-of-domain source/target corpora to train the 4 language model
#
# in-domain: point either to a parallel corpus
#outdomain-stem = [CORPUS:toy:clean-split-stem]

# … or to two separate monolingual corpora
#indomain-target = [LM:toy:lowercased-corpus]
#raw-indomain-source = $toy-data/nc-5k.$input-extension

# point to out-of-domain parallel corpus
#outdomain-stem = [CORPUS:giga:clean-split-stem]

# settings: number of lines sampled from the corpora to train each language
model on
# (if used at all, should be small as a percentage of corpus)
#settings = "--line-count 100000"

#################################################################
# TRANSLATION MODEL TRAINING

[TRAINING]

### training script to be used: either a legacy script or
# current moses training script (default)
#
script = $moses-script-dir/training/train-model.perl

### general options
# these are options that are passed on to train-model.perl, for instance
# * "-mgiza -mgiza-cpus 8" to use mgiza instead of giza
# * "-sort-buffer-size 8G -sort-compress gzip" to reduce on-disk sorting
# * "-sort-parallel 8 -cores 8" to speed up phrase table building
#
#training-options = "-sort-parallel 8 -cores 8"

#training-options = "-cores 8"

### factored training: specify here which factors used
# if none specified, single factor training is assumed
```

```
# (one translation step, surface to surface)
#
#input-factors = word lemma pos morph
#output-factors = word lemma pos
#alignment-factors = "word -> word"
#translation-factors = "word -> word"
#reordering-factors = "word -> word"
#generation-factors = "word -> pos"
#decoding-steps = "t0, g0"

### parallelization of data preparation step
# the two directions of the data preparation can be run in parallel
# comment out if not needed
#
parallel = yes

### pre-computation for giza++
# giza++ has a more efficient data structure that needs to be
# initialized with snt2cooc. if run in parallel, this may reduces
# memory requirements. set here the number of parts
#
#run-giza-in-parts = 5

### symmetrization method to obtain word alignments from giza output
# (commonly used: grow-diag-final-and)
#
alignment-symmetrization-method = grow-diag-final-and

### use of Chris Dyer's fast align for word alignment
#
#fast-align-settings = "-d -o -v"

### use of berkeley aligner for word alignment
#
#use-berkeley = true
#alignment-symmetrization-method = berkeley
#berkeley-train = $moses-script-dir/ems/support/berkeley-train.sh
#berkeley-process =  $moses-script-dir/ems/support/berkeley-process.sh
#berkeley-jar = /your/path/to/berkeleyaligner-1.1/berkeleyaligner.jar
#berkeley-java-options = "-server -mx30000m -ea"
#berkeley-training-options = "-Main.iters 5 5 -EMWordAligner.numThreads 8"
#berkeley-process-options = "-EMWordAligner.numThreads 8"
#berkeley-posterior = 0.5

### use of baseline alignment model (incremental training)
#
#baseline = 68
#baseline-alignment-model = "$working-dir/training/prepared.$baseline/$input-
```

```
extension.vcb \
#  $working-dir/training/prepared.$baseline/$output-extension.vcb \
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.cooc \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.cooc \
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.thmm.5 \
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.hhmm.5 \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.thmm.5 \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.hhmm.5"

### if word alignment should be skipped,
# point to word alignment files
#
#word-alignment = $working-dir/model/aligned.1

### filtering some corpora with modified Moore-Lewis
# specify corpora to be filtered and ratio to be kept, either before or after
word alignment
#mml-filter-corpora = toy
#mml-before-wa = "-proportion 0.9"
#mml-after-wa = "-proportion 0.9"

### create a bilingual concordancer for the model
#
#biconcor = $moses-script-dir/ems/biconcor/biconcor

############################################################

### use of Operation Sequence Model
### Durrani, Schmid and Fraser. (2011): "A Joint Sequence Translation Model with
Integrated Reordering"

#operation-sequence-model = "yes"
#operation-sequence-model-order = 5
#operation-sequence-model-settings = ""

### compile Moses with --max-kenlm-order=9 if higher order is required

### if OSM training should be skipped,
# point to OSM Model
#
# osm-model =
```

```
#################################################################

### lexicalized reordering: specify orientation type
# (default: only distance-based reordering model)
#
lexicalized-reordering = msd-bidirectional-fe

### hierarchical rule set
#
#hierarchical-rule-set = true

### settings for rule extraction
#
#extract-settings = ""
max-phrase-length = 5

### add extracted phrases from baseline model
#
#baseline-extract = $working-dir/model/extract.$baseline
#
# requires aligned parallel corpus for re-estimating lexical translation
probabilities
#baseline-corpus = $working-dir/training/corpus.$baseline
#baseline-alignment = $working-dir/model/aligned.$baseline.$alignment-
symmetrization-method

### unknown word labels (target syntax only)
# enables use of unknown word labels during decoding
# label file is generated during rule extraction
#
#use-unknown-word-labels = true

### if phrase extraction should be skipped,
# point to stem for extract files
#
# extracted-phrases =

### settings for rule scoring
#
score-settings = "--GoodTuring"

### include word alignment in phrase table
#
#include-word-alignment-in-rules = yes

### sparse lexical features
#
```

```
#sparse-features = "target-word-insertion top 50, source-word-deletion top 50,
word-translation top 50 50, phrase-length"

### domain adaptation settings
# options: sparse, any of: indicator, subset, ratio
#domain-features = "subset"

### if phrase table training should be skipped,
# point to phrase translation table
#
# phrase-translation-table =

### if reordering table training should be skipped,
# point to reordering table
#
# reordering-table =

### filtering the phrase table based on significance tests
# Johnson, Martin, Foster and Kuhn. (2007): "Improving Translation Quality by
Discarding Most of the Phrasetable"
# options: -n number of translations; -l 'a+e', 'a-e', or a positive real value
-log prob threshold
#salm-index = /path/to/project/salm/Bin/Linux/Index/IndexSA.O64
#sigtest-filter = "-l a+e -n 50"

### if training should be skipped,
# point to a configuration file that contains
# pointers to all relevant model files
#
#config-with-reused-weights =

#####################################################
### TUNING: finding good weights for model components

[TUNING]

### instead of tuning with this setting, old weights may be recycled
# specify here an old configuration file with matching weights
#
#weight-config = /panfs/panltg2/users/finch/expt/moses2.1/plain.weight.ini

### tuning script to be used
#
tuning-script = $moses-script-dir/training/mert-moses.pl
tuning-settings = "-mertdir $moses-bin-dir"

### specify the corpus used for tuning
# it should contain 1000s of sentences
```

```
#
#input-sgm =
raw-input = $thmy-data/dev.$input-extension

#tokenized-input =
#factorized-input =
#input =
#
#reference-sgm =
raw-reference = $thmy-data/dev.$output-extension

#tokenized-reference =
#factorized-reference =
#reference =

### size of n-best list used (typically 100)
#
nbest = 100

### ranges for weights for random initialization
# if not specified, the tuning script will use generic ranges
# it is not clear, if this matters
#
# lambda =

### additional flags for the filter script
#
filter-settings = ""

### additional flags for the decoder
#
decoder-settings = ""

### if tuning should be skipped, specify this here
# and also point to a configuration file that contains
# pointers to all relevant model files
#
#config =

#########################################################
## RECASER: restore case, this part only trains the model

[RECASING]

#decoder = $moses-bin-dir/moses

### training data
# raw input needs to be still tokenized,
```

```
# also also tokenized input may be specified
#
#tokenized = [LM:europarl:tokenized-corpus]

# recase-config =

#lm-training = $srilm-dir/ngram-count

#######################################################
## TRUECASER: train model to truecase corpora and input

[TRUECASER]

### script to train truecaser models
#
#trainer = $moses-script-dir/recaser/train-truecaser.perl

### training data
# data on which truecaser is trained
# if no training data is specified, parallel corpus is used
#
# raw-stem =
# tokenized-stem =

### trained model
#
# truecase-model =

######################################################################
## EVALUATION: translating a test set using the tuned system and score it

[EVALUATION]

### additional flags for the filter script
#
#filter-settings = ""

### additional decoder settings
# switches for the Moses decoder
# common choices:
#   "-threads N" for multi-threading
#   "-mbr" for MBR decoding
#   "-drop-unknown" for dropping unknown source words
#   "-search-algorithm 1 -cube-pruning-pop-limit 5000 -s 5000" for cube pruning
#
decoder-settings = "-search-algorithm 1 -cube-pruning-pop-limit 5000 -s 5000"

### specify size of n-best list, if produced
```

```
#
#nbest = 100

### multiple reference translations
#
#multiref = yes

### prepare system output for scoring
# this may include detokenization and wrapping output in sgm
# (needed for nist-bleu, ter, meteor)
#
#detokenizer = "$moses-script-dir/tokenizer/detokenizer.perl -l $output-
extension"
#recaser = $moses-script-dir/recaser/recase.perl
wrapping-script = "$moses-script-dir/ems/support/wrap-xml.perl $output-
extension"
#output-sgm =

### BLEU
#
nist-bleu = $moses-script-dir/generic/mteval-v13a.pl
nist-bleu-c = "$moses-script-dir/generic/mteval-v13a.pl -c"
multi-bleu = $moses-script-dir/generic/multi-bleu.perl
#ibm-bleu =

### TER: translation error rate (BBN metric) based on edit distance
# not yet integrated
#
# ter =

### METEOR: gives credit to stem / worknet synonym matches
# not yet integrated
#
# meteor =

### Analysis: carry out various forms of analysis on the output
#
analysis = $moses-script-dir/ems/support/analysis.perl
#
# also report on input coverage
analyze-coverage = yes
#
# also report on phrase mappings used
report-segmentation = yes
#
# report precision of translations for each input word, broken down by
# count of input word in corpus and model
report-precision-by-coverage = yes
```

```
#
# further precision breakdown by factor
#precision-by-coverage-factor = pos
#
# visualization of the search graph in tree-based models
#analyze-search-graph = yes

[EVALUATION:test]

### input data
#
#input-sgm = /panfs/panmt/users/ye/cluster/test-sgm/test.$input-
extension.src.sgm
input-sgm = $thmy-data/test-sgm/test.$input-extension.src.sgm
#input-sgm = $mykc-data/test-sgm/test.$input-extension.src.sgm

# raw-input = $mykc-data/test.$input-extension
# tokenized-input =
# factorized-input =
# input =

### reference data
#
#reference-sgm = /panfs/panmt/users/ye/cluster/test-sgm/test.$output-
extension.ref.sgm
reference-sgm = $thmy-data/test-sgm/test.$output-extension.ref.sgm
#reference-sgm = $mykc-data/test-sgm/test.$output-extension.ref.sgm

# raw-reference = $mykc-data/test.$output-extension
# tokenized-reference =
# reference =

### analysis settings
# may contain any of the general evaluation analysis settings
# specific setting: base coverage statistics on earlier run
#
#precision-by-coverage-base = $working-dir/evaluation/test.analysis.5

### wrapping frame
# for nist-bleu and other scoring scripts, the output needs to be wrapped
# in sgm markup (typically like the input sgm)
#
wrapping-frame = $input-sgm

##########################################
### REPORTING: summarize evaluation scores

[REPORTING]
```

```
### currently no parameters for reporting section
```

1.3 Confirm Before Config File Generation

config ဖိုင်တွေကို perl script နဲ့ generate မလုပ်ခင်မှာ လက်ရှိ experiment လုပ်မယ့် ဖိုလ်ဒါကို တချက်ကြိုကြည့်ထားရအောင်။

[36]:
```
!tree ./pbsmt-big/
```

```
./pbsmt-big/
  config.baseline
  generate_configs.pl
  run-baseline.pl

1 directory, 3 files
```

1.4 Config File Generation

[38]:
```
%pwd
```

[38]:
```
'/home/ye/exp/SMT-NMT_tutorial'
```

[40]:
```
%cd ./pbsmt-big/
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big
```

[41]:
```
!ls
```

```
config.baseline  generate_configs.pl  run-baseline.pl
```

[43]:
```
!perl ./generate_configs.pl
```

Config ဖိုင်က run လို့ အဆင်ပြေပြေနဲ့ ပြီးသွားရင်တော့ အောက်ပါလိုမျိုး ဖိုင်တွေကို တွေ့ရပါလိမ့်မယ်။ ဒီတစ်ခေါက်က မြန်မာ-ထိုင်း၊ ထိုင်း-မြန်မာ SMT ကို လုပ်ဖို့ ပြင်ဆင်နေတာမို့လို့ my-th, th-my ဆိုတဲ့ ဖိုလ်ဒါတွေကို ဆောက်ပေးသွားပြီး အဲဒီဖိုလ်ဒါတွေ အောက်မှာ သက်ဆိုင်ရင် configuration ဖိုင်တွေကို ဝင်ဆောက်ပေးသွားပါလိမ့်မယ်။

တကယ့်လက်တွေ့မှာက ဘာသာစကား တစ်ခုကနေ တခြား ဘာသာစကားအများကြီးကို ဘာသာပြန်တဲ့ experiment တွေ လုပ်တာမို့လို့ ပြီးတော့ baseline တစ်ခုဖြစ်တဲ့ PBSMT ကိုပဲ run တာမဟုတ်ပဲ၊ တခြား SMT approach တွေဖြစ်တဲ့ HPBSMT တို့ OSM တို့ စတဲ့ experiment တွေကို လုပ်တာမျိုးလည်း ရှိလို့ အခုလိုမျိုး automatic generation တွေအတွက် ပြင်ဆင်ထားတာပါ။ တကယ်လို့ ကိုယ်က ဘာသာစကားနှစ်ခုအကြား SMT method တစ်မျိုးလောက်ပဲ လုပ်မယ် ဆိုရင်တော့ ကိုယ်ဖာသာကိုယ် manual ဝင်ပြင်တာမျိုးတွေလည်း လုပ်လို့ရပါတယ်။

[44]:
```
!tree .
```

```
.
  baseline
```

```
        my-th
            config.baseline.my-th
        th-my
            config.baseline.th-my
    config.baseline
    generate_configs.pl
    run-baseline.pl

4 directories, 5 files
```

1.5 Check my-th Config File

Generate လုပ်ပြီးထွက်လာတဲ့ my-th အတွက် configuration ဖိုင်ကို ဝင်လေ့လာပါ။ မှားနေတာ ရှိရင် ဝင်ပြင်ရလိမ့်မယ်။

[45]:
```
!cat ./baseline/my-th/config.baseline.my-th
```

```
# Config file for my-th (baseline)

[GENERAL]
working-dir = /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th
input-extension = my
output-extension = th
pair-extension = my-th


### directories that contain tools and data
#
# moses
#moses-src-dir = /home/ros/mosesdecoder
#moses-src-dir = /home/lar/tool/moses/
#moses-src-dir = /home/ye/tool/mosesbin/ubuntu-17.04/moses
moses-src-dir = /home/ye/tool/mosesbin/ubuntu-17.04/moses

# moses binaries
moses-bin-dir = $moses-src-dir/bin

# moses scripts
moses-script-dir = $moses-src-dir/scripts

# directory where GIZA++/MGIZA programs resides
#external-bin-dir = /home/lar/tool/giza-pp-master/mkcls-v2
#external-bin-dir = /home/lar/tool/giza-pp-master/GIZA++-v2
#external-bin-dir = /home/lar/tool/giza-pp-master/
#external-bin-dir = /home/ye/tool/mgiza/mgizapp/bin
external-bin-dir = /home/ye/tool/giza-pp/GIZA++-v2

# srilm
```

```
#srilm-dir = /data/lttools/training/srilm/srilm-1.6.0/bin/i686-m64
#I haven't installed SRILM yet on Deep Learning Box computer
#srilm-dir = /home/ros/tool/srilm-1.7.1/bin/i686-m64

# irstlm
#irstlm-dir = $moses-src-dir/irstlm/bin

# randlm
#randlm-dir = $moses-src-dir/randlm/bin

# kenlm
lmplz = $moses-bin-dir/lmplz

# data
#myrk-data = /home/lar/experiment/my-rk/smt1/t9
#mykc-data = /home/lar/experiment/kachin-myanmar/demo-mykc-smt/4demo/
#mykc-data = /home/lar/experiment/kachin-myanmar/demo-mykc-smt/4demo2/
#myrk-data = /media/lar/Transcend/student/lecture/mtrss/pbsmt-demo/pbsmt/data/
thmy-data = /home/ye/exp/SMT-NMT_tutorial/clean-data/big/

### basic tools
#
# moses decoder
decoder = $moses-bin-dir/moses

# conversion of phrase table into binary on-disk format
#ttable-binarizer = $moses-bin-dir/processPhraseTable
#ttable-binarizer = $moses-bin-dir/processPhraseTableMin

# conversion of rule table into binary on-disk format
#ttable-binarizer = "$moses-bin-dir/CreateOnDiskPt 1 1 4 100 2"

# tokenizers - comment out if all your data is already tokenized
#input-tokenizer = "$moses-script-dir/tokenizer/tokenizer.perl -a -l $input-
extension"
#output-tokenizer = "$moses-script-dir/tokenizer/tokenizer.perl -a -l $output-
extension"

# truecasers - comment out if you do not use the truecaser
#input-truecaser = $moses-script-dir/recaser/truecase.perl
#output-truecaser = $moses-script-dir/recaser/truecase.perl
#detruecaser = $moses-script-dir/recaser/detruecase.perl

### generic parallelizer for cluster and multi-core machines
# you may specify a script that allows the parallel execution
# parallizable steps (see meta file). you also need specify
# the number of jobs (cluster) or cores (multicore)
#
```

```
generic-parallelizer = $moses-script-dir/ems/support/generic-parallelizer.perl
#generic-parallelizer = $moses-script-dir/ems/support/generic-multicore-
parallelizer.perl

### cluster settings (if run on a cluster machine)
# number of jobs to be submitted in parallel
#
#jobs = 10

# arguments to qsub when scheduling a job
#qsub-settings = ""

# project for priviledges and usage accounting
#qsub-project = iccs_smt

# memory and time
#qsub-memory = 4
#qsub-hours = 48

### multi-core settings
# when the generic parallelizer is used, the number of cores
# specified here
#cores = 8

#################################################################
# PARALLEL CORPUS PREPARATION:
# create a tokenized, sentence-aligned corpus, ready for training

[CORPUS]

### long sentences are filtered out, since they slow down GIZA++
# and are a less reliable source of data. set here the maximum
# length of a sentence
#
max-sentence-length = 80

[CORPUS:thmy]

### command to run to get raw corpus files
#
# get-corpus-script =

### raw corpus files (untokenized, but sentence aligned)
#
raw-stem = $thmy-data/train

### tokenized corpus files (may contain long sentences)
#
```

```
#tokenized-stem = $myrk-data/train

### if sentence filtering should be skipped,
# point to the clean training data
#
#clean-stem =

### if corpus preparation should be skipped,
# point to the prepared training data
#
#lowercased-stem =

#################################################################
# LANGUAGE MODEL TRAINING

[LM]

### tool to be used for language model training
# srilm
#lm-training = /data/lttools/training/srilm/srilm-1.6.0/bin/i686-m64/ngram-count
#lm-training = /home/ros/tool/srilm-1.7.1/bin/i686-m64/ngram-count
#settings = "-interpolate -kndiscount -unk"

# irstlm training
# msb = modified kneser ney; p=0 no singleton pruning
#lm-training = "$moses-script-dir/generic/trainlm-irst2.perl -cores $cores
-irst-dir $irstlm-dir -temp-dir $working-dir/tmp"
#settings = "-s msb -p 0"

# kenlm
# additional parameters to lmplz (check lmplz help message)
#settings = "-T $working-dir/tmp -S 10G"
# this tells EMS to use lmplz and tells EMS where lmplz is located
#lm-training = "$moses-script-dir/generic/trainlm-lmplz.perl -lmplz $lmplz"
#lm-training = "$moses-script-dir/ems/support/lmplz-wrapper.perl -bin $moses-
bin-dir/lmplz"

# kenlm training
lm-training = "$moses-script-dir/ems/support/lmplz-wrapper.perl -bin $moses-bin-
dir/lmplz"
settings = "--prune '0 0 1' -T $working-dir/lm -S 20% --discount_fallback"


#lm-binarizer = $moses-bin-dir/build_binary

# order of the language model
order = 5
```

```
### tool to be used for training randomized language model from scratch
# (more commonly, a SRILM is trained)
#
#rlm-training = "$randlm-dir/buildlm -falsepos 8 -values 8"

### script to use for binary table format for irstlm or kenlm
# (default: no binarization)

# irstlm
#lm-binarizer = $irstlm-dir/compile-lm

# kenlm, also set type to 8
lm-binarizer = $moses-bin-dir/build_binary
type = 8

### script to create quantized language model format (irstlm)
# (default: no quantization)
#
#lm-quantizer = $irstlm-dir/quantize-lm

### script to use for converting into randomized table format
# (default: no randomization)
#
#lm-randomizer = "$randlm-dir/buildlm -falsepos 8 -values 8"

### each language model to be used has its own section here

[LM:thmy]

### command to run to get raw corpus files
#
#get-corpus-script = ""

### raw corpus (untokenized)
#
raw-corpus = $thmy-data/train.$output-extension

### tokenized corpus files (may contain long sentences)
#
#tokenized-corpus = $my-data/train.$output-extension

### if corpus preparation should be skipped,
# point to the prepared language model
#
#lm =

#################################################################
# INTERPOLATING LANGUAGE MODELS
```

```
[INTERPOLATED-LM]

# if multiple language models are used, these may be combined
# by optimizing perplexity on a tuning set
# see, for instance [Koehn and Schwenk, IJCNLP 2008]

### script to interpolate language models
# if commented out, no interpolation is performed
#
# script = $moses-script-dir/ems/support/interpolate-lm.perl

### tuning set
# you may use the same set that is used for mert tuning (reference set)
#
#tuning-sgm =
#raw-tuning =
#tokenized-tuning =
#factored-tuning =
#lowercased-tuning =
#split-tuning =

### group language models for hierarchical interpolation
# (flat interpolation is limited to 10 language models)
#group = "first,second fourth,fifth"

### script to use for binary table format for irstlm or kenlm
# (default: no binarization)

# irstlm
#lm-binarizer = $irstlm-dir/compile-lm

# kenlm, also set type to 8
lm-binarizer = $moses-bin-dir/build_binary
type = 8

### script to create quantized language model format (irstlm)
# (default: no quantization)
#
#lm-quantizer = $irstlm-dir/quantize-lm

### script to use for converting into randomized table format
# (default: no randomization)
#
#lm-randomizer = "$randlm-dir/buildlm -falsepos 8 -values 8"

#################################################################
# MODIFIED MOORE LEWIS FILTERING
```

```
[MML] IGNORE

### specifications for language models to be trained
#
#lm-training = $srilm-dir/ngram-count
#lm-settings = "-interpolate -kndiscount -unk"
#lm-binarizer = $moses-src-dir/bin/build_binary
#lm-query = $moses-src-dir/bin/query
#order = 5

### in-/out-of-domain source/target corpora to train the 4 language model
#
# in-domain: point either to a parallel corpus
#outdomain-stem = [CORPUS:toy:clean-split-stem]

# … or to two separate monolingual corpora
#indomain-target = [LM:toy:lowercased-corpus]
#raw-indomain-source = $toy-data/nc-5k.$input-extension

# point to out-of-domain parallel corpus
#outdomain-stem = [CORPUS:giga:clean-split-stem]

# settings: number of lines sampled from the corpora to train each language
model on
# (if used at all, should be small as a percentage of corpus)
#settings = "--line-count 100000"

#################################################################
# TRANSLATION MODEL TRAINING

[TRAINING]

### training script to be used: either a legacy script or
# current moses training script (default)
#
script = $moses-script-dir/training/train-model.perl

### general options
# these are options that are passed on to train-model.perl, for instance
# * "-mgiza -mgiza-cpus 8" to use mgiza instead of giza
# * "-sort-buffer-size 8G -sort-compress gzip" to reduce on-disk sorting
# * "-sort-parallel 8 -cores 8" to speed up phrase table building
#
#training-options = "-sort-parallel 8 -cores 8"

#training-options = "-cores 8"
```

```
### factored training: specify here which factors used
# if none specified, single factor training is assumed
# (one translation step, surface to surface)
#
#input-factors = word lemma pos morph
#output-factors = word lemma pos
#alignment-factors = "word -> word"
#translation-factors = "word -> word"
#reordering-factors = "word -> word"
#generation-factors = "word -> pos"
#decoding-steps = "t0, g0"

### parallelization of data preparation step
# the two directions of the data preparation can be run in parallel
# comment out if not needed
#
parallel = yes

### pre-computation for giza++
# giza++ has a more efficient data structure that needs to be
# initialized with snt2cooc. if run in parallel, this may reduces
# memory requirements. set here the number of parts
#
#run-giza-in-parts = 5

### symmetrization method to obtain word alignments from giza output
# (commonly used: grow-diag-final-and)
#
alignment-symmetrization-method = grow-diag-final-and

### use of Chris Dyer's fast align for word alignment
#
#fast-align-settings = "-d -o -v"

### use of berkeley aligner for word alignment
#
#use-berkeley = true
#alignment-symmetrization-method = berkeley
#berkeley-train = $moses-script-dir/ems/support/berkeley-train.sh
#berkeley-process =  $moses-script-dir/ems/support/berkeley-process.sh
#berkeley-jar = /your/path/to/berkeleyaligner-1.1/berkeleyaligner.jar
#berkeley-java-options = "-server -mx30000m -ea"
#berkeley-training-options = "-Main.iters 5 5 -EMWordAligner.numThreads 8"
#berkeley-process-options = "-EMWordAligner.numThreads 8"
#berkeley-posterior = 0.5

### use of baseline alignment model (incremental training)
#
```

```
#baseline = 68
#baseline-alignment-model = "$working-dir/training/prepared.$baseline/$input-
extension.vcb \
#  $working-dir/training/prepared.$baseline/$output-extension.vcb \
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.cooc \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.cooc \
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.thmm.5 \
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.hhmm.5 \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.thmm.5 \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.hhmm.5"

### if word alignment should be skipped,
# point to word alignment files
#
#word-alignment = $working-dir/model/aligned.1

### filtering some corpora with modified Moore-Lewis
# specify corpora to be filtered and ratio to be kept, either before or after
word alignment
#mml-filter-corpora = toy
#mml-before-wa = "-proportion 0.9"
#mml-after-wa = "-proportion 0.9"

### create a bilingual concordancer for the model
#
#biconcor = $moses-script-dir/ems/biconcor/biconcor

############################################################

### use of Operation Sequence Model
### Durrani, Schmid and Fraser. (2011): "A Joint Sequence Translation Model with
Integrated Reordering"

#operation-sequence-model = "yes"
#operation-sequence-model-order = 5
#operation-sequence-model-settings = ""

### compile Moses with --max-kenlm-order=9 if higher order is required

### if OSM training should be skipped,
# point to OSM Model
#
```

```
# osm-model =

#############################################################


### lexicalized reordering: specify orientation type
# (default: only distance-based reordering model)
#
lexicalized-reordering = msd-bidirectional-fe

### hierarchical rule set
#
#hierarchical-rule-set = true

### settings for rule extraction
#
#extract-settings = ""
max-phrase-length = 5

### add extracted phrases from baseline model
#
#baseline-extract = $working-dir/model/extract.$baseline
#
# requires aligned parallel corpus for re-estimating lexical translation
probabilities
#baseline-corpus = $working-dir/training/corpus.$baseline
#baseline-alignment = $working-dir/model/aligned.$baseline.$alignment-
symmetrization-method

### unknown word labels (target syntax only)
# enables use of unknown word labels during decoding
# label file is generated during rule extraction
#
#use-unknown-word-labels = true

### if phrase extraction should be skipped,
# point to stem for extract files
#
# extracted-phrases =

### settings for rule scoring
#
score-settings = "--GoodTuring"

### include word alignment in phrase table
#
#include-word-alignment-in-rules = yes
```

```
### sparse lexical features
#
#sparse-features = "target-word-insertion top 50, source-word-deletion top 50,
word-translation top 50 50, phrase-length"

### domain adaptation settings
# options: sparse, any of: indicator, subset, ratio
#domain-features = "subset"

### if phrase table training should be skipped,
# point to phrase translation table
#
# phrase-translation-table =

### if reordering table training should be skipped,
# point to reordering table
#
# reordering-table =

### filtering the phrase table based on significance tests
# Johnson, Martin, Foster and Kuhn. (2007): "Improving Translation Quality by
Discarding Most of the Phrasetable"
# options: -n number of translations; -l 'a+e', 'a-e', or a positive real value
-log prob threshold
#salm-index = /path/to/project/salm/Bin/Linux/Index/IndexSA.O64
#sigtest-filter = "-l a+e -n 50"

### if training should be skipped,
# point to a configuration file that contains
# pointers to all relevant model files
#
#config-with-reused-weights =

#####################################################
### TUNING: finding good weights for model components

[TUNING]

### instead of tuning with this setting, old weights may be recycled
# specify here an old configuration file with matching weights
#
#weight-config = /panfs/panltg2/users/finch/expt/moses2.1/plain.weight.ini

### tuning script to be used
#
tuning-script = $moses-script-dir/training/mert-moses.pl
tuning-settings = "-mertdir $moses-bin-dir"
```

```
### specify the corpus used for tuning
# it should contain 1000s of sentences
#
#input-sgm =
raw-input = $thmy-data/dev.$input-extension

#tokenized-input =
#factorized-input =
#input =
#
#reference-sgm =
raw-reference = $thmy-data/dev.$output-extension

#tokenized-reference =
#factorized-reference =
#reference =

### size of n-best list used (typically 100)
#
nbest = 100

### ranges for weights for random initialization
# if not specified, the tuning script will use generic ranges
# it is not clear, if this matters
#
# lambda =

### additional flags for the filter script
#
filter-settings = ""

### additional flags for the decoder
#
decoder-settings = ""

### if tuning should be skipped, specify this here
# and also point to a configuration file that contains
# pointers to all relevant model files
#
#config =

#########################################################
## RECASER: restore case, this part only trains the model

[RECASING]

#decoder = $moses-bin-dir/moses
```

```
### training data
# raw input needs to be still tokenized,
# also also tokenized input may be specified
#
#tokenized = [LM:europarl:tokenized-corpus]

# recase-config =

#lm-training = $srilm-dir/ngram-count

#######################################################
## TRUECASER: train model to truecase corpora and input

[TRUECASER]

### script to train truecaser models
#
#trainer = $moses-script-dir/recaser/train-truecaser.perl

### training data
# data on which truecaser is trained
# if no training data is specified, parallel corpus is used
#
# raw-stem =
# tokenized-stem =

### trained model
#
# truecase-model =

######################################################################
## EVALUATION: translating a test set using the tuned system and score it

[EVALUATION]

### additional flags for the filter script
#
#filter-settings = ""

### additional decoder settings
# switches for the Moses decoder
# common choices:
#   "-threads N" for multi-threading
#   "-mbr" for MBR decoding
#   "-drop-unknown" for dropping unknown source words
#   "-search-algorithm 1 -cube-pruning-pop-limit 5000 -s 5000" for cube pruning
#
decoder-settings = "-search-algorithm 1 -cube-pruning-pop-limit 5000 -s 5000"
```

```
### specify size of n-best list, if produced
#
#nbest = 100

### multiple reference translations
#
#multiref = yes

### prepare system output for scoring
# this may include detokenization and wrapping output in sgm
# (needed for nist-bleu, ter, meteor)
#
#detokenizer = "$moses-script-dir/tokenizer/detokenizer.perl -l $output-
extension"
#recaser = $moses-script-dir/recaser/recase.perl
wrapping-script = "$moses-script-dir/ems/support/wrap-xml.perl $output-
extension"
#output-sgm =

### BLEU
#
nist-bleu = $moses-script-dir/generic/mteval-v13a.pl
nist-bleu-c = "$moses-script-dir/generic/mteval-v13a.pl -c"
multi-bleu = $moses-script-dir/generic/multi-bleu.perl
#ibm-bleu =

### TER: translation error rate (BBN metric) based on edit distance
# not yet integrated
#
# ter =

### METEOR: gives credit to stem / worknet synonym matches
# not yet integrated
#
# meteor =

### Analysis: carry out various forms of analysis on the output
#
analysis = $moses-script-dir/ems/support/analysis.perl
#
# also report on input coverage
analyze-coverage = yes
#
# also report on phrase mappings used
report-segmentation = yes
#
# report precision of translations for each input word, broken down by
```

```
# count of input word in corpus and model
report-precision-by-coverage = yes
#
# further precision breakdown by factor
#precision-by-coverage-factor = pos
#
# visualization of the search graph in tree-based models
#analyze-search-graph = yes

[EVALUATION:test]

### input data
#
#input-sgm = /panfs/panmt/users/ye/cluster/test-sgm/test.$input-
extension.src.sgm
input-sgm = $thmy-data/test-sgm/test.$input-extension.src.sgm
#input-sgm = $mykc-data/test-sgm/test.$input-extension.src.sgm

# raw-input = $mykc-data/test.$input-extension
# tokenized-input =
# factorized-input =
# input =

### reference data
#
#reference-sgm = /panfs/panmt/users/ye/cluster/test-sgm/test.$output-
extension.ref.sgm
reference-sgm = $thmy-data/test-sgm/test.$output-extension.ref.sgm
#reference-sgm = $mykc-data/test-sgm/test.$output-extension.ref.sgm

# raw-reference = $mykc-data/test.$output-extension
# tokenized-reference =
# reference =

### analysis settings
# may contain any of the general evaluation analysis settings
# specific setting: base coverage statistics on earlier run
#
#precision-by-coverage-base = $working-dir/evaluation/test.analysis.5

### wrapping frame
# for nist-bleu and other scoring scripts, the output needs to be wrapped
# in sgm markup (typically like the input sgm)
#
wrapping-frame = $input-sgm

##########################################
### REPORTING: summarize evaluation scores
```

```
[REPORTING]

### currently no parameters for reporting section
```

1.6 Check th-my Config File

[46]:
```
!cat ./baseline/th-my/config.baseline.th-my
```

```
# Config file for th-my (baseline)

[GENERAL]
working-dir = /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/th-my
input-extension = th
output-extension = my
pair-extension = th-my


### directories that contain tools and data
#
# moses
#moses-src-dir = /home/ros/mosesdecoder
#moses-src-dir = /home/lar/tool/moses/
#moses-src-dir = /home/ye/tool/mosesbin/ubuntu-17.04/moses
moses-src-dir = /home/ye/tool/mosesbin/ubuntu-17.04/moses

# moses binaries
moses-bin-dir = $moses-src-dir/bin

# moses scripts
moses-script-dir = $moses-src-dir/scripts

# directory where GIZA++/MGIZA programs resides
#external-bin-dir = /home/lar/tool/giza-pp-master/mkcls-v2
#external-bin-dir = /home/lar/tool/giza-pp-master/GIZA++-v2
#external-bin-dir = /home/lar/tool/giza-pp-master/
#external-bin-dir = /home/ye/tool/mgiza/mgizapp/bin
external-bin-dir = /home/ye/tool/giza-pp/GIZA++-v2

# srilm
#srilm-dir = /data/lttools/training/srilm/srilm-1.6.0/bin/i686-m64
#I haven't installed SRILM yet on Deep Learning Box computer
#srilm-dir = /home/ros/tool/srilm-1.7.1/bin/i686-m64

# irstlm
#irstlm-dir = $moses-src-dir/irstlm/bin

# randlm
```

```
#randlm-dir = $moses-src-dir/randlm/bin

# kenlm
lmplz = $moses-bin-dir/lmplz

# data
#myrk-data = /home/lar/experiment/my-rk/smt1/t9
#mykc-data = /home/lar/experiment/kachin-myanmar/demo-mykc-smt/4demo/
#mykc-data = /home/lar/experiment/kachin-myanmar/demo-mykc-smt/4demo2/
#myrk-data = /media/lar/Transcend/student/lecture/mtrss/pbsmt-demo/pbsmt/data/
thmy-data = /home/ye/exp/SMT-NMT_tutorial/clean-data/big/

### basic tools
#
# moses decoder
decoder = $moses-bin-dir/moses

# conversion of phrase table into binary on-disk format
#ttable-binarizer = $moses-bin-dir/processPhraseTable
#ttable-binarizer = $moses-bin-dir/processPhraseTableMin

# conversion of rule table into binary on-disk format
#ttable-binarizer = "$moses-bin-dir/CreateOnDiskPt 1 1 4 100 2"

# tokenizers - comment out if all your data is already tokenized
#input-tokenizer = "$moses-script-dir/tokenizer/tokenizer.perl -a -l $input-
extension"
#output-tokenizer = "$moses-script-dir/tokenizer/tokenizer.perl -a -l $output-
extension"

# truecasers - comment out if you do not use the truecaser
#input-truecaser = $moses-script-dir/recaser/truecase.perl
#output-truecaser = $moses-script-dir/recaser/truecase.perl
#detruecaser = $moses-script-dir/recaser/detruecase.perl

### generic parallelizer for cluster and multi-core machines
# you may specify a script that allows the parallel execution
# parallizable steps (see meta file). you also need specify
# the number of jobs (cluster) or cores (multicore)
#
generic-parallelizer = $moses-script-dir/ems/support/generic-parallelizer.perl
#generic-parallelizer = $moses-script-dir/ems/support/generic-multicore-
parallelizer.perl

### cluster settings (if run on a cluster machine)
# number of jobs to be submitted in parallel
#
#jobs = 10
```

```
# arguments to qsub when scheduling a job
#qsub-settings = ""

# project for priviledges and usage accounting
#qsub-project = iccs_smt

# memory and time
#qsub-memory = 4
#qsub-hours = 48

### multi-core settings
# when the generic parallelizer is used, the number of cores
# specified here
#cores = 8

#################################################################
# PARALLEL CORPUS PREPARATION:
# create a tokenized, sentence-aligned corpus, ready for training

[CORPUS]

### long sentences are filtered out, since they slow down GIZA++
# and are a less reliable source of data. set here the maximum
# length of a sentence
#
max-sentence-length = 80

[CORPUS:thmy]

### command to run to get raw corpus files
#
# get-corpus-script =

### raw corpus files (untokenized, but sentence aligned)
#
raw-stem = $thmy-data/train

### tokenized corpus files (may contain long sentences)
#
#tokenized-stem = $myrk-data/train

### if sentence filtering should be skipped,
# point to the clean training data
#
#clean-stem =

### if corpus preparation should be skipped,
```

```
# point to the prepared training data
#
#lowercased-stem =

#################################################################
# LANGUAGE MODEL TRAINING

[LM]

### tool to be used for language model training
# srilm
#lm-training = /data/lttools/training/srilm/srilm-1.6.0/bin/i686-m64/ngram-count
#lm-training = /home/ros/tool/srilm-1.7.1/bin/i686-m64/ngram-count
#settings = "-interpolate -kndiscount -unk"

# irstlm training
# msb = modified kneser ney; p=0 no singleton pruning
#lm-training = "$moses-script-dir/generic/trainlm-irst2.perl -cores $cores
-irst-dir $irstlm-dir -temp-dir $working-dir/tmp"
#settings = "-s msb -p 0"

# kenlm
# additional parameters to lmplz (check lmplz help message)
#settings = "-T $working-dir/tmp -S 10G"
# this tells EMS to use lmplz and tells EMS where lmplz is located
#lm-training = "$moses-script-dir/generic/trainlm-lmplz.perl -lmplz $lmplz"
#lm-training = "$moses-script-dir/ems/support/lmplz-wrapper.perl -bin $moses-
bin-dir/lmplz"

# kenlm training
lm-training = "$moses-script-dir/ems/support/lmplz-wrapper.perl -bin $moses-bin-
dir/lmplz"
settings = "--prune '0 0 1' -T $working-dir/lm -S 20% --discount_fallback"


#lm-binarizer = $moses-bin-dir/build_binary

# order of the language model
order = 5

### tool to be used for training randomized language model from scratch
# (more commonly, a SRILM is trained)
#
#rlm-training = "$randlm-dir/buildlm -falsepos 8 -values 8"

### script to use for binary table format for irstlm or kenlm
# (default: no binarization)
```

```
# irstlm
#lm-binarizer = $irstlm-dir/compile-lm

# kenlm, also set type to 8
lm-binarizer = $moses-bin-dir/build_binary
type = 8

### script to create quantized language model format (irstlm)
# (default: no quantization)
#
#lm-quantizer = $irstlm-dir/quantize-lm

### script to use for converting into randomized table format
# (default: no randomization)
#
#lm-randomizer = "$randlm-dir/buildlm -falsepos 8 -values 8"

### each language model to be used has its own section here

[LM:thmy]

### command to run to get raw corpus files
#
#get-corpus-script = ""

### raw corpus (untokenized)
#
raw-corpus = $thmy-data/train.$output-extension

### tokenized corpus files (may contain long sentences)
#
#tokenized-corpus = $my-data/train.$output-extension

### if corpus preparation should be skipped,
# point to the prepared language model
#
#lm =

#################################################################
# INTERPOLATING LANGUAGE MODELS

[INTERPOLATED-LM]

# if multiple language models are used, these may be combined
# by optimizing perplexity on a tuning set
# see, for instance [Koehn and Schwenk, IJCNLP 2008]

### script to interpolate language models
```

```
# if commented out, no interpolation is performed
#
# script = $moses-script-dir/ems/support/interpolate-lm.perl

### tuning set
# you may use the same set that is used for mert tuning (reference set)
#
#tuning-sgm =
#raw-tuning =
#tokenized-tuning =
#factored-tuning =
#lowercased-tuning =
#split-tuning =

### group language models for hierarchical interpolation
# (flat interpolation is limited to 10 language models)
#group = "first,second fourth,fifth"

### script to use for binary table format for irstlm or kenlm
# (default: no binarization)

# irstlm
#lm-binarizer = $irstlm-dir/compile-lm

# kenlm, also set type to 8
lm-binarizer = $moses-bin-dir/build_binary
type = 8

### script to create quantized language model format (irstlm)
# (default: no quantization)
#
#lm-quantizer = $irstlm-dir/quantize-lm

### script to use for converting into randomized table format
# (default: no randomization)
#
#lm-randomizer = "$randlm-dir/buildlm -falsepos 8 -values 8"

#################################################################
# MODIFIED MOORE LEWIS FILTERING

[MML] IGNORE

### specifications for language models to be trained
#
#lm-training = $srilm-dir/ngram-count
#lm-settings = "-interpolate -kndiscount -unk"
#lm-binarizer = $moses-src-dir/bin/build_binary
```

```
#lm-query = $moses-src-dir/bin/query
#order = 5

### in-/out-of-domain source/target corpora to train the 4 language model
#
# in-domain: point either to a parallel corpus
#outdomain-stem = [CORPUS:toy:clean-split-stem]

# … or to two separate monolingual corpora
#indomain-target = [LM:toy:lowercased-corpus]
#raw-indomain-source = $toy-data/nc-5k.$input-extension

# point to out-of-domain parallel corpus
#outdomain-stem = [CORPUS:giga:clean-split-stem]

# settings: number of lines sampled from the corpora to train each language
model on
# (if used at all, should be small as a percentage of corpus)
#settings = "--line-count 100000"

#################################################################
# TRANSLATION MODEL TRAINING

[TRAINING]

### training script to be used: either a legacy script or
# current moses training script (default)
#
script = $moses-script-dir/training/train-model.perl

### general options
# these are options that are passed on to train-model.perl, for instance
# * "-mgiza -mgiza-cpus 8" to use mgiza instead of giza
# * "-sort-buffer-size 8G -sort-compress gzip" to reduce on-disk sorting
# * "-sort-parallel 8 -cores 8" to speed up phrase table building
#
#training-options = "-sort-parallel 8 -cores 8"

#training-options = "-cores 8"

### factored training: specify here which factors used
# if none specified, single factor training is assumed
# (one translation step, surface to surface)
#
#input-factors = word lemma pos morph
#output-factors = word lemma pos
#alignment-factors = "word -> word"
#translation-factors = "word -> word"
```

```
#reordering-factors = "word -> word"
#generation-factors = "word -> pos"
#decoding-steps = "t0, g0"

### parallelization of data preparation step
# the two directions of the data preparation can be run in parallel
# comment out if not needed
#
parallel = yes

### pre-computation for giza++
# giza++ has a more efficient data structure that needs to be
# initialized with snt2cooc. if run in parallel, this may reduces
# memory requirements. set here the number of parts
#
#run-giza-in-parts = 5

### symmetrization method to obtain word alignments from giza output
# (commonly used: grow-diag-final-and)
#
alignment-symmetrization-method = grow-diag-final-and

### use of Chris Dyer's fast align for word alignment
#
#fast-align-settings = "-d -o -v"

### use of berkeley aligner for word alignment
#
#use-berkeley = true
#alignment-symmetrization-method = berkeley
#berkeley-train = $moses-script-dir/ems/support/berkeley-train.sh
#berkeley-process =  $moses-script-dir/ems/support/berkeley-process.sh
#berkeley-jar = /your/path/to/berkeleyaligner-1.1/berkeleyaligner.jar
#berkeley-java-options = "-server -mx30000m -ea"
#berkeley-training-options = "-Main.iters 5 5 -EMWordAligner.numThreads 8"
#berkeley-process-options = "-EMWordAligner.numThreads 8"
#berkeley-posterior = 0.5

### use of baseline alignment model (incremental training)
#
#baseline = 68
#baseline-alignment-model = "$working-dir/training/prepared.$baseline/$input-
extension.vcb \
#  $working-dir/training/prepared.$baseline/$output-extension.vcb \
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.cooc \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.cooc \
```

```
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.thmm.5 \
#  $working-dir/training/giza.$baseline/${output-extension}-$input-
extension.hhmm.5 \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.thmm.5 \
#  $working-dir/training/giza-inverse.$baseline/${input-extension}-$output-
extension.hhmm.5"

### if word alignment should be skipped,
# point to word alignment files
#
#word-alignment = $working-dir/model/aligned.1

### filtering some corpora with modified Moore-Lewis
# specify corpora to be filtered and ratio to be kept, either before or after
word alignment
#mml-filter-corpora = toy
#mml-before-wa = "-proportion 0.9"
#mml-after-wa = "-proportion 0.9"

### create a bilingual concordancer for the model
#
#biconcor = $moses-script-dir/ems/biconcor/biconcor

############################################################

### use of Operation Sequence Model
### Durrani, Schmid and Fraser. (2011): "A Joint Sequence Translation Model with
Integrated Reordering"

#operation-sequence-model = "yes"
#operation-sequence-model-order = 5
#operation-sequence-model-settings = ""

### compile Moses with --max-kenlm-order=9 if higher order is required

### if OSM training should be skipped,
# point to OSM Model
#
# osm-model =

############################################################


### lexicalized reordering: specify orientation type
# (default: only distance-based reordering model)
#
```

```
lexicalized-reordering = msd-bidirectional-fe

### hierarchical rule set
#
#hierarchical-rule-set = true

### settings for rule extraction
#
#extract-settings = ""
max-phrase-length = 5

### add extracted phrases from baseline model
#
#baseline-extract = $working-dir/model/extract.$baseline
#
# requires aligned parallel corpus for re-estimating lexical translation
probabilities
#baseline-corpus = $working-dir/training/corpus.$baseline
#baseline-alignment = $working-dir/model/aligned.$baseline.$alignment-
symmetrization-method

### unknown word labels (target syntax only)
# enables use of unknown word labels during decoding
# label file is generated during rule extraction
#
#use-unknown-word-labels = true

### if phrase extraction should be skipped,
# point to stem for extract files
#
# extracted-phrases =

### settings for rule scoring
#
score-settings = "--GoodTuring"

### include word alignment in phrase table
#
#include-word-alignment-in-rules = yes

### sparse lexical features
#
#sparse-features = "target-word-insertion top 50, source-word-deletion top 50,
word-translation top 50 50, phrase-length"

### domain adaptation settings
# options: sparse, any of: indicator, subset, ratio
#domain-features = "subset"
```

```
### if phrase table training should be skipped,
# point to phrase translation table
#
# phrase-translation-table =

### if reordering table training should be skipped,
# point to reordering table
#
# reordering-table =

### filtering the phrase table based on significance tests
# Johnson, Martin, Foster and Kuhn. (2007): "Improving Translation Quality by
Discarding Most of the Phrasetable"
# options: -n number of translations; -l 'a+e', 'a-e', or a positive real value
-log prob threshold
#salm-index = /path/to/project/salm/Bin/Linux/Index/IndexSA.O64
#sigtest-filter = "-l a+e -n 50"

### if training should be skipped,
# point to a configuration file that contains
# pointers to all relevant model files
#
#config-with-reused-weights =

#####################################################
### TUNING: finding good weights for model components

[TUNING]

### instead of tuning with this setting, old weights may be recycled
# specify here an old configuration file with matching weights
#
#weight-config = /panfs/panltg2/users/finch/expt/moses2.1/plain.weight.ini

### tuning script to be used
#
tuning-script = $moses-script-dir/training/mert-moses.pl
tuning-settings = "-mertdir $moses-bin-dir"

### specify the corpus used for tuning
# it should contain 1000s of sentences
#
#input-sgm =
raw-input = $thmy-data/dev.$input-extension

#tokenized-input =
#factorized-input =
```

```
#input =
#
#reference-sgm =
raw-reference = $thmy-data/dev.$output-extension

#tokenized-reference =
#factorized-reference =
#reference =

### size of n-best list used (typically 100)
#
nbest = 100

### ranges for weights for random initialization
# if not specified, the tuning script will use generic ranges
# it is not clear, if this matters
#
# lambda =

### additional flags for the filter script
#
filter-settings = ""

### additional flags for the decoder
#
decoder-settings = ""

### if tuning should be skipped, specify this here
# and also point to a configuration file that contains
# pointers to all relevant model files
#
#config =

#########################################################
## RECASER: restore case, this part only trains the model

[RECASING]

#decoder = $moses-bin-dir/moses

### training data
# raw input needs to be still tokenized,
# also also tokenized input may be specified
#
#tokenized = [LM:europarl:tokenized-corpus]

# recase-config =
```

```
#lm-training = $srilm-dir/ngram-count

#######################################################
## TRUECASER: train model to truecase corpora and input

[TRUECASER]

### script to train truecaser models
#
#trainer = $moses-script-dir/recaser/train-truecaser.perl

### training data
# data on which truecaser is trained
# if no training data is specified, parallel corpus is used
#
# raw-stem =
# tokenized-stem =

### trained model
#
# truecase-model =

#####################################################################
## EVALUATION: translating a test set using the tuned system and score it

[EVALUATION]

### additional flags for the filter script
#
#filter-settings = ""

### additional decoder settings
# switches for the Moses decoder
# common choices:
#   "-threads N" for multi-threading
#   "-mbr" for MBR decoding
#   "-drop-unknown" for dropping unknown source words
#   "-search-algorithm 1 -cube-pruning-pop-limit 5000 -s 5000" for cube pruning
#
decoder-settings = "-search-algorithm 1 -cube-pruning-pop-limit 5000 -s 5000"

### specify size of n-best list, if produced
#
#nbest = 100

### multiple reference translations
#
#multiref = yes
```

```
### prepare system output for scoring
# this may include detokenization and wrapping output in sgm
# (needed for nist-bleu, ter, meteor)
#
#detokenizer = "$moses-script-dir/tokenizer/detokenizer.perl -l $output-
extension"
#recaser = $moses-script-dir/recaser/recase.perl
wrapping-script = "$moses-script-dir/ems/support/wrap-xml.perl $output-
extension"
#output-sgm =

### BLEU
#
nist-bleu = $moses-script-dir/generic/mteval-v13a.pl
nist-bleu-c = "$moses-script-dir/generic/mteval-v13a.pl -c"
multi-bleu = $moses-script-dir/generic/multi-bleu.perl
#ibm-bleu =

### TER: translation error rate (BBN metric) based on edit distance
# not yet integrated
#
# ter =

### METEOR: gives credit to stem / worknet synonym matches
# not yet integrated
#
# meteor =

### Analysis: carry out various forms of analysis on the output
#
analysis = $moses-script-dir/ems/support/analysis.perl
#
# also report on input coverage
analyze-coverage = yes
#
# also report on phrase mappings used
report-segmentation = yes
#
# report precision of translations for each input word, broken down by
# count of input word in corpus and model
report-precision-by-coverage = yes
#
# further precision breakdown by factor
#precision-by-coverage-factor = pos
#
# visualization of the search graph in tree-based models
#analyze-search-graph = yes
```

```
[EVALUATION:test]

### input data
#
#input-sgm = /panfs/panmt/users/ye/cluster/test-sgm/test.$input-
extension.src.sgm
input-sgm = $thmy-data/test-sgm/test.$input-extension.src.sgm
#input-sgm = $mykc-data/test-sgm/test.$input-extension.src.sgm

# raw-input = $mykc-data/test.$input-extension
# tokenized-input =
# factorized-input =
# input =

### reference data
#
#reference-sgm = /panfs/panmt/users/ye/cluster/test-sgm/test.$output-
extension.ref.sgm
reference-sgm = $thmy-data/test-sgm/test.$output-extension.ref.sgm
#reference-sgm = $mykc-data/test-sgm/test.$output-extension.ref.sgm

# raw-reference = $mykc-data/test.$output-extension
# tokenized-reference =
# reference =

### analysis settings
# may contain any of the general evaluation analysis settings
# specific setting: base coverage statistics on earlier run
#
#precision-by-coverage-base = $working-dir/evaluation/test.analysis.5

### wrapping frame
# for nist-bleu and other scoring scripts, the output needs to be wrapped
# in sgm markup (typically like the input sgm)
#
wrapping-frame = $input-sgm

##########################################
### REPORTING: summarize evaluation scores

[REPORTING]

### currently no parameters for reporting section
```

1.7 Start PBSMT

Myanmar-Thai, Thai-Myanmar ဘာသာပြန်ဖို့အတွက် Phrase-based Statistical Machine Translatin (PBSMT) နည်းလမ်းနဲ့ training/tuning/testing တွေကို လုပ်ခိုင်းကြည့်ရအောင်။

[47]:
```
%pwd
```

[47]: '/home/ye/exp/SMT-NMT_tutorial/pbsmt-big'

[48]:
```
!time perl ./run-baseline.pl
```

```
my-th-baseline /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/config.baseline.my-th
th-my-baseline /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/th-
my/config.baseline.th-my

real    143m6.061s
user    0m10.019s
sys     0m9.106s
```

1.8 Let's Check

၂ နာရီခွဲ နီးပါး ကြာပါတယ်။
တကယ်တမ်း မြန်မာ-ထိုင်း၊ ထိုင်း-မြန်မာ နှစ်ဘက်စလုံးကို အဆင်ပြေပြေနဲ့ PBSMT လုပ်သွားသလား ဆိုတာကို စစ်ကြည့်ကြရအောင်။

အရင်ဆုံး folder structure တွေကနေ ကြည့်ကြရအောင်။

[50]:
```
!tree -L 3 .
```

```
.
  baseline
      my-th
          config.baseline.my-th
          corpus
          evaluation
          lm
          model
          steps
          training
      th-my
          config.baseline.th-my
          corpus
          evaluation
          lm
          model
          steps
          training
          tuning
```

```
config.baseline
generate_configs.pl
run1.log
run-baseline.pl
steps

18 directories, 6 files
```

အထက်ပါ folder structure ကနေ th-my အတွက်က အဆင့်အားလုံး ပြီးသွားတဲ့ ပုံစံရှိပေမဲ့ my-th အတွက်က tuning ဆိုတဲ့ ဖိုလ်ဒါ မပါလာတာကို တွေ့ရပါလိမ့်မယ်။

ဆိုတော့ကာ my-th အတွက်က training အဆင့်မှာ error ဖြစ်ခဲ့တယ်လို့ အကြမ်းပြောလို့ ရပါတယ်။

my-th ရဲ့ Error log တွေကို သွားကြည့်ကြရအောင်။

[51]:
```
%cd ./baseline/my-th/
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th
```

[52]:
```
!tree .
```

```
.
  config.baseline.my-th
  corpus
     thmy.clean.1.lines-retained
     thmy.clean.1.my
     thmy.clean.1.th
  evaluation
     test.input.txt.1
     test.reference.txt.1
  lm
     thmy.binlm.1
     thmy.lm.1
  model
     aligned.1.grow-diag-final-and
     extract.1.inv.sorted.gz
     extract.1.o.sorted.gz
     extract.1.sorted.gz
     lex.1.e2f
     lex.1.f2e
  steps
     1
        config.1
        CORPUS_thmy_clean.1
        CORPUS_thmy_clean.1.DONE
        CORPUS_thmy_clean.1.INFO
        CORPUS_thmy_clean.1.STDERR
        CORPUS_thmy_clean.1.STDERR.digest
        CORPUS_thmy_clean.1.STDOUT
```

```
EVALUATION_test_input-from-sgm.1
EVALUATION_test_input-from-sgm.1.DONE
EVALUATION_test_input-from-sgm.1.INFO
EVALUATION_test_input-from-sgm.1.STDERR
EVALUATION_test_input-from-sgm.1.STDERR.digest
EVALUATION_test_input-from-sgm.1.STDOUT
EVALUATION_test_reference-from-sgm.1
EVALUATION_test_reference-from-sgm.1.DONE
EVALUATION_test_reference-from-sgm.1.INFO
EVALUATION_test_reference-from-sgm.1.STDERR
EVALUATION_test_reference-from-sgm.1.STDERR.digest
EVALUATION_test_reference-from-sgm.1.STDOUT
graph.1.dot
graph.1.png
graph.1.ps
LM_thmy_binarize.1
LM_thmy_binarize.1.DONE
LM_thmy_binarize.1.INFO
LM_thmy_binarize.1.STDERR
LM_thmy_binarize.1.STDERR.digest
LM_thmy_binarize.1.STDOUT
LM_thmy_train.1
LM_thmy_train.1.DONE
LM_thmy_train.1.INFO
LM_thmy_train.1.STDERR
LM_thmy_train.1.STDERR.digest
LM_thmy_train.1.STDOUT
parameter.1
re-use.1
running.1
TRAINING_build-lex-trans.1
TRAINING_build-lex-trans.1.DONE
TRAINING_build-lex-trans.1.INFO
TRAINING_build-lex-trans.1.STDERR
TRAINING_build-lex-trans.1.STDERR.digest
TRAINING_build-lex-trans.1.STDOUT
TRAINING_consolidate.1
TRAINING_consolidate.1.DONE
TRAINING_consolidate.1.INFO
TRAINING_consolidate.1.STDERR
TRAINING_consolidate.1.STDERR.digest
TRAINING_consolidate.1.STDOUT
TRAINING_extract-phrases.1
TRAINING_extract-phrases.1.DONE
TRAINING_extract-phrases.1.INFO
TRAINING_extract-phrases.1.STDERR
TRAINING_extract-phrases.1.STDERR.digest
TRAINING_extract-phrases.1.STDOUT
```

```
        TRAINING_prepare-data.1
        TRAINING_prepare-data.1.DONE
        TRAINING_prepare-data.1.INFO
        TRAINING_prepare-data.1.STDERR
        TRAINING_prepare-data.1.STDERR.digest
        TRAINING_prepare-data.1.STDOUT
        TRAINING_run-giza.1
        TRAINING_run-giza.1.DONE
        TRAINING_run-giza.1.INFO
        TRAINING_run-giza.1.STDERR
        TRAINING_run-giza.1.STDERR.digest
        TRAINING_run-giza.1.STDOUT
        TRAINING_run-giza-inverse.1
        TRAINING_run-giza-inverse.1.DONE
        TRAINING_run-giza-inverse.1.INFO
        TRAINING_run-giza-inverse.1.STDERR
        TRAINING_run-giza-inverse.1.STDERR.digest
        TRAINING_run-giza-inverse.1.STDOUT
        TRAINING_symmetrize-giza.1
        TRAINING_symmetrize-giza.1.DONE
        TRAINING_symmetrize-giza.1.INFO
        TRAINING_symmetrize-giza.1.STDERR
        TRAINING_symmetrize-giza.1.STDERR.digest
        TRAINING_symmetrize-giza.1.STDOUT
   training
       corpus.1.my -> /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/corpus/thmy.clean.1.my
       corpus.1.th -> /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/corpus/thmy.clean.1.th
       giza.1
           th-my.A3.final.gz
           th-my.cooc
           th-my.gizacfg
       giza-inverse.1
           my-th.A3.final.gz
           my-th.cooc
           my-th.gizacfg
       prepared.1
           my-th-int-train.snt
           my.vcb
           my.vcb.classes
           my.vcb.classes.cats
           th-my-int-train.snt
           th.vcb
           th.vcb.classes
           th.vcb.classes.cats

11 directories, 109 files
```

[53]:
```
%cd steps/1
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/steps/1
```

အတန်းထဲမှာလည်း ပြောခဲ့သလိုပဲ။ STDERR ဆိုတဲ့ ဖိုင် extension တွေထဲမှာ error ရှိရင် အဲဒီ error ဖြစ်နိုင်တဲ့ information ကို ရေးပေးထားတာမို့ အဲဒီကနေ SMT debugging လုပ်ကြရပါတယ်လို့။ ကြည့်ကြည့်ရအောင်။

[54]:
```
!tail *.STDERR
```

```
==> CORPUS_thmy_clean.1.STDERR <==
clean-corpus.perl: processing /home/ye/exp/SMT-NMT_tutorial/clean-
data/big//train.my & .th to /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/corpus/thmy.clean.1, cutoff 1-80, ratio 9
…
Input sentences: 62625  Output sentences:  62569

==> EVALUATION_test_input-from-sgm.1.STDERR <==

==> EVALUATION_test_reference-from-sgm.1.STDERR <==

==> LM_thmy_binarize.1.STDERR <==
Reading /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/lm/thmy.lm.1
----5---10---15---20---25---30---35---40---45---50---55---60---65---70---75---80
---85---90---95--100
********************************************************************************
********************
SUCCESS

==> LM_thmy_train.1.STDERR <==
trie     2785 without quantization
trie     1429 assuming -q 8 -b 8 quantization
trie     2547 assuming -a 22 array pointer compression
trie     1191 assuming -a 22 -q 8 -b 8 array pointer compression and quantization
=== 3/5 Calculating and sorting initial probabilities ===
Chain sizes: 1:52968 2:623872 3:1489600 4:2045256 5:2224656
=== 4/5 Calculating and writing order-interpolated probabilities ===
Chain sizes: 1:52968 2:623872 3:1489600 4:2045256 5:2224656
=== 5/5 Writing ARPA model ===
Name:lmplz      VmPeak:13285628 kB      VmRSS:12204 kB  RSSMax:2331964 kB
user:0.231548   sys:0.539943    CPU:0.771491    real:0.693065

==> TRAINING_build-lex-trans.1.STDERR <==
Using SCRIPTS_ROOTDIR: /home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts
using pigz
(4) generate lexical translation table 0-0 @ Thu Apr 30 13:10:39 +07 2026
(/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/corpus.1.my,/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/corpus.1.th,/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
```

```
th/model/lex.1)
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
Saved: /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model/lex.1.f2e
and /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model/lex.1.e2f

==> TRAINING_consolidate.1.STDERR <==

==> TRAINING_extract-phrases.1.STDERR <==
.
.ERROR: some opened tags were never closed:                 <
    >









==> TRAINING_prepare-data.1.STDERR <==
(1.1) running mkcls  @ Thu Apr 30 13:09:33 +07 2026
/home/ye/tool/giza-pp/GIZA++-v2/mkcls -c50 -n2 -p/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.my -V/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/prepared.1/my.vcb.classes opt
(1.2) creating vcb file /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/prepared.1/my.vcb @ Thu Apr 30 13:09:33 +07 2026
Executing: /home/ye/tool/giza-pp/GIZA++-v2/mkcls -c50 -n2 -p/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.my -V/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/prepared.1/my.vcb.classes opt
(1.1) running mkcls  @ Thu Apr 30 13:09:33 +07 2026
/home/ye/tool/giza-pp/GIZA++-v2/mkcls -c50 -n2 -p/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.th -V/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/prepared.1/th.vcb.classes opt
Executing: /home/ye/tool/giza-pp/GIZA++-v2/mkcls -c50 -n2 -p/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.th -V/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/prepared.1/th.vcb.classes opt
(1.2) creating vcb file /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/prepared.1/th.vcb @ Thu Apr 30 13:09:33 +07 2026
(1.3) numberizing corpus /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/prepared.1/my-th-int-train.snt @ Thu Apr 30 13:09:33 +07 2026
(1.3) numberizing corpus /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/prepared.1/th-my-int-train.snt @ Thu Apr 30 13:09:33 +07 2026

==> TRAINING_run-giza.1.STDERR <==
NTable contains 19730 parameter.
10000
20000
```

```
30000
40000
50000
60000
NTable contains 19730 parameter.
Executing: rm -f /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/giza.1/th-my.A3.final.gz
Executing: pigz /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/giza.1/th-my.A3.final

==> TRAINING_run-giza-inverse.1.STDERR <==
NTable contains 44040 parameter.
10000
20000
30000
40000
50000
60000
NTable contains 44040 parameter.
Executing: rm -f /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/giza-inverse.1/my-th.A3.final.gz
Executing: pigz /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/giza-inverse.1/my-th.A3.final

==> TRAINING_symmetrize-giza.1.STDERR <==
Using SCRIPTS_ROOTDIR: /home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts
using pigz
(3) generate word alignment @ Thu Apr 30 13:10:27 +07 2026
Combining forward and inverted alignment from files:
  /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/giza-
inverse.1/my-th.A3.final.{bz2,gz}
  /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/giza.1/th-
my.A3.final.{bz2,gz}
Executing: mkdir -p ./model
Executing:
/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/training/giza2bal.pl -d "pigz
-cd /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/giza.1/th-
my.A3.final.gz" -i "pigz -cd /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/training/giza-inverse.1/my-th.A3.final.gz"
|/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/../bin/symal
-alignment="grow" -diagonal="yes" -final="yes" -both="yes" > /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/aligned.1.grow-diag-final-and
symal: computing grow alignment: diagonal (1) final (1)both-uncovered (1)
skip=<0> counts=<62569>
```

1.9 FYI

အထက်ပါအတိုင်း .STDERR နဲ့ ဆုံးတဲ့ ဖိုင်အားလုံးရဲ့ နောက်ဆုံးပိုင်းတွေကို ရိုက်ထုတ်ကြည့်ခိုင်းလိုက်တဲ့အခါမှာ

ERROR နဲ့ ပတ်သက်ပြီး TRAINING_extract-phrases.1.STDERR ဆိုတဲ့ ဖိုင်မှာ တခုခုကို ပြောနေတာကို သွားတွေ့ရပါတယ်။ အဲဒါကြောင့် အဲဒီဖိုင် တစ်ခုလုံးကို ရိုက်ထုတ်ကြည့်ပြီး လေ့လာကြရအောင်။

[56]:
```
!cat TRAINING_extract-phrases.1.STDERR
```

```
Using SCRIPTS_ROOTDIR: /home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts
using pigz
(5) extract phrases @ Thu Apr 30 13:10:34 +07 2026
/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/generic/extract-parallel.perl
24 split "sort    "
/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/../bin/extract
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.th
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.my
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model/aligned.1.grow-
diag-final-and /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/extract.1 5 orientation --model wbe-msd --GZOutput
Executing: /home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/generic/extract-
parallel.perl 24 split "sort    "
/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/../bin/extract
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.th
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.my
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model/aligned.1.grow-
diag-final-and /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/extract.1 5 orientation --model wbe-msd --GZOutput
using pigz
isBSDSplit=0
Executing: mkdir -p /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347; ls -l /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347
split -d -l 2608 -a 7 /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/training/corpus.1.th /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/target.split -d -l 2608 -a 7 /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/training/corpus.1.my /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/source.split -d -l 2608
-a 7 /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/aligned.1.grow-diag-final-and /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/align.merging extract / extract.inv
gunzip -c /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000000.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000001.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000002.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000003.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000004.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000005.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000006.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000007.gz /home/ye/exp/SMT-
```

```
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000008.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000009.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000010.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000011.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000012.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000013.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000014.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000015.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000016.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000017.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000018.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000019.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000020.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000021.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000022.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000023.gz  |
LC_ALL=C sort     -T /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347 2>> /dev/stderr | pigz -c > /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/extract.1.sorted.gz 2>> /dev/stderr
gunzip -c /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000000.inv.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000001.inv.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000002.inv.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000003.inv.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000004.inv.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000005.inv.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000006.inv.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000007.inv.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000008.inv.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000009.inv.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000010.inv.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000011.inv.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000012.inv.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000013.inv.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000014.inv.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000015.inv.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000016.inv.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000017.inv.gz
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000018.inv.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000019.inv.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000020.inv.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000021.inv.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000022.inv.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000023.inv.gz
| LC_ALL=C sort     -T /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347 2>> /dev/stderr | pigz -c > /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/extract.1.inv.sorted.gz 2>>
/dev/stderr
gunzip -c /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000000.o.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000001.o.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000002.o.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000003.o.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000004.o.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000005.o.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000006.o.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000007.o.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000008.o.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000009.o.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000010.o.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000011.o.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000012.o.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000013.o.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000014.o.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000015.o.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000016.o.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000017.o.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000018.o.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000019.o.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000020.o.gz
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347/extract.0000021.o.gz /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/model/tmp.3357347/extract.0000022.o.gz /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp.3357347/extract.0000023.o.gz  |
LC_ALL=C sort     -T /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/model/tmp.3357347 2>> /dev/stderr | pigz -c > /home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/extract.1.o.sorted.gz 2>>
/dev/stderr
```

```
Finished Thu Apr 30 13:10:35 2026
                 __/__/__ -
                             <    >   <    >
ERROR: some opened tags were never closed:
       < repeat ' best contact number '        >
.ERROR: some opened tags were never closed:
                         <    >   <    >
ERROR: some opened tags were never closed:
       < repeat ' best contact number '        >
ERROR: some opened tags were never closed:
          <    >   <    >
ERROR: malformed XML:         < 100 , 000    /  . .
no target (0) or source (18) words << end insentence 27407
T:        < 100 , 000    /  . .
S: သွေး ခဲ အ ရေ အ တွက် 1 0 0 , 0 0 0 ဆဲလ် / m m 3
ERROR: malformed XML:         < 100 , 000    /  . .
no target (0) or source (21) words << end insentence 48379
T:        < 100 , 000    /  . .
S: သွေး ခဲ တွေ ရဲ့ အ ရေ အ တွက် က 1 0 0 , 0 0 0 ဆဲလ် / m m 3
ERROR: malformed XML:                < 3 . 5
no target (0) or source (22) words << end insentence 17134
T:              < 3 . 5
S: သွေး ထဲ က အမ် ဘန် a l b u m i n အ ဆ င့် 3 . 5 g % နည်း သည်
ERROR: some opened tags were never closed:
                                        <
 >   <    >
ERROR: some opened tags were never closed:
           <                 >
     COVID - 19
ERROR: some opened tags were never closed:
                         <    >   <    >
ERROR: some opened tags were never closed:                <
>
ERROR: some opened tags were never closed:
     <     >
ERROR: some opened tags were never closed:          <     >
               <            >
ERROR: some opened tags were never closed:    <
>
ERROR: some opened tags were never closed:
                                        <
 >   <    >
ERROR: malformed XML: ERROR: some opened tags were never closed:
          < 3 . 5
                              <    >   <    >

no target (0) or source (22) words << end insentence 28040
```

```
T:                < 3 . 5
S: သွေး ထဲ တွင် အမ် ဘန် a l b u m i n အ ဆ င့် 3 . 5 g % နည်း သည်
ERROR: some opened tags were never closed:
    <      >
ERROR: some opened tags were never closed:
      < repeat ' best contact number '        >
.ERROR: some opened tags were never closed:
    <      >
ERROR: some opened tags were never closed:
                        <     >   <     >
ERROR: some opened tags were never closed:
                                       <
 >   <     >

ERROR: malformed XML:                < 3 . 5
no target (0) or source (23) words << end insentence 46789
T:                < 3 . 5
S: သွေး ထဲ တွင် အမ် ဘန် a l b u m i n အ ဆ င့် 3 . 5 g % နည်း နေ သည်
ERROR: some opened tags were never closed:                 <
>

ERROR: some opened tags were never closed:           <     >
                <            >
ERROR: some opened tags were never closed:     <
>



ERROR: some opened tags were never closed:
           <                  >
     COVID - 19
ERROR: some opened tags were never closed:
          <     >   <     >










.
.ERROR: some opened tags were never closed:                 <
   >
```

1.10 ERROR Information

အထက်က log ဖိုင်မှာ ပြောနေတဲ့ စာကြောင်းတွေကနေ ဆရာ နားလည်တာက ထောင့်ကွင်းတွေကို escape လုပ်ပေးဖို့ လိုအပ်တယ် ဆိုတဲ့ အချက်ပါ။

T: < 3 . 5 S: သွေး ထဲ တွင် အမ် ဘန် a l b u m i n အ ဆ င့် 3 . 5 g % နည်း နေ သည်

ဒီနေရာမှာ T ဆိုတာက target language ဖြစ်တဲ့ ထိုင်းကို ပြောတာပါ။
s ဆိုတာကတော့ source language ဖြစ်တဲ့ မြန်မာကို ဆိုလိုတာပါ။

Error အကြောင်းအရင်းကိုတော့ သိရပြီ။

အတန်းထဲမှာလည်း သင်ပေးခဲ့သလို SMT process အားလုံးအတွက် ထုတ်ပေးတဲ့ graph ပုံကနေလည်း debug လုပ်လို့ ရပါတယ်။ လက်ရှိ ခုလို အနေအထားမှာ my-th အတွက် ဘယ်လိုမျိုး graph ပုံ ဆွဲပေးထားသလဲ ဆိုတာကို လေ့လာကြည့်ကြရအောင်။

[57]:
```
%pwd
```

[57]: '/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/steps/1'

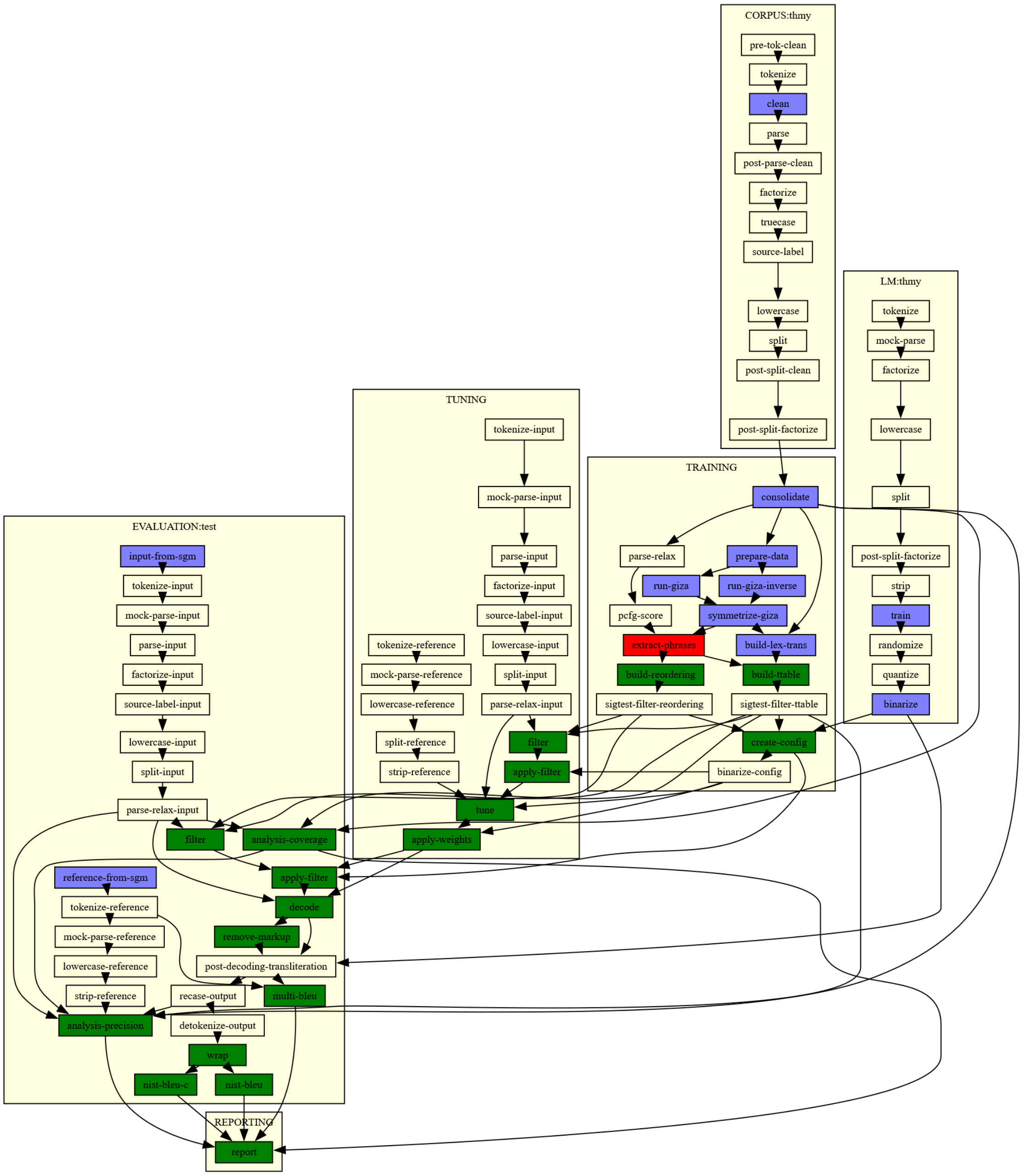
[58]:
```
!ls graph*
```

```
graph.1.dot   graph.1.png   graph.1.ps
```

graph.1.png ဖိုင်ကို နောက်ပိုင်းမှာလည်း လေ့လာလို့ ရအောင် ဆရာ error-eg-graph/ ဆိုတဲ့ ဖိုလ်ဒါထဲမှာ နာမည်ပြောင်းပေးပြီး သိမ်းပေးထားပါမယ်။

[61]:
```
from IPython import display
display.Image("/home/ye/exp/SMT-NMT_tutorial/error-examples/graph.1.
↪myth-big-error1.png", width=800)
```

[61]:



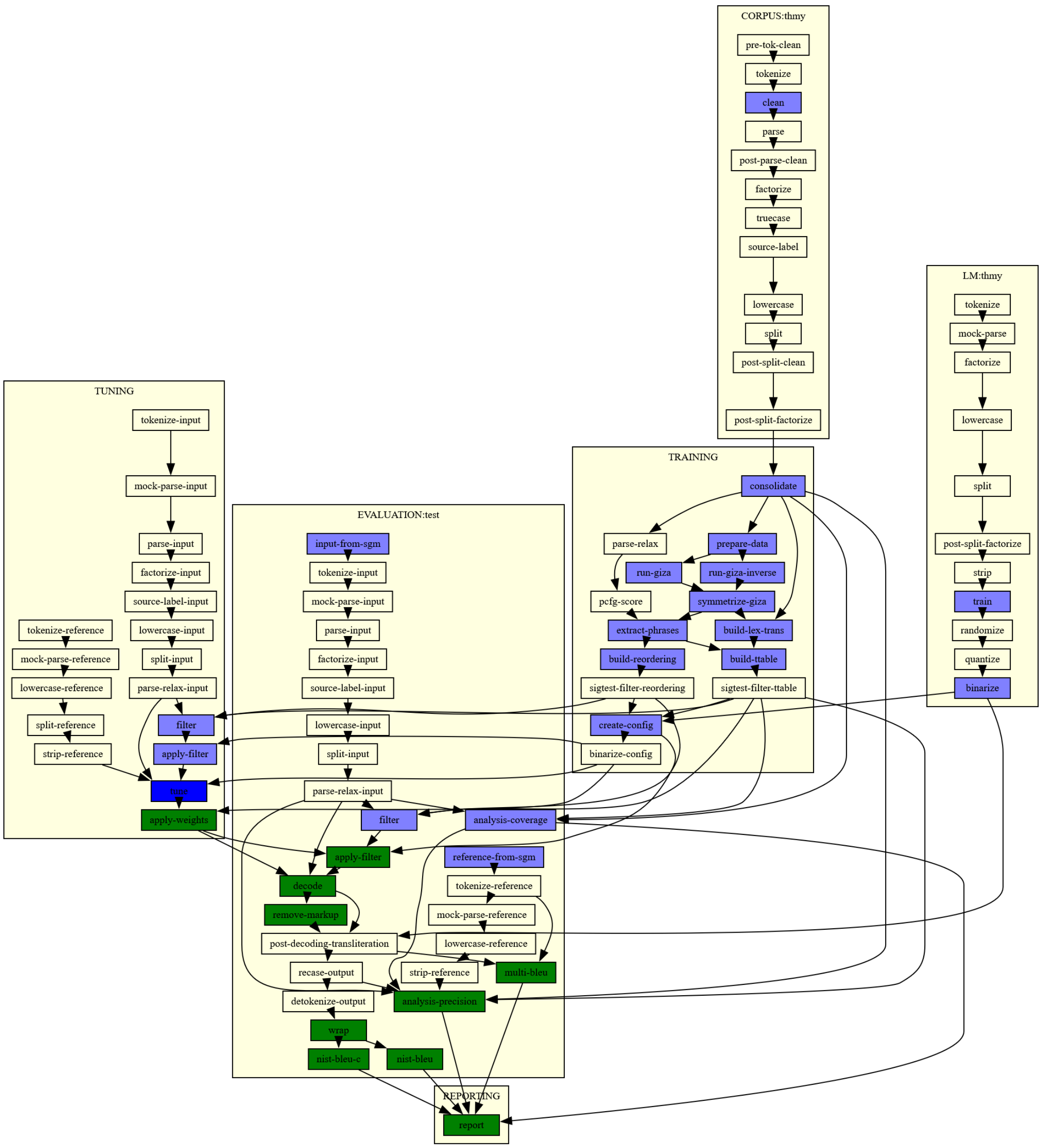
အထက်မှာ မြင်ရတဲ့အတိုင်းပဲ မြန်မာ-ထိုင်း အတွက် PBSMT experiment လုပ်ရာမှာ training လုပ်နေတဲ့အပိုင်း၊ အသေးစိတ် ပြောရရင် extract-phrases ဆိုတဲ့ alignment table ကနေ phrase pair တွေကို ဆွဲထုတ်ယူနေတဲ့ နေရာမှာ error ဖြစ်ခဲ့တယ်လို့ ပြနေပါတယ်။ အဲဒါကြောင့် အဲဒီအပိုင်းကို ဆရာတို့ ပြင်မှသာ နောက်အဆင့်တွေဖြစ်တဲ့ tuning တို့ evaluation တို့ဆီကို ဆက်သွားမှာ ဖြစ်ပါတယ်။

ဒါပေမဲ့ အထက်က run ခဲ့တဲ့ experiment မှာ ထိုင်းကနေမြန်မာ ဘာသာပြန်ဖို့အတွက် မော်ဒယ်ဆောက်တဲ့အပိုင်းမှာတော့ error မပေးတာကို အောက်ပါအတိုင်း တွေ့ရှိရပါတယ်။ ဒီကနေလည်း လေ့လာလို့ ရနိုင်တာက parallel corpus က အတူတူပေမဲ့လည်း ဘာသာပြန်တဲ့ direction ပေါ်ကို မူတည်ပြီးတော့

error မပေးတာကိုလည်း သိရပါလိမ့်မယ်။

[63]:
```
from IPython import display
display.Image("/home/ye/exp/SMT-NMT_tutorial/error-examples/graph.1.
↪thmy-success.png", width=800)
```

[63]:



1.11 Error Fixing

အထက်က error ကို ပြင်ဖို့အတွက်က moses SMT toolkit မှာ perl script က ရှိပြီးသားပါ။ အတန်းထဲမှာလည်း ပြောပြခဲ့သလိုပါပဲ SMT model မဆောက်ခင် မှာ တကယ်တန်းက preprocessing

အလုပ်တွေကို လုပ်ဖို့ လိုအပ်တယ်လို့။ အဲဒါကလည်း ကိုယ်သုံးမယ် ဘာသာစကား၊ ပြီးတော့ ဒိုမိန်း၊ ပြီးတော့ ဒေတာက ဘယ်လောက်မသန့်ရှင်းသလဲ စတဲ့အပေါ်ကို မူတည်ပြီး လုပ်ဆောင်ရတာက အမျိုးမျိုးပါပဲ။ ဒီ turorial မှာတော့ model training လုပ်ကြည့်ပြီး ပေးလာတဲ့ special character escaping အဆင့်ကိုပဲ လုပ်ကြည့်ကြရအောင်ပါ။

moses ကို ဆရာ့ လက်ရှိစက်ထဲမှာက အောက်ပါ path မှာ ထည့်ထားတာမို့ အဲဒီအောက်မှာရှိတဲ့ escape-special-chars.perl ကို ရှာကြည့်ကြရအောင်။

[64]:
```
%pwd
```

[64]: '/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/steps/1'

[65]:
```
%cd /home/ye/tool/mosesbin/ubuntu-17.04/moses/
```

```
/home/ye/tool/mosesbin/ubuntu-17.04/moses
```

[66]:
```
!ls
```

```
bin  scripts
```

[67]:
```
%cd scripts
```

```
/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts
```

[68]:
```
!ls
```

```
analysis     generic         other               server    training
docker       Jamfile         README              share     Transliteration
ems          nbest-rescore   recaser             tests
fuzzy-match  OSM             regression-testing  tokenizer
```

[69]:
```
%cd tokenizer
```

```
/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/tokenizer
```

[70]:
```
!ls *.perl
```

```
deescape-special-chars.perl      pre-tok-clean.perl
deescape-special-chars-PTB.perl  pre-tokenizer.perl
delete-long-words.perl           remove-non-printing-char.perl
detokenizer.perl                 replace-unicode-punctuation.perl
escape-special-chars.perl        tokenizer.perl
lowercase.perl                   tokenizer_PTB.perl
normalize-punctuation.perl
```

escape-special-chars.perl ကို တွေ့ပါပြီ။

[71]:
```
!cat ./escape-special-chars.perl
```

```
#!/usr/bin/env perl
#
# This file is part of moses.  Its use is licensed under the GNU Lesser General
```

```
# Public License version 2.1 or, at your option, any later version.

use warnings;
use strict;

while(<STDIN>) {
  chop;

  # avoid general madness
  s/[\000-\037]//g;
  s/\s+/ /g;
        s/^ //g;
        s/ $//g;

  # special characters in moses
  s/\&/\&amp;/g;   # escape escape
  s/\|/\&#124;/g;  # factor separator
  s/\</\&lt;/g;    # xml
  s/\>/\&gt;/g;    # xml
  s/\'/\&apos;/g;  # xml
  s/\"/\&quot;/g;  # xml
  s/\[/\&#91;/g;   # syntax non-terminal
  s/\]/\&#93;/g;   # syntax non-terminal

  # restore xml instructions
  s/\&lt;(\S+) translation=&quot;(.+?)&quot;&gt; (.+?) &lt;\/(\S+)&gt;/\<$1
translation=\"$2\"> $3 <\/$4>/g;
  print $_."\n";
}
```

အထက်က perl script က အောက်ပါစာလုံးတွေ၊ သင်္ကေတတွေကို moses ရဲ့ translation engine နဲ့ run တဲ့အခါမှာ error မတက်အောင် ပြောင်းပေးတာပါ။

- & → &
- → | (Very important, as Moses uses “|” for factors)
- < → <
- > → >
- ’ → '
- ” → "
- [→ [
-] →]

`s/[\000-\037]//g;` ကတော့ non-printable control characters တွေဖြစ်တဲ့ ဥပမာ “Null,” “Bell,” သို့မဟုတ် “Escape.” တို့ကို ဖယ်ပေးတာပါ။

`s/\s+/ /g;`, `s/^ //g;` နဲ့ `s/ $//g;` တို့ကတော့ space တွေကို ရှင်းပေးတာပါ။

ဒီ `s/\<(\S+) translation="(.+?)"> (.+?) <\/(\S+)>/\<$1 translation=\"$2\"> $3 <\/$4>/g;` လိုင်းကတော့ တခါတလေ XML tag အနေနဲ့ မပြောင်းပဲ ထားချင်တာတွေလည်း ရှိတာမို့ အဲဒီအတွက် opening/closing tag တွေကို detect လုပ်ပြီး specific XML markup mode အနေနဲ့ ပြောင်းဖို့ ရေးထားတာပါ။

ကောင်းပြီ။ ဒီ perl script ကို သုံးပြီး လက်ရှိ error ပေးနေတဲ့ corpus ကို cleaning လုပ်ကြရအောင်။

[72]:
```
%pwd
```

[72]: '/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/tokenizer'

[73]:
```
%cd /home/ye/exp/SMT-NMT_tutorial/clean-data/big/
```

```
/home/ye/exp/SMT-NMT_tutorial/clean-data/big
```

[74]:
```
!ls
```

```
dev.my  dev.th  test.my  test-sgm  test.th  train.my  train.th
```

[75]:
```
!mkdir escape
```

[76]:
```
!mv *.my ./escape/
```

[77]:
```
!mv *.th ./escape/
```

[78]:
```
%cd ./escape/
```

```
/home/ye/exp/SMT-NMT_tutorial/clean-data/big/escape
```

Shell script တစ်ပုဒ်ကို အောက်ပါအတိုင်း ပြင်ခဲ့တယ်။

[79]:
```
!cat ./run-escape.sh
```

```
#!/bin/bash

# Define the path to the Moses tokenizer scripts
SCRIPTS_DIR="/home/ye/tool/mosesbin/ubuntu-17.04/moses/scripts/tokenizer"
ESCAPE_SCRIPT="$SCRIPTS_DIR/escape-special-chars.perl"

# Check if the script exists before running
if [ ! -f "$ESCAPE_SCRIPT" ]; then
    echo "Error: escape-special-chars.perl not found at $ESCAPE_SCRIPT"
    exit 1
fi

echo "Starting character escaping…"
```

```
# Loop through all files ending in .my and .th
for file in *.my *.th; do
    # Skip if the file doesn't exist (in case no files match)
    [ -e "$file" ] || continue

    echo "Processing: $file"

    # Run the escaping script and output to <filename>.escape
    cat "$file" | perl "$ESCAPE_SCRIPT" > "$file.escape"
done

echo "Done! Escaped files have been created with the .escape suffix."
```

Executable mode အဖြစ်ပြောင်းမယ်။

[80]:
```
!chmod +x ./run-escape.sh
```

Run မယ်။

[81]:
```
!./run-escape.sh
```

```
Starting character escaping…
Processing: dev.my
Processing: test.my
Processing: train.my
Processing: dev.th
Processing: test.th
Processing: train.th
Done! Escaped files have been created with the .escape suffix.
```

[82]:
```
!ls *
```

```
dev.my         dev.th.escape  test.my.escape  train.my         train.th.escape
dev.my.escape  run-escape.sh  test.th         train.my.escape
dev.th         test.my        test.th.escape  train.th
```

Escape character တွေအဖြစ် မပြောင်းရသေးခင်က အောက်ပါအတိုင်းပါ။

[87]:
```
!grep -n "<" train.th
```

```
149:                              <     >   <
 >
492:                         <      >
4349:
                    <     >   <     >
7100:
      <     >   <     >
8159:                                <
          >                     COVID - 19
```

```
8965:                          < repeat ' best contact
number '         >
9744:    <                 >
10078:                         <      >
11595:
                      <     >   <     >
12901:                     <     >
14003:      [         ]    <          xxxx >    {
    ] "
16251:
                      <     >   <     >
17152:                     < 3 . 5
19181:                              <     >   <
   >
20936:      [         ]    <          xxxx >    {
    ] "
21413:                              <
          >                  COVID - 19
25479:
      <     >   <     >
26033:                     <     >
27434:            < 100 , 000    /   . .
28068:                     < 3 . 5
28161:                         <      >
31524:                       __/__/__ _
                                   <     >
<     >
33218:
                      <     >   <     >
33510:
      <     >   <     >
33812:                              <     >   <
   >
35254:                     <     >
35270:          <     >                   <
         >
36664:      [         ]    <          xxxx >    {
    ] "
36919:                     < 3 . 5
37387:            < 100 , 000    /   . .
39050:                              <
          >                  COVID - 19
39265:          <     >                   <
         >
40441:                              <
          >                  COVID - 19
41112:     <                 >
46831:                     < 3 . 5
```

```
46915:          <     >                   <
     >
47166:                          < repeat ' best contact
number '        >
48298:                          < repeat ' best contact
number '        >
48424:        < 100 , 000    /  . .
50672:                     __/__/__ _
                                 <    >
<    >
51462:                         <      >
51760:                          < repeat ' best contact
number '        >
52627:                 < 3 . 5
53412:
      <    >   <    >
55103:     [         ]    <         xxxx >    {
    ] "
60536:    <                 >
```

[90]:
```
!grep -n "&lt" train.th.escape
```

```
149:                             &lt;     &gt;
&lt;     &gt;
492:                        &lt;      &gt;
4349:
               &lt;     &gt;   &lt;     &gt;
7100:
      &lt;     &gt;   &lt;     &gt;
8159:                            &lt;
    &gt;                    COVID - 19
8965:                         &lt; repeat &apos; best
contact number &apos;        &gt;
9744:    &lt;                 &gt;

10078:                        &lt;      &gt;
11595:
                      &lt;     &gt;   &lt;     &gt;
12901:                &lt;     &gt;
14003:     &#91;         &#93;    &lt;          xxxx
&gt;    {        &#93; &quot;
16251:
                      &lt;     &gt;   &lt;     &gt;
17152:                &lt; 3 . 5
19181:                           &lt;     &gt;
  &lt;     &gt;
20936:     &#91;         &#93;    &lt;          xxxx
&gt;    {        &#93; &quot;
```

21413: <
> COVID - 19
25479:
< > < >
26033: < >
27434: < 100 , 000 / . .
28068: < 3 . 5
28161: < >
31524: __/__/__ _
<
> < >
33218:
< > < >
33510:
< > < >
33812: < >
< >
35254: < >
35270: < > <
>
36664: [ ] < xxxx
> {] "
36919: < 3 . 5
37387: < 100 , 000 / . .
39050: <
> COVID - 19
39265: < > <
>
40441: <
> COVID - 19
41112: < >

46831: < 3 . 5
46915: < > <
>
47166: < repeat ' best
contact number ' >
48298: < repeat ' best
contact number ' >
48424: < 100 , 000 / . .
50672: __/__/__ _
<
> < >
51462: < >
51760: < repeat ' best
contact number ' >
52627: < 3 . 5
53412:

```
      &lt;     &gt;    &lt;     &gt;
55103:      &#91;         &#93;    &lt;          xxxx
&gt;    {        &#93; &quot;
60536:    &lt;                &gt;
```

1.12 Prepare SGM File for Testset

ဖိုင်တွေအားလုံးကို အသစ်ပြောင်းလိုက်သလို ဖြစ်တာမို့ လက်ရှိ .escape တွေကို ဖြုတ်ပြီး ဒေတာဖိုလ်ဒါထဲကို ပြန်ရွှေ့ရဦးမယ်။ ပြီးတော့ SGM ဖိုင်အဖြစ် ပြောင်းပေးရပါဦးမယ်။

[91]:
```
!cp *.escape ../
```

[92]:
```
%pwd
```

[92]:
```
'/home/ye/exp/SMT-NMT_tutorial/clean-data/big/escape'
```

[93]:
```
%cd ..
```

```
/home/ye/exp/SMT-NMT_tutorial/clean-data/big
```

[94]:
```
!ls
```

```
dev.my.escape  escape          test-sgm        train.my.escape
dev.th.escape  test.my.escape  test.th.escape  train.th.escape
```

[95]:
```
!mv train.my.escape train.my
```

[96]:
```
!mv train.th.escape train.th
```

[97]:
```
!mv dev.my.escape dev.my
```

[98]:
```
!mv dev.th.escape dev.th
```

[99]:
```
!mv test.my.escape test.my
```

[100]:
```
!mv test.th.escape test.th
```

[101]:
```
!ls *.my *.th
```

```
dev.my  dev.th  test.my  test.th  train.my  train.th
```

[102]:
```
!wc *.my *.th
```

```
   10000   117871   1127057 dev.my
    8000    93908    896438 test.my
   62625   734667   7029791 train.my
   10000    71013    826135 dev.th
    8000    56113    652038 test.th
```

```
  62625   439436  5111625 train.th
 161250  1513008 15643084 total
```

[103]:
```
%cd ./test-sgm
```

```
/home/ye/exp/SMT-NMT_tutorial/clean-data/big/test-sgm
```

[104]:
```
!ls
```

```
generate_sgms.pl  src2sgm.pl        test.my.src.sgm  test.th.src.sgm
ref2sgm.pl        test.my.ref.sgm  test.th.ref.sgm
```

[105]:
```
!rm *.sgm
```

[106]:
```
!ls
```

```
generate_sgms.pl  ref2sgm.pl  src2sgm.pl
```

[107]:
```
!perl ./generate_sgms.pl
```

[108]:
```
!ls *.sgm
```

```
test.my.ref.sgm  test.my.src.sgm  test.th.ref.sgm  test.th.src.sgm
```

[109]:
```
!wc *.sgm
```

```
  8004  109920 1071469 test.my.ref.sgm
  8004  109918 1071443 test.my.src.sgm
  8004   72124  827069 test.th.ref.sgm
  8004   72122  827043 test.th.src.sgm
 32016  364084 3797024 total
```

[110]:
```
!head *.sgm
```

```
==> test.my.ref.sgm <==
<refset trglang="my" setid="Thai-Myanmar_data" srclang="any">
<doc sysid="ref" docid="none" genre="8000" origlang="any">
<seg id="1">ခွဲ မှ ရ မှာ လား </seg>
<seg id="2">မျက် ခံ လှုပ် တဲ့ လက္ခ ဏာ ရှိ မယ် </seg>
<seg id="3">အ ရက် ကို စွ နဲ့ လွှတ် ရ မည် ဖြစ် ပြီး ပ ရို တင်း နှ င့် ဗီ တာ မင်
ဓာတ် ကြွယ် ဝ သော အာ ဟာ ရ ရှိ သော အ စား အ စာ များ ကို စား သ င့် သည် </
↪seg>
<seg id="4">ဟုတ် ကဲ့ သူ ဖျား ပါ တယ် သူ သီး သန့် တော့ မ နေ ခဲ့ ဘူး </seg>
<seg id="5">မောင် လေး နောက် ဆုံး အ ကြိမ် ဆေး သောက် ခဲ့ တာ ဘယ် ချိန် လဲ </seg>
<seg id="6">ကိုယ် မှာ ကူး စက် တာ က နေ ဦး နှောက် ထိ ပါ ကူး စက် လာ တယ် </
↪seg>
<seg id="7">မစ် စ တာ ပ ရာ မို့ ရဲ့ ဆွေ မျိုး ဝင် လာ ဖို့ တစ် ချက် ခေါ် ပေး ပါ
ရှင် </seg>
<seg id="8">အ ချို ကို လျော့ ချ ကြ ည့် လိုက် ပါ </seg>
```

```
==> test.my.src.sgm <==
<srcset setid="Thai-Myanmar_data" srclang="any">
<doc docid="none" genre="8000" origlang="my">
<seg id="1">ခွဲ မှ ရ မှာ လား </seg>
<seg id="2">မျက် ခံ လှုပ် တဲ့ လက္ခ ဏာ ရှိ မယ် </seg>
<seg id="3">အ ရက် ကို စွ န့် လွှတ် ရ မည် ဖြစ် ပြီး ပ ရို တင်း နှ င့် ဗီ တာ မင် ဓာတ် ကြွယ် ဝ သော အာ ဟာ ရ ရှိ သော အ စား အ စာ များ ကို စား သ င့် သည် </seg>
<seg id="4">ဟုတ် ကဲ့ သူ ဖျား ပါ တယ် သူ သီး သန့် တော့ မ နေ ခဲ့ ဘူး </seg>
<seg id="5">မောင် လေး နောက် ဆုံး အ ကြိမ် ဆေး သောက် ခဲ့ တာ ဘယ် ချိန် လဲ </seg>
<seg id="6">ကိုယ် မှာ ကူး စက် တာ က နေ ဦး နှောက် ထိ ပါ ကူး စက် လာ တယ် </seg>
<seg id="7">မစ် စ တာ ပ ရာ မို့ ရဲ့ ဆွေ မျိုး ဝင် လာ ဖို့ တစ် ချက် ခေါ် ပေး ပါ ရှင် </seg>
<seg id="8">အ ချို့ ကို လျော့ ချ ကြ ည့် လိုက် ပါ </seg>

==> test.th.ref.sgm <==
<refset trglang="th" setid="Thai-Myanmar_data" srclang="any">
<doc sysid="ref" docid="none" genre="8000" origlang="any">
<seg id="1">        </seg>
<seg id="2">        </seg>
<seg id="3">
              </seg>
<seg id="4">                      </seg>
<seg id="5">                   </seg>
<seg id="6">                       </seg>
<seg id="7">                      </seg>
<seg id="8">             </seg>

==> test.th.src.sgm <==
<srcset setid="Thai-Myanmar_data" srclang="any">
<doc docid="none" genre="8000" origlang="th">
<seg id="1">        </seg>
<seg id="2">        </seg>
<seg id="3">
              </seg>
<seg id="4">                      </seg>
<seg id="5">                   </seg>
<seg id="6">                       </seg>
<seg id="7">                      </seg>
<seg id="8">             </seg>
```

1.13 Train/Tune/Evaluation Again

နောက်တစ်ခေါက် PBSMT experiment ကို ထပ် run ကြည့်ကြမယ်။

ဒီတစ်ခါတော့ my-th မှာ error မတက်လောက်တော့ဘူးလို့ ယူဆပါတယ်။

တစ်ခု သတိရလို့။ ဆရာတို့ th-my ရဲ့ BLEU score ကို မကြည့်ရသေးလို့ တချက် ကြည့်ထားလိုက်ကြမယ်။

[111]:
```
%pwd
```

[111]: '/home/ye/exp/SMT-NMT_tutorial/clean-data/big/test-sgm'

[112]:
```
%cd ../../..
```

```
/home/ye/exp/SMT-NMT_tutorial
```

[113]:
```
!ls ./pbsmt-big/baseline/th-my/evaluation/
```

```
test.analysis.1        test.filtered.ini.1  test.nist-bleu-c.1
test.cleaned.1         test.input.txt.1     test.output.1
test.detokenized.sgm.1 test.multi-bleu.1    test.output.1.wa
test.filtered.1        test.nist-bleu.1     test.reference.txt.1
```

မော်ဒယ်ဆောက်တာ၊ tuning လုပ်တာနဲ့ နောက်ဆုံး အကောင်းဆုံး မော်ဒယ်နဲ့ testing လုပ်တာက အဆင်ပြေရင် test.multi-bleu.1 ဆိုတဲ့ ဖိုင်ထဲမှာ BLEU score ရလဒ်တွေကို တွေ့ရလိမ့်မယ်။ အဲဒီဖိုင်ကို ရိုက်ထုတ်ကြည့်ကြရအောင်။

[114]:
```
!cat ./pbsmt-big/baseline/th-my/evaluation/test.multi-bleu.1
```

```
BLEU = 32.73, 59.2/38.8/26.8/19.4 (BP=0.989, ratio=0.989, hyp_len=92915,
ref_len=93908)
```

**Training/Tuning/Testing ကို special character တွေကို escape လုပ်ထားပြီးသား ဖိုင်တွေနဲ့ နောက်တစ်ခေါက် လုပ်ကြည့်ကြရအောင်။

[115]:
```
%cd ./pbsmt-big/
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big
```

[116]:
```
!ls
```

```
baseline         generate_configs.pl  run-baseline.pl
config.baseline  run1.log             steps
```

[117]:
```
!rm -r baseline
```

[118]:
```
!rm run1.log
```

[119]:
```
!ls
```

```
config.baseline  generate_configs.pl  run-baseline.pl  steps
```

ပထမအကြိမ် run ထားတုန်းက ဖိုလ်ဒါနဲ့ log ဖိုင်ကို ရှင်းထားပြီးသွားပြီ။ အရင်ဆုံး config ဖိုင် ထုတ်မယ်။

[120]:
```
!perl ./generate_configs.pl
```

[121]:
```
!ls
```

```
baseline  config.baseline  generate_configs.pl  run-baseline.pl  steps
```

[122]:
```
!tree .
```

```
.
  baseline
     my-th
         config.baseline.my-th
     th-my
          config.baseline.th-my
  config.baseline
  generate_configs.pl
  run-baseline.pl
  steps

5 directories, 5 files
```

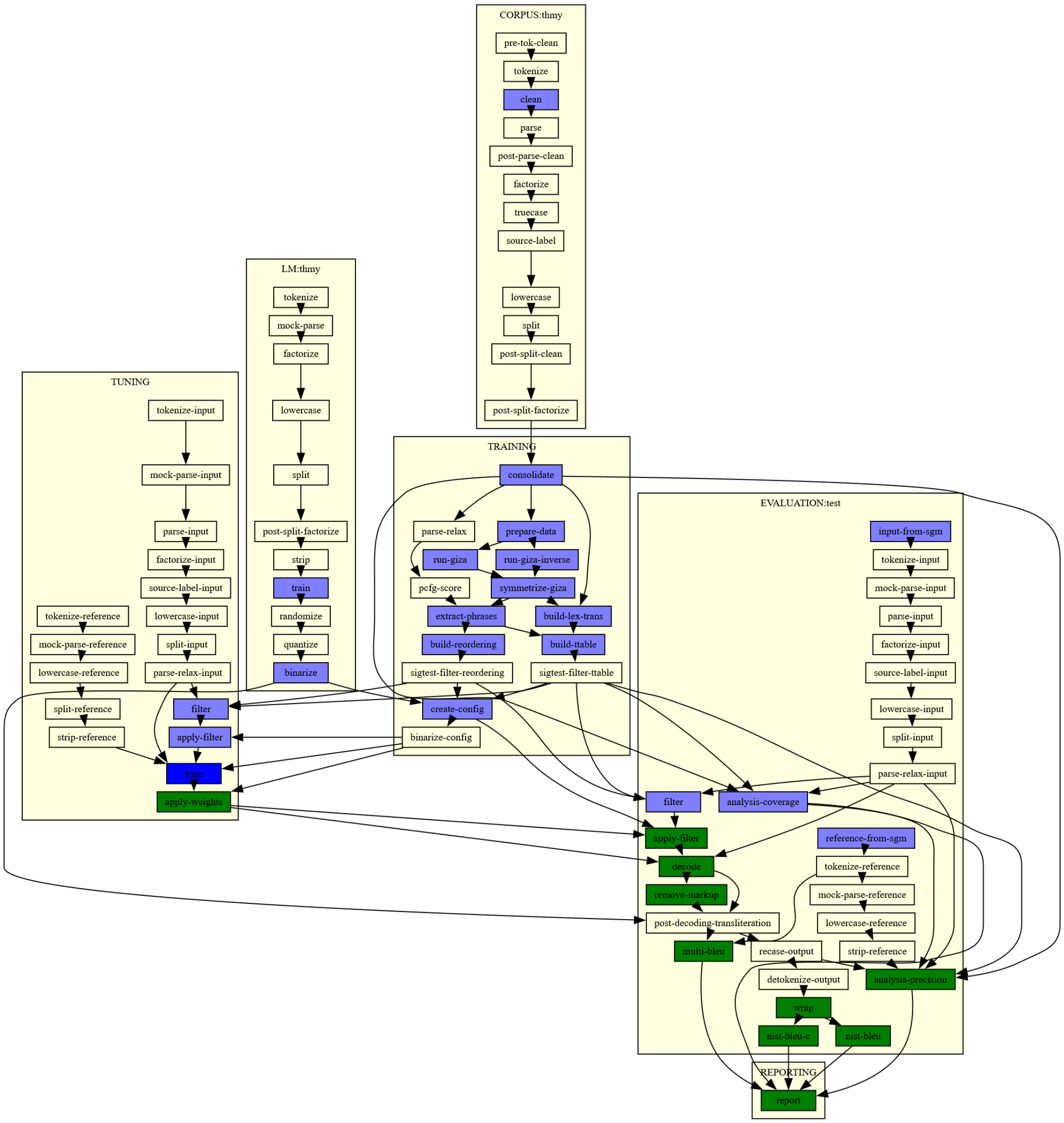
[ ]:
```
!time perl ./run-baseline.pl
```

```
my-th-baseline /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/config.baseline.my-th
```

စောင့်နေရင်းနဲ့ လက်ရှိ မြန်မာ-ထိုင်း **PBSMT experiment** ရဲ့ **graph** ကို လေ့လာကြည့်တော့ အောက်ပါအတိုင်း **tuning** လုပ်နေဆဲ ဆိုတာကို တွေ့ရလို့ ပထမအကြိမ် **run** တုန်းက ဖြစ်တဲ့ **error** ကိုတော့ **special character** တွေ **escape** လုပ်လိုက်ခြင်းအားဖြင့် ဖြေရှင်းလိုက်နိုင်တယ် ဆိုတာ သေချာသွားပါပြီ :)

[127]:
```
from IPython import display
display.Image("/home/ye/exp/SMT-NMT_tutorial/error-examples/myth-big-tuning.
 ↪png", width=800)
```

[127]:



1.14 Let's See the PBSMT System

Notebook ကို ညဘက် run ထားခဲ့ပြီး မနက်မိုးလင်းမှ ပြန်ကြည့်တယ်။
Training/Testing/Tuning/Evaluation အကုန် အဆင်ပြေပြေနဲ့ ပြီးမပြီးကို စစ်ဆေးကြည့်ကြရအောင်။

[134]:
```
%pwd
```

[134]: '/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline'

[135]:
```
!ls
```

```
my-th  th-my
```

[136]:
```
!cd my-th
```

[137]:
```
!tree .
```

```
.
  my-th
      config.baseline.my-th
      corpus
          thmy.clean.1.lines-retained
          thmy.clean.1.my
          thmy.clean.1.th
      evaluation
          test.analysis.1
              bleu-annotation
              corpus-coverage-by-phrase
              corpus-coverage-summary
              corpus-unknown
              input-annotation
              n-gram-precision.1
              n-gram-precision.2
              n-gram-precision.3
              n-gram-precision.4
              n-gram-recall.1
              n-gram-recall.2
              n-gram-recall.3
              n-gram-recall.4
              precision-by-corpus-coverage
              precision-by-input-word
              precision-by-ttable-coverage
              segmentation
              segmentation-annotation
              summary
              ttable-coverage-by-phrase
              ttable-coverage-summary
              ttable-unknown
          test.cleaned.1
          test.detokenized.sgm.1
          test.filtered.1
              info
              input.3413985
              moses.ini
              phrase-table.0-0.1.1.gz
              reordering-table.1.wbe-msd-bidirectional-fe.0-0.1
          test.filtered.ini.1
          test.input.txt.1
          test.multi-bleu.1
```

```
    test.nist-bleu.1
    test.nist-bleu-c.1
    test.output.1
    test.output.1.wa
    test.reference.txt.1
lm
    thmy.binlm.1
    thmy.lm.1
model
    aligned.1.grow-diag-final-and
    extract.1.inv.sorted.gz
    extract.1.o.sorted.gz
    extract.1.sorted.gz
    lex.1.e2f
    lex.1.f2e
    moses.ini.1
    phrase-table.1.gz
    reordering-table.1.wbe-msd-bidirectional-fe.gz
steps
    1
        config.1
        CORPUS_thmy_clean.1
        CORPUS_thmy_clean.1.DONE
        CORPUS_thmy_clean.1.INFO
        CORPUS_thmy_clean.1.STDERR
        CORPUS_thmy_clean.1.STDERR.digest
        CORPUS_thmy_clean.1.STDOUT
        EVALUATION_test_analysis-coverage.1
        EVALUATION_test_analysis-coverage.1.DONE
        EVALUATION_test_analysis-coverage.1.INFO
        EVALUATION_test_analysis-coverage.1.STDERR
        EVALUATION_test_analysis-coverage.1.STDERR.digest
        EVALUATION_test_analysis-coverage.1.STDOUT
        EVALUATION_test_analysis-precision.1
        EVALUATION_test_analysis-precision.1.DONE
        EVALUATION_test_analysis-precision.1.INFO
        EVALUATION_test_analysis-precision.1.STDERR
        EVALUATION_test_analysis-precision.1.STDERR.digest
        EVALUATION_test_analysis-precision.1.STDOUT
        EVALUATION_test_apply-filter.1
        EVALUATION_test_apply-filter.1.DONE
        EVALUATION_test_apply-filter.1.INFO
        EVALUATION_test_apply-filter.1.STDERR
        EVALUATION_test_apply-filter.1.STDERR.digest
        EVALUATION_test_apply-filter.1.STDOUT
        EVALUATION_test_decode.1
        EVALUATION_test_decode.1.DONE
        EVALUATION_test_decode.1.INFO
```

```
EVALUATION_test_decode.1.STDERR
EVALUATION_test_decode.1.STDERR.digest
EVALUATION_test_decode.1.STDOUT
EVALUATION_test_filter.1
EVALUATION_test_filter.1.DONE
EVALUATION_test_filter.1.INFO
EVALUATION_test_filter.1.STDERR
EVALUATION_test_filter.1.STDERR.digest
EVALUATION_test_filter.1.STDOUT
EVALUATION_test_input-from-sgm.1
EVALUATION_test_input-from-sgm.1.DONE
EVALUATION_test_input-from-sgm.1.INFO
EVALUATION_test_input-from-sgm.1.STDERR
EVALUATION_test_input-from-sgm.1.STDERR.digest
EVALUATION_test_input-from-sgm.1.STDOUT
EVALUATION_test_multi-bleu.1
EVALUATION_test_multi-bleu.1.DONE
EVALUATION_test_multi-bleu.1.INFO
EVALUATION_test_multi-bleu.1.STDERR
EVALUATION_test_multi-bleu.1.STDERR.digest
EVALUATION_test_multi-bleu.1.STDOUT
EVALUATION_test_nist-bleu.1
EVALUATION_test_nist-bleu.1.DONE
EVALUATION_test_nist-bleu.1.INFO
EVALUATION_test_nist-bleu.1.STDERR
EVALUATION_test_nist-bleu.1.STDERR.digest
EVALUATION_test_nist-bleu.1.STDOUT
EVALUATION_test_nist-bleu-c.1
EVALUATION_test_nist-bleu-c.1.DONE
EVALUATION_test_nist-bleu-c.1.INFO
EVALUATION_test_nist-bleu-c.1.STDERR
EVALUATION_test_nist-bleu-c.1.STDERR.digest
EVALUATION_test_nist-bleu-c.1.STDOUT
EVALUATION_test_reference-from-sgm.1
EVALUATION_test_reference-from-sgm.1.DONE
EVALUATION_test_reference-from-sgm.1.INFO
EVALUATION_test_reference-from-sgm.1.STDERR
EVALUATION_test_reference-from-sgm.1.STDERR.digest
EVALUATION_test_reference-from-sgm.1.STDOUT
EVALUATION_test_remove-markup.1
EVALUATION_test_remove-markup.1.DONE
EVALUATION_test_remove-markup.1.INFO
EVALUATION_test_remove-markup.1.STDERR
EVALUATION_test_remove-markup.1.STDERR.digest
EVALUATION_test_remove-markup.1.STDOUT
EVALUATION_test_wrap.1
EVALUATION_test_wrap.1.DONE
EVALUATION_test_wrap.1.INFO
```

```
EVALUATION_test_wrap.1.STDERR
EVALUATION_test_wrap.1.STDERR.digest
EVALUATION_test_wrap.1.STDOUT
graph.1.dot
graph.1.png
graph.1.ps
LM_thmy_binarize.1
LM_thmy_binarize.1.DONE
LM_thmy_binarize.1.INFO
LM_thmy_binarize.1.STDERR
LM_thmy_binarize.1.STDERR.digest
LM_thmy_binarize.1.STDOUT
LM_thmy_train.1
LM_thmy_train.1.DONE
LM_thmy_train.1.INFO
LM_thmy_train.1.STDERR
LM_thmy_train.1.STDERR.digest
LM_thmy_train.1.STDOUT
parameter.1
re-use.1
running.1
TRAINING_build-lex-trans.1
TRAINING_build-lex-trans.1.DONE
TRAINING_build-lex-trans.1.INFO
TRAINING_build-lex-trans.1.STDERR
TRAINING_build-lex-trans.1.STDERR.digest
TRAINING_build-lex-trans.1.STDOUT
TRAINING_build-reordering.1
TRAINING_build-reordering.1.DONE
TRAINING_build-reordering.1.INFO
TRAINING_build-reordering.1.STDERR
TRAINING_build-reordering.1.STDERR.digest
TRAINING_build-reordering.1.STDOUT
TRAINING_build-ttable.1
TRAINING_build-ttable.1.DONE
TRAINING_build-ttable.1.INFO
TRAINING_build-ttable.1.STDERR
TRAINING_build-ttable.1.STDERR.digest
TRAINING_build-ttable.1.STDOUT
TRAINING_consolidate.1
TRAINING_consolidate.1.DONE
TRAINING_consolidate.1.INFO
TRAINING_consolidate.1.STDERR
TRAINING_consolidate.1.STDERR.digest
TRAINING_consolidate.1.STDOUT
TRAINING_create-config.1
TRAINING_create-config.1.DONE
TRAINING_create-config.1.INFO
```

```
TRAINING_create-config.1.STDERR
TRAINING_create-config.1.STDERR.digest
TRAINING_create-config.1.STDOUT
TRAINING_extract-phrases.1
TRAINING_extract-phrases.1.DONE
TRAINING_extract-phrases.1.INFO
TRAINING_extract-phrases.1.STDERR
TRAINING_extract-phrases.1.STDERR.digest
TRAINING_extract-phrases.1.STDOUT
TRAINING_prepare-data.1
TRAINING_prepare-data.1.DONE
TRAINING_prepare-data.1.INFO
TRAINING_prepare-data.1.STDERR
TRAINING_prepare-data.1.STDERR.digest
TRAINING_prepare-data.1.STDOUT
TRAINING_run-giza.1
TRAINING_run-giza.1.DONE
TRAINING_run-giza.1.INFO
TRAINING_run-giza.1.STDERR
TRAINING_run-giza.1.STDERR.digest
TRAINING_run-giza.1.STDOUT
TRAINING_run-giza-inverse.1
TRAINING_run-giza-inverse.1.DONE
TRAINING_run-giza-inverse.1.INFO
TRAINING_run-giza-inverse.1.STDERR
TRAINING_run-giza-inverse.1.STDERR.digest
TRAINING_run-giza-inverse.1.STDOUT
TRAINING_symmetrize-giza.1
TRAINING_symmetrize-giza.1.DONE
TRAINING_symmetrize-giza.1.INFO
TRAINING_symmetrize-giza.1.STDERR
TRAINING_symmetrize-giza.1.STDERR.digest
TRAINING_symmetrize-giza.1.STDOUT
TUNING_apply-filter.1
TUNING_apply-filter.1.DONE
TUNING_apply-filter.1.INFO
TUNING_apply-filter.1.STDERR
TUNING_apply-filter.1.STDERR.digest
TUNING_apply-filter.1.STDOUT
TUNING_apply-weights.1
TUNING_apply-weights.1.DONE
TUNING_apply-weights.1.INFO
TUNING_apply-weights.1.STDERR
TUNING_apply-weights.1.STDERR.digest
TUNING_apply-weights.1.STDOUT
TUNING_filter.1
TUNING_filter.1.DONE
TUNING_filter.1.INFO
```

```
            TUNING_filter.1.STDERR
            TUNING_filter.1.STDERR.digest
            TUNING_filter.1.STDOUT
            TUNING_tune.1
            TUNING_tune.1.DONE
            TUNING_tune.1.INFO
            TUNING_tune.1.STDERR
            TUNING_tune.1.STDERR.digest
            TUNING_tune.1.STDOUT
    training
        corpus.1.my -> /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/corpus/thmy.clean.1.my
        corpus.1.th -> /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/corpus/thmy.clean.1.th
        giza.1
            th-my.A3.final.gz
            th-my.cooc
            th-my.gizacfg
        giza-inverse.1
            my-th.A3.final.gz
            my-th.cooc
            my-th.gizacfg
        prepared.1
            my-th-int-train.snt
            my.vcb
            my.vcb.classes
            my.vcb.classes.cats
            th-my-int-train.snt
            th.vcb
            th.vcb.classes
            th.vcb.classes.cats
    tuning
        filtered.1
            info
            input.3414026
            moses.ini
            phrase-table.0-0.1.1.gz
            reordering-table.1.wbe-msd-bidirectional-fe.0-0.1
        moses.filtered.ini.1
        moses.ini.1
        moses.tuned.ini.1
        tmp.1
            extract.err
            extractor.sh
            extract.out
            features.list
            finished_step.txt
            init.opt
```

```
mert.log
mert.out
moses.ini
run10.best100.out.gz
run10.dense
run10.extract.err
run10.extract.out
run10.features.dat
run10.init.opt
run10.mert.log
run10.mert.out
run10.moses.ini
run10.out
run10.scores.dat
run10.weights.txt
run11.best100.out.gz
run11.dense
run11.extract.err
run11.extract.out
run11.features.dat
run11.init.opt
run11.mert.log
run11.mert.out
run11.moses.ini
run11.out
run11.scores.dat
run11.weights.txt
run12.best100.out.gz
run12.dense
run12.extract.err
run12.extract.out
run12.features.dat
run12.init.opt
run12.mert.log
run12.mert.out
run12.moses.ini
run12.out
run12.scores.dat
run12.weights.txt
run1.best100.out.gz
run1.dense
run1.extract.err
run1.extract.out
run1.features.dat
run1.init.opt
run1.mert.log
run1.mert.out
run1.moses.ini
```

```
run1.out
run1.scores.dat
run1.weights.txt
run2.best100.out.gz
run2.dense
run2.extract.err
run2.extract.out
run2.features.dat
run2.init.opt
run2.mert.log
run2.mert.out
run2.moses.ini
run2.out
run2.scores.dat
run2.weights.txt
run3.best100.out.gz
run3.dense
run3.extract.err
run3.extract.out
run3.features.dat
run3.init.opt
run3.mert.log
run3.mert.out
run3.moses.ini
run3.out
run3.scores.dat
run3.weights.txt
run4.best100.out.gz
run4.dense
run4.extract.err
run4.extract.out
run4.features.dat
run4.init.opt
run4.mert.log
run4.mert.out
run4.moses.ini
run4.out
run4.scores.dat
run4.weights.txt
run5.best100.out.gz
run5.dense
run5.extract.err
run5.extract.out
run5.features.dat
run5.init.opt
run5.mert.log
run5.mert.out
run5.moses.ini
```

```
run5.out
run5.scores.dat
run5.weights.txt
run6.best100.out.gz
run6.dense
run6.extract.err
run6.extract.out
run6.features.dat
run6.init.opt
run6.mert.log
run6.mert.out
run6.moses.ini
run6.out
run6.scores.dat
run6.weights.txt
run7.best100.out.gz
run7.dense
run7.extract.err
run7.extract.out
run7.features.dat
run7.init.opt
run7.mert.log
run7.mert.out
run7.moses.ini
run7.out
run7.scores.dat
run7.weights.txt
run8.best100.out.gz
run8.dense
run8.extract.err
run8.extract.out
run8.features.dat
run8.init.opt
run8.mert.log
run8.mert.out
run8.moses.ini
run8.out
run8.scores.dat
run8.weights.txt
run9.best100.out.gz
run9.dense
run9.extract.err
run9.extract.out
run9.features.dat
run9.init.opt
run9.mert.log
run9.mert.out
run9.moses.ini
```

```
            run9.out
            run9.scores.dat
            run9.weights.txt
            weights.txt
th-my
    config.baseline.th-my
    corpus
        thmy.clean.1.lines-retained
        thmy.clean.1.my
        thmy.clean.1.th
    evaluation
        test.analysis.1
            bleu-annotation
            corpus-coverage-by-phrase
            corpus-coverage-summary
            corpus-unknown
            input-annotation
            n-gram-precision.1
            n-gram-precision.2
            n-gram-precision.3
            n-gram-precision.4
            n-gram-recall.1
            n-gram-recall.2
            n-gram-recall.3
            n-gram-recall.4
            precision-by-corpus-coverage
            precision-by-input-word
            precision-by-ttable-coverage
            segmentation
            segmentation-annotation
            summary
            ttable-coverage-by-phrase
            ttable-coverage-summary
            ttable-unknown
        test.cleaned.1
        test.detokenized.sgm.1
        test.filtered.1
            info
            input.3452065
            moses.ini
            phrase-table.0-0.1.1.gz
            reordering-table.1.wbe-msd-bidirectional-fe.0-0.1
        test.filtered.ini.1
        test.input.txt.1
        test.multi-bleu.1
        test.nist-bleu.1
        test.nist-bleu-c.1
        test.output.1
```

```
    test.output.1.wa
    test.reference.txt.1
lm
    thmy.binlm.1
    thmy.lm.1
model
    aligned.1.grow-diag-final-and
    extract.1.inv.sorted.gz
    extract.1.o.sorted.gz
    extract.1.sorted.gz
    lex.1.e2f
    lex.1.f2e
    moses.ini.1
    phrase-table.1.gz
    reordering-table.1.wbe-msd-bidirectional-fe.gz
steps
    1
        config.1
        CORPUS_thmy_clean.1
        CORPUS_thmy_clean.1.DONE
        CORPUS_thmy_clean.1.INFO
        CORPUS_thmy_clean.1.STDERR
        CORPUS_thmy_clean.1.STDERR.digest
        CORPUS_thmy_clean.1.STDOUT
        EVALUATION_test_analysis-coverage.1
        EVALUATION_test_analysis-coverage.1.DONE
        EVALUATION_test_analysis-coverage.1.INFO
        EVALUATION_test_analysis-coverage.1.STDERR
        EVALUATION_test_analysis-coverage.1.STDERR.digest
        EVALUATION_test_analysis-coverage.1.STDOUT
        EVALUATION_test_analysis-precision.1
        EVALUATION_test_analysis-precision.1.DONE
        EVALUATION_test_analysis-precision.1.INFO
        EVALUATION_test_analysis-precision.1.STDERR
        EVALUATION_test_analysis-precision.1.STDERR.digest
        EVALUATION_test_analysis-precision.1.STDOUT
        EVALUATION_test_apply-filter.1
        EVALUATION_test_apply-filter.1.DONE
        EVALUATION_test_apply-filter.1.INFO
        EVALUATION_test_apply-filter.1.STDERR
        EVALUATION_test_apply-filter.1.STDERR.digest
        EVALUATION_test_apply-filter.1.STDOUT
        EVALUATION_test_decode.1
        EVALUATION_test_decode.1.DONE
        EVALUATION_test_decode.1.INFO
        EVALUATION_test_decode.1.STDERR
        EVALUATION_test_decode.1.STDERR.digest
        EVALUATION_test_decode.1.STDOUT
```

```
EVALUATION_test_filter.1
EVALUATION_test_filter.1.DONE
EVALUATION_test_filter.1.INFO
EVALUATION_test_filter.1.STDERR
EVALUATION_test_filter.1.STDERR.digest
EVALUATION_test_filter.1.STDOUT
EVALUATION_test_input-from-sgm.1
EVALUATION_test_input-from-sgm.1.DONE
EVALUATION_test_input-from-sgm.1.INFO
EVALUATION_test_input-from-sgm.1.STDERR
EVALUATION_test_input-from-sgm.1.STDERR.digest
EVALUATION_test_input-from-sgm.1.STDOUT
EVALUATION_test_multi-bleu.1
EVALUATION_test_multi-bleu.1.DONE
EVALUATION_test_multi-bleu.1.INFO
EVALUATION_test_multi-bleu.1.STDERR
EVALUATION_test_multi-bleu.1.STDERR.digest
EVALUATION_test_multi-bleu.1.STDOUT
EVALUATION_test_nist-bleu.1
EVALUATION_test_nist-bleu.1.DONE
EVALUATION_test_nist-bleu.1.INFO
EVALUATION_test_nist-bleu.1.STDERR
EVALUATION_test_nist-bleu.1.STDERR.digest
EVALUATION_test_nist-bleu.1.STDOUT
EVALUATION_test_nist-bleu-c.1
EVALUATION_test_nist-bleu-c.1.DONE
EVALUATION_test_nist-bleu-c.1.INFO
EVALUATION_test_nist-bleu-c.1.STDERR
EVALUATION_test_nist-bleu-c.1.STDERR.digest
EVALUATION_test_nist-bleu-c.1.STDOUT
EVALUATION_test_reference-from-sgm.1
EVALUATION_test_reference-from-sgm.1.DONE
EVALUATION_test_reference-from-sgm.1.INFO
EVALUATION_test_reference-from-sgm.1.STDERR
EVALUATION_test_reference-from-sgm.1.STDERR.digest
EVALUATION_test_reference-from-sgm.1.STDOUT
EVALUATION_test_remove-markup.1
EVALUATION_test_remove-markup.1.DONE
EVALUATION_test_remove-markup.1.INFO
EVALUATION_test_remove-markup.1.STDERR
EVALUATION_test_remove-markup.1.STDERR.digest
EVALUATION_test_remove-markup.1.STDOUT
EVALUATION_test_wrap.1
EVALUATION_test_wrap.1.DONE
EVALUATION_test_wrap.1.INFO
EVALUATION_test_wrap.1.STDERR
EVALUATION_test_wrap.1.STDERR.digest
EVALUATION_test_wrap.1.STDOUT
```

```
graph.1.dot
graph.1.png
graph.1.ps
LM_thmy_binarize.1
LM_thmy_binarize.1.DONE
LM_thmy_binarize.1.INFO
LM_thmy_binarize.1.STDERR
LM_thmy_binarize.1.STDERR.digest
LM_thmy_binarize.1.STDOUT
LM_thmy_train.1
LM_thmy_train.1.DONE
LM_thmy_train.1.INFO
LM_thmy_train.1.STDERR
LM_thmy_train.1.STDERR.digest
LM_thmy_train.1.STDOUT
parameter.1
re-use.1
running.1
TRAINING_build-lex-trans.1
TRAINING_build-lex-trans.1.DONE
TRAINING_build-lex-trans.1.INFO
TRAINING_build-lex-trans.1.STDERR
TRAINING_build-lex-trans.1.STDERR.digest
TRAINING_build-lex-trans.1.STDOUT
TRAINING_build-reordering.1
TRAINING_build-reordering.1.DONE
TRAINING_build-reordering.1.INFO
TRAINING_build-reordering.1.STDERR
TRAINING_build-reordering.1.STDERR.digest
TRAINING_build-reordering.1.STDOUT
TRAINING_build-ttable.1
TRAINING_build-ttable.1.DONE
TRAINING_build-ttable.1.INFO
TRAINING_build-ttable.1.STDERR
TRAINING_build-ttable.1.STDERR.digest
TRAINING_build-ttable.1.STDOUT
TRAINING_consolidate.1
TRAINING_consolidate.1.DONE
TRAINING_consolidate.1.INFO
TRAINING_consolidate.1.STDERR
TRAINING_consolidate.1.STDERR.digest
TRAINING_consolidate.1.STDOUT
TRAINING_create-config.1
TRAINING_create-config.1.DONE
TRAINING_create-config.1.INFO
TRAINING_create-config.1.STDERR
TRAINING_create-config.1.STDERR.digest
TRAINING_create-config.1.STDOUT
```

```
TRAINING_extract-phrases.1
TRAINING_extract-phrases.1.DONE
TRAINING_extract-phrases.1.INFO
TRAINING_extract-phrases.1.STDERR
TRAINING_extract-phrases.1.STDERR.digest
TRAINING_extract-phrases.1.STDOUT
TRAINING_prepare-data.1
TRAINING_prepare-data.1.DONE
TRAINING_prepare-data.1.INFO
TRAINING_prepare-data.1.STDERR
TRAINING_prepare-data.1.STDERR.digest
TRAINING_prepare-data.1.STDOUT
TRAINING_run-giza.1
TRAINING_run-giza.1.DONE
TRAINING_run-giza.1.INFO
TRAINING_run-giza.1.STDERR
TRAINING_run-giza.1.STDERR.digest
TRAINING_run-giza.1.STDOUT
TRAINING_run-giza-inverse.1
TRAINING_run-giza-inverse.1.DONE
TRAINING_run-giza-inverse.1.INFO
TRAINING_run-giza-inverse.1.STDERR
TRAINING_run-giza-inverse.1.STDERR.digest
TRAINING_run-giza-inverse.1.STDOUT
TRAINING_symmetrize-giza.1
TRAINING_symmetrize-giza.1.DONE
TRAINING_symmetrize-giza.1.INFO
TRAINING_symmetrize-giza.1.STDERR
TRAINING_symmetrize-giza.1.STDERR.digest
TRAINING_symmetrize-giza.1.STDOUT
TUNING_apply-filter.1
TUNING_apply-filter.1.DONE
TUNING_apply-filter.1.INFO
TUNING_apply-filter.1.STDERR
TUNING_apply-filter.1.STDERR.digest
TUNING_apply-filter.1.STDOUT
TUNING_apply-weights.1
TUNING_apply-weights.1.DONE
TUNING_apply-weights.1.INFO
TUNING_apply-weights.1.STDERR
TUNING_apply-weights.1.STDERR.digest
TUNING_apply-weights.1.STDOUT
TUNING_filter.1
TUNING_filter.1.DONE
TUNING_filter.1.INFO
TUNING_filter.1.STDERR
TUNING_filter.1.STDERR.digest
TUNING_filter.1.STDOUT
```

```
            TUNING_tune.1
            TUNING_tune.1.DONE
            TUNING_tune.1.INFO
            TUNING_tune.1.STDERR
            TUNING_tune.1.STDERR.digest
            TUNING_tune.1.STDOUT
    training
        corpus.1.my -> /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/th-my/corpus/thmy.clean.1.my
        corpus.1.th -> /home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/th-my/corpus/thmy.clean.1.th
        giza.1
            my-th.A3.final.gz
            my-th.cooc
            my-th.gizacfg
        giza-inverse.1
            th-my.A3.final.gz
            th-my.cooc
            th-my.gizacfg
        prepared.1
            my-th-int-train.snt
            my.vcb
            my.vcb.classes
            my.vcb.classes.cats
            th-my-int-train.snt
            th.vcb
            th.vcb.classes
            th.vcb.classes.cats
    tuning
        filtered.1
            info
            input.3452140
            moses.ini
            phrase-table.0-0.1.1.gz
            reordering-table.1.wbe-msd-bidirectional-fe.0-0.1
        moses.filtered.ini.1
        moses.ini.1
        moses.tuned.ini.1
        tmp.1
            extract.err
            extractor.sh
            extract.out
            features.list
            finished_step.txt
            init.opt
            mert.log
            mert.out
            moses.ini
```

```
run1.best100.out.gz
run1.dense
run1.extract.err
run1.extract.out
run1.features.dat
run1.init.opt
run1.mert.log
run1.mert.out
run1.moses.ini
run1.out
run1.scores.dat
run1.weights.txt
run2.best100.out.gz
run2.dense
run2.extract.err
run2.extract.out
run2.features.dat
run2.init.opt
run2.mert.log
run2.mert.out
run2.moses.ini
run2.out
run2.scores.dat
run2.weights.txt
run3.best100.out.gz
run3.dense
run3.extract.err
run3.extract.out
run3.features.dat
run3.init.opt
run3.mert.log
run3.mert.out
run3.moses.ini
run3.out
run3.scores.dat
run3.weights.txt
run4.best100.out.gz
run4.dense
run4.extract.err
run4.extract.out
run4.features.dat
run4.init.opt
run4.mert.log
run4.mert.out
run4.moses.ini
run4.out
run4.scores.dat
run4.weights.txt
```

```
                run5.best100.out.gz
                run5.dense
                run5.extract.err
                run5.extract.out
                run5.features.dat
                run5.init.opt
                run5.mert.log
                run5.mert.out
                run5.moses.ini
                run5.out
                run5.scores.dat
                run5.weights.txt
                run6.best100.out.gz
                run6.dense
                run6.extract.err
                run6.extract.out
                run6.features.dat
                run6.init.opt
                run6.mert.log
                run6.mert.out
                run6.moses.ini
                run6.out
                run6.scores.dat
                run6.weights.txt
                run7.best100.out.gz
                run7.dense
                run7.extract.err
                run7.extract.out
                run7.features.dat
                run7.init.opt
                run7.mert.log
                run7.mert.out
                run7.moses.ini
                run7.out
                run7.scores.dat
                run7.weights.txt
                weights.txt

33 directories, 762 files
```

Alignment Table ကို ဝင်ကြည့်ရအောင်

[152]:
```
%cd my-th/training
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training
```

[153]:
```
!ls -F --color=auto
```

```
corpus.1.my@  corpus.1.th@  giza.1/
```

giza-inverse.1/ prepared.1/

[154]:
```
%cd giza.1
```

/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/giza.1

[157]:
```
!ls -F
```

th-my.A3.final.gz th-my.cooc th-my.gizacfg tmp/

.gz ဖိုင် မို့လို့ temporary folder တစ်ခုဆောက်ပြီး အဲဒီထဲမှာပဲ ဖြေပြီး ဖွင့်ကြည့်ကြရအောင်

[156]:
```
!mkdir tmp
```

[168]:
```
%cd tmp
```

/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/giza.1/tmp

[ ]:
```
!cp ../th-my.A3.final.gz .
```

gunzip command နဲ့ ပုံမှန် text ဖိုင်ဖြစ်အောင် ဖြေလိုက်တယ်

[171]:
```
!gunzip ./th-my.A3.final.gz
```

[172]:
```
!ls
```

th-my.A3.final

[175]:
```
!head -n 50 ./th-my.A3.final
```

```
# Sentence pair (1) source length 10 target length 10 alignment score :
2.24352e-26

NULL ({ 9 }) အောက် ({ 7 8 10 }) ပါ ({ }) ရော ({ }) ဂါ ({ }) လက္ခ ({ 1 4 5 6 })
ဏာ ({ 3 }) တွေ ({ }) များ ({ }) ရှိ ({ 2 }) ရင် ({ })
# Sentence pair (2) source length 8 target length 5 alignment score :
1.08954e-11

NULL ({ }) စိတ် ({ }) ကို ({ }) အေး ({ 1 }) အေး ({ 2 }) ဆေး ({ }) ဆေး ({ }) ␣
  ↪ထား
({ 3 }) ပါ ({ 4 5 })
# Sentence pair (3) source length 12 target length 5 alignment score :
3.80172e-09

NULL ({ }) နာ ({ }) တာ ({ }) ရှည် ({ }) ရော ({ }) ဂါ ({ 3 4 5 }) ရယ် ({ }) လို့
({ }) မ ({ 1 }) ရှိ ({ 2 }) ပါ ({ }) ဘူး ({ }) ဗျ ({ })
# Sentence pair (4) source length 4 target length 3 alignment score :
1.53981e-08

NULL ({ }) လက္ခ ({ }) ဏာ ({ 1 2 3 }) တွေ ({ }) နဲ့ ({ })
```

```
# Sentence pair (5) source length 18 target length 14 alignment score :
2.38628e-34

NULL ({ }) ပြီး ({ 1 }) တဲ့ ({ }) အ ({ }) ခါ ({ }) နား ({ 3 }) ကြပ် ({ 2 11 12
}) ကို ({ }) တံ ({ }) တော ({ }) င့် ({ }) ဆစ် ({ 7 8 }) နား ({ 9 }) မှာ ({ 10 })
ထား ({ }) မှာ ({ }) မို့ ({ 4 5 6 }) ပါ ({ }) ရှင် ({ 13 14 })
# Sentence pair (6) source length 7 target length 5 alignment score :
2.19945e-09

NULL ({ }) ပု ({ }) စွန် ({ 3 4 5 }) တော့ ({ }) ကျွန် ({ 1 }) တော် ({ }) စား ({
2 }) တယ် ({ })
# Sentence pair (7) source length 4 target length 1 alignment score : 0.00273058

NULL ({ }) သို့ ({ }) ပေ ({ 1 }) မဲ့ ({ }) လည်း ({ })
# Sentence pair (8) source length 11 target length 8 alignment score :
2.36294e-17

NULL ({ }) ကျွန် ({ 1 }) တော် ({ }) ခေါင်း ({ 7 }) ပေါ် ({ 6 }) ရေ ({ 4 }) ခဲ ({
3 5 }) တင် ({ }) ထား ({ }) ရ ({ 2 }) မ ({ }) လား ({ 8 })
# Sentence pair (9) source length 12 target length 9 alignment score :
2.51468e-15

NULL ({ }) ကွန် ({ }) ပြူ ({ 1 2 3 4 5 6 7 }) တာ ({ }) ရောင် ({ }) ခြည် ({ })
နဲ့ ({ }) ထပ် ({ }) စစ် ({ }) ဆေး ({ }) ပါ ({ }) မယ် ({ }) နော် ({ 8 9 })
# Sentence pair (10) source length 14 target length 9 alignment score :
3.78696e-23

NULL ({ }) ဒီ ({ 2 5 6 }) ကြား ({ 1 }) မှာ ({ }) ခ ({ }) ဏ ({ }) လေး ({ }) ထိုင်
({ 3 }) စော ({ 4 }) င့် ({ }) ပေး ({ 7 8 }) လို့ ({ }) ရ ({ }) မ ({ }) လား ({ 9
})
# Sentence pair (11) source length 10 target length 6 alignment score :
4.52493e-14

NULL ({ }) မိ ({ }) သား ({ }) စု ({ 2 3 4 5 6 }) နဲ့ ({ 1 }) စ ({ }) ကား ({ })
ပြော ({ }) ဆက် ({ }) ဆံ ({ }) တာ ({ })
# Sentence pair (12) source length 10 target length 7 alignment score :
1.08517e-11

NULL ({ }) ငါ ({ 1 }) မ ({ }) နက် ({ 7 }) တွေ ({ }) တိုင်း ({ 6 }) နှာ ({ 2 })
ခေါင်း ({ }) သွေး ({ 3 }) လျှံ ({ 4 5 }) တယ် ({ })
# Sentence pair (13) source length 2 target length 2 alignment score : 0.0008606

NULL ({ }) ဒါ ({ }) ပေါ့ ({ 1 2 })
# Sentence pair (14) source length 23 target length 11 alignment score :
```

```
4.03246e-24
                  3 - 5
NULL ({ }) တို့ ({ 1 2 }) တွေ ({ }) အ ({ }) နည်း ({ 7 }) ဆုံး ({ }) သွား ({ 5 })
ကို ({ }) ၃ ({ 8 }) - ({ 9 }) ၅ ({ 10 }) မိ ({ }) နစ် ({ }) အ ({ }) ထိ ({ }) အ
({ }) နည်း ({ }) ဆုံး ({ 6 }) တိုက် ({ 4 }) သ ({ }) ငို့ ({ 3 }) တယ် ({ }) ရှ ({
11 }) ငို့ ({ })
# Sentence pair (15) source length 10 target length 6 alignment score :
4.49226e-16
       32
NULL ({ }) ကိုယ် ({ }) ဝန် ({ }) က ({ }) ၃ ({ 1 }) ၂ ({ 4 }) ပတ် ({ 5 }) တော့␣
 ↪({
}) ရှိ ({ }) ပါ ({ 6 }) ပြီ ({ 2 3 })
# Sentence pair (16) source length 9 target length 7 alignment score :
1.52952e-18

NULL ({ 4 }) အဲ့ ({ 1 }) ဒါ ({ }) လုပ် ({ 3 5 }) ပါ ({ }) မယ် ({ 2 }) ဆ ({ 6 7
}) ရာ ({ }) မ ({ }) လေး ({ })
# Sentence pair (17) source length 17 target length 13 alignment score :
5.29433e-30
```

- The NULL token handles words that don't translate to anything.
- The numbers in ({ }) represent the position of the source word that aligns to that target word.

[3]:
```
%cd ..
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/giza.1
```

[4]:
```
!ls
```

```
th-my.A3.final.gz  th-my.cooc  th-my.gizacfg  tmp
```

[5]:
```
%cd ..
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training
```

[7]:
```
!ls --color=auto
```

```
corpus.1.my  corpus.1.th  giza.1
giza-inverse.1  prepared.1
```

ဒီတစ်ခါတော့ inverse alignment ဖိုင်ကို လေ့လာကြည့်ရအောင်။

[10]:
```
%cd ./giza-inverse.1/
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/giza-inverse.1
```

[11]:
```
!ls --color=auto
```

```
my-th.A3.final.gz  my-th.cooc  my-th.gizacfg
```

[12]:
```
!mkdir tmp
```

[13]:
```
!cp my-th.A3.final.gz ./tmp/
```

[14]:
```
%cd ./tmp/
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/training/giza-
inverse.1/tmp
```

[15]:
```
!gunzip ./my-th.A3.final.gz
```

[16]:
```
!ls --color=auto
```

```
my-th.A3.final
```

[17]:
```
!head -n 50 ./my-th.A3.final
```

```
# Sentence pair (1) source length 10 target length 10 alignment score :
2.16558e-24
အောက် ပါ ရော ဂါ လက္ခ ဏာ တွေ များ ရှိ ရင်
NULL ({ })   ({ })   ({ 2 9 })    ({ 3 4 5 6 7 8 })    ({ })   ({ 1 10
})    ({ })    ({ })    ({ })    ({ })   ({ })
# Sentence pair (2) source length 5 target length 8 alignment score :
5.24945e-14
စိတ် ကို အေး အေး ဆေး ဆေး ထား ပါ
NULL ({ })    ({ 1 2 3 4 5 6 7 })  ({ })    ({ })   ({ 8 })   ({ })
# Sentence pair (3) source length 5 target length 12 alignment score :
5.83236e-31
နာ တာ ရှည် ရော ဂါ ရယ် လို့ မ ရှိ ပါ ဘူး ဗျ
NULL ({ })   ({ 8 11 })   ({ 7 9 10 })    ({ 4 5 })     ({ 3 6 12 })
({ 1 2 })
# Sentence pair (4) source length 3 target length 4 alignment score :
1.54375e-08
လက္ခ ဏာ တွေ နဲ့
NULL ({ })    ({ 1 2 3 4 })    ({ })   ({ })
# Sentence pair (5) source length 14 target length 18 alignment score :
6.26777e-38
ပြီး တဲ့ အ ခါ နား ကြပ် ကို တံ တော င့် ဆစ် နား မှာ ထား မှာ မို့ ပါ ရှင်
NULL ({ 3 })    ({ 1 2 4 })    ({ })   ({ 5 })    ({ })    ({ 7 14 })
     ({ 6 8 9 10 12 13 15 16 })   ({ 11 })    ({ })    ({ })   ({ })
  ({ })    ({ })    ({ 17 })    ({ 18 })
# Sentence pair (6) source length 5 target length 7 alignment score :
2.92134e-13
ပု စွန် တော့ ကျွန် တော် စား တယ်
NULL ({ })   ({ 4 5 })   ({ 6 })   ({ })    ({ })   ({ 1 2 3 7 })
# Sentence pair (7) source length 1 target length 4 alignment score :
```

```
1.37022e-06
သို့ ပေ မဲ့ လည်း
NULL ({ })    ({ 1 2 3 4 })
# Sentence pair (8) source length 8 target length 11 alignment score :
6.03475e-24
ကျွန် တော် ခေါင်း ပေါ် ရေ ခဲ တင် ထား ရ မ လား
NULL ({ })   ({ 1 2 })   ({ 8 9 10 })    ({ })   ({ 5 })   ({ 6 7 })
  ({ 4 })   ({ 3 })   ({ 11 })
# Sentence pair (9) source length 9 target length 12 alignment score :
6.08648e-23
ကွန် ပြူ တာ ရောင် ခြည် နဲ့ ထပ် စစ် ဆေး ပါ မယ် နော်
NULL ({ })    ({ })    ({ })   ({ })     ({ 2 3 4 5 6 7 8 9 })
      ({ 1 })   ({ })   ({ })   ({ 10 11 12 })   ({ })
# Sentence pair (10) source length 9 target length 14 alignment score :
2.07035e-33
ဒီ ကြား မှာ ခ ဏ လေး ထိုင် စော င့် ပေး လို့ ရ မ လား
NULL ({ })     ({ 2 3 })  ({ 1 })   ({ 4 5 6 7 })   ({ 8 9 10 })
({ })   ({ 14 })   ({ 13 })   ({ })   ({ 11 12 })
# Sentence pair (11) source length 6 target length 10 alignment score :
9.24302e-18
မိ သား စု နဲ့ စ ကား ပြော ဆက် ဆံ တာ
NULL ({ })   ({ })     ({ 4 5 6 7 8 9 10 })   ({ })    ({ })
({ })     ({ 1 2 3 })
# Sentence pair (12) source length 7 target length 10 alignment score :
7.40657e-17
ငါ မ နက် တွေ တိုင်း နှာ ခေါင်း သွေး လျှံ တယ်
NULL ({ })   ({ 1 })  ({ 4 })    ({ 8 })    ({ 6 7 9 10 })   ({ })
  ({ 5 })   ({ 2 3 })
# Sentence pair (13) source length 2 target length 2 alignment score :
3.47546e-05
ဒါ ပေါ့
NULL ({ })   ({ 1 2 })   ({ })
# Sentence pair (14) source length 11 target length 23 alignment score :
2.34497e-41
တို့ တွေ အ နည်း ဆုံး သွား ကို ၃ - ၅ မိ နစ် အ ထိ အ နည်း ဆုံး တိုက် သ င့် တယ် ရှ
င့်
NULL ({ 15 })   ({ })   ({ 1 2 })   ({ 19 20 })    ({ 18 })   ({ 6 7 })
   ({ 3 })   ({ 4 5 16 17 }) 3 ({ 8 }) - ({ 9 }) 5  ({ 10 11 12 13 14
})   ({ 21 22 23 })
# Sentence pair (15) source length 6 target length 10 alignment score :
1.93235e-17
ကိုယ် ဝန် က ၃ ၂ ပတ် တော့ ရှိ ပါ ပြီ
NULL ({ })   ({ })   ({ })   ({ }) 32 ({ 1 2 3 4 5 7 8 })    ({ 6 })
   ({ 9 10 })
# Sentence pair (16) source length 7 target length 9 alignment score :
```

```
2.55062e-17
အဲ့ ဒါ လုပ် ပါ မယ် ဆ ရာ မ လေး
NULL ({ }) ({ }) ({ }) ({ }) ({ }) ({ 1 2 3 4 5 }) ({ })
  ({ 6 7 8 9 })
# Sentence pair (17) source length 13 target length 17 alignment score :
1.90262e-33
သား လေး က ရေ အိုင် များ များ ရှိ တဲ့ နေ ရာ က စား ရ တာ ကြိုက် တယ်
```

1.15 For Your Information

အထက်မှာ ကြည့်နေတဲ့ ဖိုင်တွေက IBM Models (1-5) နဲ့ HMM training ရဲ့ alignment ဖိုင်တွေပါပဲ။ GIZA++ က အခြေခံအားဖြင့် asymmetric နည်းလမ်းပါ။ အဲဒါကြောင့် သူက multiple source words ကနေ single word ကိုပဲ mapping လုပ်နိုင်ပြီး အသွားအပြန် မလုပ်ပေးနိုင်ဘူး။ အဲဒါကြောင့် လက်တွေ့ alignment လုပ်တဲ့အခါမှာ ပထမဆုံး source ကနေ target ကို alignment အရင်လုပ်တယ်။ ပြီးတော့ target ကနေ source ကိုလည်း alignment လုပ်တယ်။ အဲဒီ alignment နှစ်ခုကနေမှ ပြန်ပေါင်းယူတဲ့ ပုံစံပါ။

1.15.1 The prepared.1 Folder: The "Setup"

Vocab ဆောက်တာကို အရင်ဆုံးလုပ်ပါတယ်။ စာလုံးတွေကို integer အဖြစ် mapping လုပ်တဲ့အပိုင်းပါ။

File	Purpose
th.vcb / my.vcb	Vocabulary files. Maps every unique word in Thai and Myanmar to a unique integer ID. GIZA++ doesn't process strings; it processes these IDs.
th-my-int-train.snt	Integerized sentences. This is your parallel corpus where every word has been replaced by its ID from the .vcb files.
.vcb.classes	Word Classes. Generated by a tool called mkcls. It groups words into clusters (e.g., "days of the week" might be in one cluster) to help the alignment model generalize better, especially for IBM Model 4.

1.15.2 2. `giza.1` and `giza-inverse.1`: The "Learning"

အထက်မှာ ပြောခဲ့တဲ့အတိုင်းပဲ GIZA++ က "many-to-one" alignment နှစ်ခါ ပြီးသွားတဲ့အခါမှာတော့ အောက်ပါဖိုင်နှစ်ဖိုင်ကို ရလာပါလိမ့်မယ်။

- giza.1 (Forward): Usually Thai → Myanmar.
- giza-inverse.1 (Backward): Myanmar → Thai.

အထက်က နှစ်ဖိုင်ကနေမှ ပြန်ပေါင်းယူထားတဲ့ alignment ဖိုင်က အောက်ပါအတိုင်းပါ။ ဒါက မြန်မာ-ထိုင်း အတွဲအတွက် နောက်ဆုံး alignment ရလဒ်ဖိုင်ပါပဲ။

- th-my.A3.final.gz

တကယ်တမ်း အသေးစိတ် ဘယ်လို လုပ်သလဲ ဆိုတာကတော့ အောက်ပါအတိုင်းပါပဲ။

1.16 The Workflow

The process follows a very specific sequence of scripts (usually managed by the Moses `train-model.perl` script):

1. Word Clustering (`mkcls`): It runs on both languages to create the `.classes` files. This helps the model handle rare words by looking at the behavior of their "class."
2. Vocabulary Extraction: It scans your corpus to create the `.vcb` maps.
3. Sentence Integerization: It converts your raw text into `.snt` files.
4. GIZA++ Training:
    - It starts with **IBM Model 1** (lexical translation probabilities, no word order).
    - It moves to **HMM** (captures local word order).
    - It finishes with **IBM Model 4** (handles "fertility"—the idea that one word can produce two words).
    - The .A3.final is the output of the very last iteration of the highest model used.

Debugging အတွက်က လိုအပ်တဲ့အခါမှာ `.gizacfg` ဖိုင်တွေကို ဝင်ကြည့်ပါ။

1.17 Phrase Table

ဆက်လက်ပြီး phrase table ကို လေ့လာရအောင်။
model ဖိုလ်ဒါအောက်ကို directory change ပါမယ်။

[18]:
```
%cd /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model
```

[19]:
```
!ls --color=auto
```

```
aligned.1.grow-diag-final-and  lex.1.f2e
extract.1.inv.sorted.gz        moses.ini.1
extract.1.o.sorted.gz          phrase-table.1.gz
extract.1.sorted.gz            reordering-table.1.wbe-msd-

bidirectional-fe.gz
lex.1.e2f
```

ပြောရရင် မော်ဒယ် ဖိုလ်ဒါက PBSMT ရဲ့ translation အတွက် အဓိကသုံးတဲ့ အသိဉာဏ်ပိုင်းလို့ ပြောလို့ရပါတယ်။ GIZA++ က word-level alignment ပဲ အကြမ်းလုပ်ပေးတာပါ။ ဒီ model ဖိုလ်ဒါအထဲမှာတော့ phrase-level mapping နဲ့ probability score တွေ ရအောင်တွက်ထုတ်ထားတဲ့ ဖိုင်တွေပါဝင်လာပါပြီ။ အဲဒီ phrase pair, probability score တွေကိုပဲ သုံးပြီးတော့ moses decoder က ဘာသာပြန်တဲ့အလုပ်ကို လုပ်ပေးသွားမှာပါ။ ဖိုင်တစ်ဖိုင်ချင်းစီ ကိုရှင်းပြထားတဲ့ အောက်ပါ ဇယားကို လေ့လာပါ။

File	Description
`aligned.1.grow-diag-final-and`	The symmetrized word alignment file. It combines the giza.1 and giza-inverse.1 outputs using the grow-diag-final-and heuristic to create a many-to-many alignment grid.
`lex.1.e2f` / `lex.1.f2e`	Lexical translation tables. These contain word-level translation probabilities

File	Description
`extract.1.sorted.gz`	An intermediate file containing all extracted phrase pairs that are "consistent" with the word alignment, sorted for efficient scoring.
`extract.1.inv.sorted.gz`	The inverse version of the extract file, used to calculate the "backward" translation probabilities
`extract.1.o.sorted.gz`	Contains orientation information for every phrase pair (monotonic, swapped, or discontinuous), used to build the reordering table.
`phrase-table.1.gz`	The core model. It lists every source phrase, its possible target translations, and 5 scores (usually: inverse phrase prob, lexical weight, direct phrase prob, inverse lexical weight, and a phrase penalty).
`reordering-table.1..gz`	The Lexicalized Reordering Model. This tells Moses how likely a phrase is to change its position relative to the previous phrase (e.g., in Thai-Myanmar, how likely is an adjective-noun swap?).
`moses.ini.1`	The Configuration file. This is the "map" for the decoder. It tells Moses which phrase table to load, which reordering model to use, and what weights to apply to each feature.

[20]:
```
%mkdir tmp
```

[21]:
```
!cp phrase-table.1.gz ./tmp/
```

[23]:
```
!cp reordering-table.1.wbe-msd-bidirectional-fe.gz ./tmp/
```

[24]:
```
%cd tmp
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp
```

[25]:
```
!gunzip ./phrase-table.1.gz
```

[26]:
```
!ls --color=auto
```

```
phrase-table.1  reordering-table.1.wbe-msd-bidirectional-fe.gz
```

[28]:
```
!head -n 300 ./phrase-table.1
```

```
! ! ! ||| !!! ||| 0.331844 0.041477 0.331844 0.75 ||| 0-0 1-0 2-0 ||| 1 1 1 |||
|||
% ||| % ||| 0.306122 0.328767 0.833333 0.657534 ||| 0-0 ||| 49 18 15 ||| |||
% ||| %   ||| 0.653084 0.179768 0.108847 0.0360293 ||| 0-0 0-1 ||| 3 18 3 |||
```

|||
% က ||| ||| 0.000932147 1.09161e-06 0.331844 0.0073358 ||| 1-0 ||| 356 1 1
||| |||
% က နည်း သည် ||| ||| 0.0474063 0.000164062 0.331844 0.0260911 |||
0-0 1-0 2-0 3-0 ||| 7 1 1 ||| |||
% ဆို ||| % ||| 0.00677233 0.00272104 0.165922 0.657534 ||| 0-0 ||| 49 2 1 |||
|||
% ဆို ||| % ||| 0.0829611 0.0114478 0.165922 0.00245314 ||| 0-0 0-1 1-2
||| 4 2 1 ||| |||
% ဆို တော့ ||| % ||| 0.489813 0.000154099 0.489813 0.000819808 ||| 0-0
1-0 2-0 0-1 1-2 ||| 4 4 3 ||| |||
% ဆို တော့ ||| % ||| 0.331844 0.000477638 0.0829611 4.65487e-07
||| 0-0 1-0 2-0 0-1 1-2 ||| 1 4 1 ||| |||
% ဆို တော့ ပြင်း ထန် ||| % ||| 0.576972 4.33677e-06 0.576972
2.94149e-05 ||| 0-0 0-1 1-2 2-2 3-3 4-3 3-4 ||| 2 2 2 ||| |||
% ဆို ရင် ||| % ||| 0.00677233 3.27548e-05 0.331844 0.657534 ||| 0-0 ||| 49 1 1
||| |||
% ဆို ရင် အ ||| % ||| 0.00677233 1.94243e-06 0.331844 0.657534 ||| 0-0 ||| 49 1
1 ||| |||
% ဆို ရင် အ ဆ ||| % ||| 0.00677233 5.38325e-09 0.331844 0.657534 ||| 0-0 ||| 49
1 1 ||| |||
% တော့ ||| % ||| 0.0235499 0.0247701 0.576972 0.32947 ||| 0-0 1-0 ||| 49 2 2 |||
|||
% နည်း ||| ||| 0.0474063 0.021875 0.331844 0.0489303 ||| 0-0 1-0 |||
7 1 1 ||| |||
% နည်း နေ ||| ||| 0.0474063 0.000813682 0.331844 0.0489303 ||| 0-0
1-0 ||| 7 1 1 ||| |||
% နည်း နေ သည် ||| ||| 0.0474063 1.44966e-06 0.331844 0.0489303 |||
0-0 1-0 ||| 7 1 1 ||| |||
% နည်း လာ သည် ||| ||| 0.0474063 5.46875e-05 0.331844 0.0260917 |||
0-0 1-0 2-0 3-0 ||| 7 1 1 ||| |||
% နည်း သည် ||| ||| 0.164849 0.0021875 0.576972 0.0347299 ||| 0-0 1-0
2-0 ||| 7 2 2 ||| |||
% ပါ ||| % ||| 0.00677233 0.0247701 0.331844 0.32895 ||| 0-0 1-0 ||| 49 1 1 |||
|||
% ဘဲ ||| % ||| 0.0235499 0.000598126 0.576972 0.657534 ||| 0-0 ||| 49 2 2 |||
|||
% ဘဲ ရှိ ||| % ||| 0.0235499 5.51981e-06 0.576972 0.657534 ||| 0-0 ||| 49 2 2
||| |||
% ဘဲ ရှိ နေ ||| % ||| 0.0235499 2.0532e-07 0.576972 0.657534 ||| 0-0 ||| 49 2 2
||| |||
% ဘဲ ရှိ နေ တယ် ||| % ||| 0.0235499 8.74241e-09 0.576972 0.657534 ||| 0-0 ||| 49
2 2 ||| |||

```
% လောက် ||| % ||| 0.0235499 0.0112592 0.576972 0.329555 ||| 0-0 1-0 ||| 49 2 2
||| |||
% လောက် ရှိ တယ် ||| % ||| 0.00677233 4.22562e-05 0.331844 0.165193 ||| 0-0 1-0
2-0 3-0 ||| 49 1 1 ||| |||
% အ ||| % ||| 0.00677233 0.0194966 0.331844 0.657534 ||| 0-0 ||| 49 1 1 ||| |||
% အ သေ ||| % ||| 0.00677233 2.24211e-05 0.331844 0.657534 ||| 0-0 ||| 49 1 1␣
 ↪|||
|||
% အ သေ အ ||| % ||| 0.00677233 1.32962e-06 0.331844 0.657534 ||| 0-0 ||| 49 1␣
 ↪1
||| |||
&#91; ||| &#91; ||| 0.0553074 0.321429 0.00809376 0.243243 ||| 0-0 ||| 6 41 1
||| |||
&#91; ||| &#91;         &#93; ||| 0.653084 0.225807 0.0637155
3.63421e-15 ||| 0-4 ||| 4 41 4 ||| |||
&#91; ||| &#93; ||| 0.130617 0.225807 0.0637155 0.189189 ||| 0-0 ||| 20 41 4 |||
|||
&#91; ||| &#93;    ||| 0.653084 0.225807 0.0637155 0.00192648 ||| 0-0 ||| 4 41
4 ||| |||
&#91; ||| &#93;    ||| 0.279893 0.113177 0.0477866 0.0153397 ||| 0-0 0-1 ||| 7
41 3 ||| |||
&#91; ||| , ||| 0.00873275 0.0291262 0.00809376 0.162162 ||| 0-0 ||| 38 41 1 |||
|||
&#91; |||         &#93; ||| 0.653084 0.225807 0.0637155 2.7428e-11 |||
0-3 ||| 4 41 4 ||| |||
&#91; |||         &#93;    ||| 0.653084 0.225807 0.0637155
2.79294e-13 ||| 0-3 ||| 4 41 4 ||| |||
&#91; |||       &#93; ||| 0.653084 0.225807 0.0637155 6.58694e-08 ||| 0-2
||| 4 41 4 ||| |||
&#91; |||       &#93;    ||| 0.653084 0.225807 0.0637155 6.70735e-10 |||
0-2 ||| 4 41 4 ||| |||
&#91; |||     &#93; ||| 0.653084 0.225807 0.0637155 1.43216e-05 ||| 0-1 |||
4 41 4 ||| |||
&#91; |||     &#93;    ||| 0.653084 0.225807 0.0637155 1.45834e-07 ||| 0-1
||| 4 41 4 ||| |||
&#91; စစ် ဆေး |||    &#93;    ||| 0.279893 0.00839246 0.653084 0.00474528 |||
1-0 2-0 0-1 0-2 ||| 7 3 3 ||| |||
&#91; အ ||| &#91;          ||| 0.326542 0.055321 0.653084 9.10826e-05
||| 0-0 0-1 1-1 0-2 0-3 ||| 6 3 3 ||| |||
&#91; အ မည် ||| &#91;   , ||| 0.279893 0.00556384 0.244906 0.0064196 ||| 0-0
1-1 2-1 0-2 ||| 7 8 3 ||| |||
&#91; အ မည် ||| &#91;   ,   ||| 0.164849 0.00556384 0.144243 0.000248235
||| 0-0 1-1 2-1 0-2 ||| 7 8 2 ||| |||
&#91; အ မည် ||| &#91;   ,     ||| 0.164849 0.00556384 0.144243
4.22744e-08 ||| 0-0 1-1 2-1 0-2 ||| 7 8 2 ||| |||
&#91; အ မည် |||   , ||| 0.331844 0.000924554 0.0414805 0.0263917 ||| 1-0 2-0
```

0-1 ||| 1 8 1 ||| |||
[အ မည် တည် ||| [, ||| 0.331844 0.000777301 0.331844
0.000440334 ||| 3-0 0-1 1-2 2-2 0-3 ||| 1 1 1 ||| |||
[အ မည် တည် နေ ||| [, ||| 0.165922 4.32452e-05 0.331844
1.35024e-05 ||| 3-0 0-1 1-2 2-2 0-3 4-4 ||| 2 1 1 ||| |||
[အ မည် နေ ||| [, ||| 0.164849 0.000206957 0.128216 0.0064196 |||
0-0 1-1 2-1 0-2 ||| 7 9 2 ||| |||
[အ မည် နေ ||| [, ||| 0.164849 0.000206957 0.128216
0.000248235 ||| 0-0 1-1 2-1 0-2 ||| 7 9 2 ||| |||
[အ မည် နေ ||| [, ||| 0.164849 0.000206957 0.128216
4.22744e-08 ||| 0-0 1-1 2-1 0-2 ||| 7 9 2 ||| |||
[အ မည် နေ ||| , ||| 0.331844 5.14377e-05 0.0368716 0.000809273 |||
1-0 2-0 0-1 3-2 ||| 1 9 1 ||| |||
[အ မည် နေ ||| , , ||| 0.331844 5.14377e-05 0.0368716
8.3477e-11 ||| 1-0 2-0 0-1 3-2 ||| 1 9 1 ||| |||
[အ မည် နေ ||| , ||| 0.331844 5.14377e-05 0.0368716
1.37819e-07 ||| 1-0 2-0 0-1 3-2 ||| 1 9 1 ||| |||
[အ မည် နေ ရပ် ||| [, ||| 0.164849 1.58012e-07 0.128216 0.0064196
||| 0-0 1-1 2-1 0-2 ||| 7 9 2 ||| |||
[အ မည် နေ ရပ် ||| [, ||| 0.279893 1.58012e-07 0.217695
0.000248235 ||| 0-0 1-1 2-1 0-2 ||| 7 9 3 ||| |||
[အ မည် နေ ရပ် ||| [, ||| 0.279893 1.58012e-07 0.217695
4.22744e-08 ||| 0-0 1-1 2-1 0-2 ||| 7 9 3 ||| |||
[အ မည် နေ ရပ် ||| [, ||| 0.165922 1.83706e-06
0.0368716 1.26629e-10 ||| 4-1 1-2 2-2 0-3 3-4 ||| 2 9 1 ||| |||
] ||| [||| 0.0553074 0.107143 0.0110615 0.09375 ||| 0-0 ||| 6 30 1 |||
|||
] |||] " ||| 0.653084 0.387097 0.0870778 0.000127762 ||| 0-0 ||| 4
30 4 ||| |||
] |||] ||| 0.163271 0.387097 0.108847 0.375 ||| 0-0 ||| 20 30 5 ||| |||
] |||] ||| 0.022123 0.0977987 0.0110615 0.000514984 ||| 0-0
0-1 0-2 0-3 ||| 15 30 1 ||| |||
] |||] ||| 0.0384648 0.0783271 0.0384648 3.21865e-05
||| 0-0 0-1 0-2 0-3 0-4 ||| 30 30 2 ||| |||
] |||] " ||| 0.653084 0.387097 0.0870778 4.35287e-08 ||| 0-1
||| 4 30 4 ||| |||
] |||] ||| 0.653084 0.387097 0.0870778 0.000127762 ||| 0-1 ||| 4 30
4 ||| |||
] ||| [||| 0.331844 0.107143 0.0110615 0.00362515 ||| 0-1 ||| 1 30
1 ||| |||
] |||] " ||| 0.653084 0.387097 0.0870778 2.00201e-10 |||
0-2 ||| 4 30 4 ||| |||
] |||] ||| 0.653084 0.387097 0.0870778 5.87618e-07 ||| 0-2 |||
4 30 4 ||| |||
] ကို ||| [||| 0.0829611 0.00935213 0.165922 2.82267e-07 |||

1-1 0-3 ||| 4 2 1 ||| |||
] ကို ||| [||| 0.0829611 0.00935213 0.165922 0.00012325 ||| 1-0
0-2 ||| 4 2 1 ||| |||
] ကို သွား ||| [||| 0.0829611 5.03378e-05 0.165922
2.82267e-07 ||| 1-1 0-3 ||| 4 2 1 ||| |||
] ကို သွား ||| [||| 0.0829611 5.03378e-05 0.165922 0.00012325
||| 1-0 0-2 ||| 4 2 1 ||| |||
] ကို သွား ခဲ့ ||| [||| 0.0829611 1.89804e-07 0.165922
2.82267e-07 ||| 1-1 0-3 ||| 4 2 1 ||| |||
] ကို သွား ခဲ့ ||| [||| 0.0829611 1.89804e-07 0.165922
0.00012325 ||| 1-0 0-2 ||| 4 2 1 ||| |||
] ကို သွား ခဲ့ တာ ||| [||| 0.0829611 5.83453e-09 0.165922
2.82267e-07 ||| 1-1 0-3 ||| 4 2 1 ||| |||
] ကို သွား ခဲ့ တာ ||| [||| 0.0829611 5.83453e-09 0.165922
0.00012325 ||| 1-0 0-2 ||| 4 2 1 ||| |||
] တို့ |||] ||| 0.0979626 0.049948 0.653084 0.18892 ||| 0-0 1-0 ||| 20
3 3 ||| |||
] တွေ |||] ||| 0.0165922 0.00559022 0.0829611 0.375 ||| 0-0 ||| 20 4 1
||| |||
] တွေ |||] ||| 0.022123 0.00141235 0.0829611 0.000514984
||| 0-0 0-1 0-2 0-3 ||| 15 4 1 ||| |||
] တွေ |||] ||| 0.0384648 0.00113115 0.288486
3.21865e-05 ||| 0-0 0-1 0-2 0-3 0-4 ||| 30 4 2 ||| |||
] တွေ ရော |||] ||| 0.0110615 7.45328e-06 0.331844
3.21865e-05 ||| 0-0 0-1 0-2 0-3 0-4 ||| 30 1 1 ||| |||
] တွေ ရော ရှိ |||] ||| 0.0110615 6.87826e-08 0.331844
3.21865e-05 ||| 0-0 0-1 0-2 0-3 0-4 ||| 30 1 1 ||| |||
] တွေ ရော ရှိ သ |||] ||| 0.0110615 8.27331e-10
0.331844 3.21865e-05 ||| 0-0 0-1 0-2 0-3 0-4 ||| 30 1 1 ||| |||
] တွေ သ |||] ||| 0.022123 1.6988e-05 0.165922 0.000514984
||| 0-0 0-1 0-2 0-3 ||| 15 2 1 ||| |||
] တွေ သ |||] ||| 0.0110615 1.36057e-05 0.165922
3.21865e-05 ||| 0-0 0-1 0-2 0-3 0-4 ||| 30 2 1 ||| |||
] တွေ သ င့် |||] ||| 0.022123 2.04176e-07 0.165922
0.000514984 ||| 0-0 0-1 0-2 0-3 ||| 15 2 1 ||| |||
] တွေ သ င့် |||] ||| 0.0110615 1.63525e-07 0.165922
3.21865e-05 ||| 0-0 0-1 0-2 0-3 0-4 ||| 30 2 1 ||| |||
] တွေ သ င့် မှာ |||] ||| 0.022123 2.14023e-09 0.165922
0.000514984 ||| 0-0 0-1 0-2 0-3 ||| 15 2 1 ||| |||
] တွေ သ င့် မှာ |||] ||| 0.0110615 1.71411e-09
0.165922 3.21865e-05 ||| 0-0 0-1 0-2 0-3 0-4 ||| 30 2 1 ||| |||
" B " ||| " D" ||| 0.331844 0.0423341 0.331844 0.0403706 |||
0-0 0-1 1-1 2-1 ||| 1 1 1 ||| |||
" B " နဲ့ " ||| B " D" ||| 0.165922

1.11655e-06 0.110615 1.16503e-09 ||| 3-2 0-3 0-4 1-4 2-4 ||| 2 3 1 ||| |||
" B " နဲ့ " ||| " D" ||| 0.165922 1.11655e-06
0.110615 0.00685874 ||| 3-0 0-1 0-2 1-2 2-2 ||| 2 3 1 ||| |||
" B " နဲ့ " ||| " D" ||| 0.165922
1.11655e-06 0.110615 2.05111e-05 ||| 3-1 0-2 0-3 1-3 2-3 ||| 2 3 1 ||| |||
" B " နဲ့ ||| B " D" ||| 0.165922 0.0148083
0.110615 1.16503e-09 ||| 3-2 0-3 0-4 1-4 2-4 ||| 2 3 1 ||| |||
" B " နဲ့ ||| " D" ||| 0.165922 0.0148083 0.110615
0.00685874 ||| 3-0 0-1 0-2 1-2 2-2 ||| 2 3 1 ||| |||
" B " နဲ့ ||| " D" ||| 0.165922 0.0148083
0.110615 2.05111e-05 ||| 3-1 0-2 0-3 1-3 2-3 ||| 2 3 1 ||| |||
" C " က ||| O ||| 0.0165922 3.21591e-13 0.165922 0.0001747 ||| 3-0 |||
20 2 1 ||| |||
" C " က ||| O ||| 0.0165922 3.21591e-13 0.165922 5.2244e-07 |||
3-0 ||| 20 2 1 ||| |||
" C " က တူ ||| O ||| 0.0165922 4.12279e-16 0.165922 0.0001747 ||| 3-0
||| 20 2 1 ||| |||
" C " က တူ ||| O ||| 0.0165922 4.12279e-16 0.165922 5.2244e-07
||| 3-0 ||| 20 2 1 ||| |||
" F " ||| " I " ||| 0.331844 0.0807144 0.331844 0.0093985
||| 0-0 2-0 1-1 2-1 2-2 ||| 1 1 1 ||| |||
" F " နှ ||| " I " ||| 0.165922 0.00247265 0.331844
0.00173893 ||| 3-0 0-1 2-1 1-2 2-2 2-3 ||| 2 1 1 ||| |||
" F " နှ င့် ||| " I " ||| 0.165922 2.97183e-05 0.331844
0.00173893 ||| 3-0 0-1 2-1 1-2 2-2 2-3 ||| 2 1 1 ||| |||
" I " ကို ||| " F" ||| 0.0165922 4.50265e-13 0.110615
7.55332e-08 ||| 3-1 ||| 20 3 1 ||| |||
" I " ကို ||| F" ||| 0.0165922 4.50265e-13 0.110615 0.0002217 |||
3-0 ||| 20 3 1 ||| |||
" I " ကို ||| " F" ||| 0.0165922 4.50265e-13
0.110615 4.29029e-12 ||| 3-2 ||| 20 3 1 ||| |||
" I " ကို လဲ ||| " F" ||| 0.0165922 4.59247e-15 0.110615
7.55332e-08 ||| 3-1 ||| 20 3 1 ||| |||
" I " ကို လဲ ||| F" ||| 0.0165922 4.59247e-15 0.110615 0.0002217
||| 3-0 ||| 20 3 1 ||| |||
" I " ကို လဲ ||| " F" ||| 0.0165922 4.59247e-15
0.110615 4.29029e-12 ||| 3-2 ||| 20 3 1 ||| |||
" O " ||| " C" ||| 0.331844 0.0492658 0.331844 0.0464138 |||
0-0 0-1 1-1 2-1 ||| 1 1 1 ||| |||
" O " နဲ့ " ||| " C" ||| 0.165922 1.29937e-06
0.165922 0.00788544 ||| 3-0 0-1 0-2 1-2 2-2 ||| 2 2 1 ||| |||
" O " နဲ့ " ||| " C" ||| 0.165922
1.29937e-06 0.165922 2.35814e-05 ||| 3-1 0-2 0-3 1-3 2-3 ||| 2 2 1 ||| |||
" O " နဲ့ ||| " C" ||| 0.165922 0.017233 0.165922
0.00788544 ||| 3-0 0-1 0-2 1-2 2-2 ||| 2 2 1 ||| |||

```
&quot; 0 &quot; နဲ့ |||       &quot; C&quot; ||| 0.165922 0.017233 0.165922
2.35814e-05 ||| 3-1 0-2 0-3 1-3 2-3 ||| 2 2 1 ||| |||
&quot; က ||| 0 ||| 0.0165922 2.51333e-05 0.165922 0.0001747 ||| 1-0 ||| 20 2 1
||| |||
&quot; က ||| 0     ||| 0.0165922 2.51333e-05 0.165922 5.2244e-07 ||| 1-0 |||
20 2 1 ||| |||
&quot; က တူ ||| 0 ||| 0.0165922 3.22209e-08 0.165922 0.0001747 ||| 1-0 ||| 20 2
1 ||| |||
&quot; က တူ ||| 0     ||| 0.0165922 3.22209e-08 0.165922 5.2244e-07 ||| 1-0
||| 20 2 1 ||| |||
&quot; က တူ နေ ||| 0 ||| 0.0165922 1.19852e-09 0.165922 0.0001747 ||| 1-0 |||␣
 ↪20
2 1 ||| |||
&quot; က တူ နေ ||| 0     ||| 0.0165922 1.19852e-09 0.165922 5.2244e-07 |||
1-0 ||| 20 2 1 ||| |||
&quot; က တူ နေ တယ် ||| 0 ||| 0.0165922 5.10323e-11 0.165922 0.0001747 ||| 1-0
||| 20 2 1 ||| |||
&quot; က တူ နေ တယ် ||| 0     ||| 0.0165922 5.10323e-11 0.165922 5.2244e-07
||| 1-0 ||| 20 2 1 ||| |||
&quot; ကို ||| &quot; F&quot; ||| 0.0165922 4.524e-05 0.110615 7.55332e-08 |||
1-1 ||| 20 3 1 ||| |||
&quot; ကို ||| F&quot; ||| 0.0165922 4.524e-05 0.110615 0.0002217 ||| 1-0 ||| 20
3 1 ||| |||
&quot; ကို |||      &quot; F&quot; ||| 0.0165922 4.524e-05 0.110615
4.29029e-12 ||| 1-2 ||| 20 3 1 ||| |||
&quot; ကို လဲ ||| &quot; F&quot; ||| 0.0165922 4.61425e-07 0.110615 7.55332e-08
||| 1-1 ||| 20 3 1 ||| |||
&quot; ကို လဲ ||| F&quot; ||| 0.0165922 4.61425e-07 0.110615 0.0002217 ||| 1-0
||| 20 3 1 ||| |||
&quot; ကို လဲ |||      &quot; F&quot; ||| 0.0165922 4.61425e-07 0.110615
4.29029e-12 ||| 1-2 ||| 20 3 1 ||| |||
&quot; ကို လဲ ခွဲ ||| &quot; F&quot; ||| 0.0165922 1.95737e-10 0.110615
7.55332e-08 ||| 1-1 ||| 20 3 1 ||| |||
&quot; ကို လဲ ခွဲ ||| F&quot; ||| 0.0165922 1.95737e-10 0.110615 0.0002217 |||
1-0 ||| 20 3 1 ||| |||
&quot; ကို လဲ ခွဲ |||      &quot; F&quot; ||| 0.0165922 1.95737e-10 0.110615
4.29029e-12 ||| 1-2 ||| 20 3 1 ||| |||
&quot; ကို လဲ ခွဲ ခြား ||| &quot; F&quot; ||| 0.0165922 2.39856e-13 0.110615
7.55332e-08 ||| 1-1 ||| 20 3 1 ||| |||
&quot; ကို လဲ ခွဲ ခြား ||| F&quot; ||| 0.0165922 2.39856e-13 0.110615 0.0002217
||| 1-0 ||| 20 3 1 ||| |||
&quot; ကို လဲ ခွဲ ခြား |||      &quot; F&quot; ||| 0.0165922 2.39856e-13
0.110615 4.29029e-12 ||| 1-2 ||| 20 3 1 ||| |||
&quot; ကျွန် တော် |||      ||| 0.00161089 2.71984e-06 0.331844 0.000804589 |||
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1-0 2-0 2-1 ||| 206 1 1 ||| |||
" ကျွန် တော် လမ်း လျှောက် ||| ||| 0.0553074 1.79143e-07 0.331844
0.000290903 ||| 1-0 2-0 2-1 3-2 4-2 ||| 6 1 1 ||| |||
(||| ||| 0.000322989 0.0003064 0.653084 1 ||| 0-0 ||| 8088 4 4 ||| |||
* ||| * ||| 0.391892 0.663043 1 0.835616 ||| 0-0 ||| 74 29 29 ||| |||
* ထုံ ကျဉ် ||| * ||| 0.0663689 0.00805457 0.331844 0.0621493 ||| 0-0 1-0 2-0
||| 5 1 1 ||| |||
* ထုံ ကျဉ် ဆေး ထိုး ||| * ||| 0.0553074 0.000374529 0.331844 0.0122469
||| 3-0 4-0 0-1 1-1 2-1 ||| 6 1 1 ||| |||
* ထုံ ကျဉ် သော ||| * ||| 0.0663689 0.000350199 0.331844 0.0469165 ||| 0-0
1-0 2-0 3-0 ||| 5 1 1 ||| |||
* ထုံ ဆေး ||| * ||| 0.230789 0.00805457 0.576972 0.0400401 ||| 0-0 1-0 2-0
||| 5 2 2 ||| |||
* ထုံ ဆေး ထိုး ||| * ||| 0.192324 0.00368443 0.576972 0.0152372 ||| 3-0
0-1 1-1 2-1 ||| 6 2 2 ||| |||
* ထုံ ဆေး ထိုး နေ ||| * ||| 0.0553074 0.000137049 0.331844 0.0152372 |||
3-0 0-1 1-1 2-1 ||| 6 1 1 ||| |||
* ထုံ ဆေး ထိုး နှံ့ ||| * ||| 0.0553074 0.000374529 0.331844 0.00988899
||| 2-0 3-0 0-1 1-1 4-1 ||| 6 1 1 ||| |||
* ထုံ အောင် ||| * ||| 0.0663689 0.00402729 0.331844 0.0400882 ||| 0-0 1-0
2-0 ||| 5 1 1 ||| |||
* ထုံ အောင် ဆေး ထိုး ||| * ||| 0.0553074 0.000187265 0.331844 0.00789965
||| 3-0 4-0 0-1 1-1 2-1 ||| 6 1 1 ||| |||
* ပြီး ခဲ့ ပါ ပြီ ||| * ||| 0.110615 8.7388e-05 0.331844
0.00480268 ||| 0-0 1-1 2-1 3-1 4-1 4-2 ||| 3 1 1 ||| |||
* ပြီး သွား ပါ ပြီ ||| * ||| 0.110615 0.000699104 0.331844
0.00527525 ||| 0-0 1-1 2-1 3-1 4-1 4-2 ||| 3 1 1 ||| |||
* ပြီး သွား ပြီ ||| * ||| 0.110615 0.00439437 0.331844 0.00677106
||| 0-0 1-1 2-1 3-1 3-2 ||| 3 1 1 ||| |||
* ပြီး သွား ပြီ ရှင် ||| * ||| 0.331844 0.000595905 0.331844
0.00192347 ||| 0-0 1-1 2-1 3-1 3-2 4-3 ||| 1 1 1 ||| |||
* သွား ကို ခွဲ ||| * ||| 0.110615 0.00261054 0.331844 6.89171e-05
||| 0-0 2-1 3-1 1-2 3-3 ||| 3 1 1 ||| |||
* သွား ခွဲ ||| * ||| 0.110615 0.0780987 0.331844 0.000136819 ||| 0-0
2-1 1-2 2-3 ||| 3 1 1 ||| |||
* သွား ခွဲ ထုတ် ||| * ||| 0.110615 0.00705987 0.331844 0.00221426
||| 0-0 2-1 3-1 1-2 3-3 ||| 3 1 1 ||| |||
* သွား ခွဲ သည် * ||| *   * ||| 0.331844 0.00675429 0.331844 5.8463e-05
||| 0-0 2-1 1-2 2-3 3-4 4-4 ||| 1 1 1 ||| |||
* အ ||| * ||| 0.0264764 0.0393199 0.653084 0.835616 ||| 0-0 ||| 74 3 3 ||| |||
* အ နာ ||| * ||| 0.0264764 0.00032098 0.653084 0.835616 ||| 0-0 ||| 74 3 3 |||
|||
* အ နာ ကြ ည့် ||| * ||| 0.331844 0.00273231 0.331844 0.00566017 ||| 0-0

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3-1 4-1 1-2 2-2 ||| 1 1 1 ||| |||
* အ နာ ဆေး ||| * ||| 0.0264764 1.43115e-06 0.653084 0.835616 ||| 0-0 ||| 74 3
3
||| |||
* အ နာ ဆေး ချုပ် ||| * ||| 0.00448438 1.21362e-10 0.331844 0.835616 ||| 0-0 |||
74 1 1 ||| |||
* အ နာ ပြန် ချုပ် ||| *     ||| 0.331844 0.000835394 0.331844 0.00197451
||| 0-0 3-1 4-1 1-2 2-2 4-2 ||| 1 1 1 ||| |||
* အံ ဆုံး သွား ခွဲ ||| *     ||| 0.331844 0.0175002 0.331844 0.00134036
||| 0-0 4-1 3-2 1-3 2-3 ||| 1 1 1 ||| |||
, 0 0 0 - ||| , 000 - ||| 0.165922 0.00663168 0.331844 0.000744062 ||| 1-0 0-1
1-1 2-1 3-1 4-2 ||| 2 1 1 ||| |||
, 0 0 0 ||| &lt; 100 , 000 ||| 0.164849 4.9423e-05 0.144243 9.6203e-08 ||| 1-1
2-2 2-3 3-3 ||| 7 8 2 ||| |||
, 0 0 0 ||| , 000 ||| 0.0368716 0.010744 0.0414805 0.00137141 ||| 1-0 0-1 1-1
2-1 3-1 ||| 9 8 1 ||| |||
, 0 0 0 ||| 10 , 000 ||| 0.0829611 1.99133e-05 0.0414805 0.000206994 ||| 1-0 2-1
2-2 3-2 ||| 4 8 1 ||| |||
, 0 0 0 ||| 100 , 000 ||| 0.164849 4.9423e-05 0.144243 0.000195495 ||| 1-0 2-1
2-2 3-2 ||| 7 8 2 ||| |||
, 0 0 0 |||   &lt; 100 , 000 ||| 0.164849 4.9423e-05 0.144243 2.54938e-11 |||
1-2 2-3 2-4 3-4 ||| 7 8 2 ||| |||
, 0 0 0 ဆဲလ် ||| &lt; 100 , 000    ||| 0.230789 9.19498e-06 0.288486
2.33219e-08 ||| 1-1 2-2 2-3 3-3 4-4 ||| 5 4 2 ||| |||
, 0 0 0 ဆဲလ် ||| 100 , 000    ||| 0.230789 9.19498e-06 0.288486 4.73927e-05
||| 1-0 2-1 2-2 3-2 4-3 ||| 5 4 2 ||| |||
, 0 0 0 ဘတ် ||| 10 , 000    ||| 0.110615 4.56346e-06 0.331844 2.11808e-05 |||
1-0 2-1 2-2 3-2 4-3 ||| 3 1 1 ||| |||
, 0 ||| &lt; 100 ||| 0.104904 0.000243482 0.164849 5.43227e-05 ||| 1-1 ||| 11 7
2 ||| |||
, 0 ||| 10 ||| 0.00238737 9.81027e-05 0.0474063 0.116883 ||| 1-0 ||| 139 7 1 |||
|||
, 0 ||| 100 ||| 0.0501715 0.000243482 0.164849 0.11039 ||| 1-0 ||| 23 7 2 |||
|||
, 0 |||   &lt; 100 ||| 0.104904 0.000243482 0.164849 1.43955e-08 ||| 1-2 |||
11 7 2 ||| |||
, ||| , ||| 0.165252 0.0485437 0.22427 0.0221729 ||| 0-0 ||| 38 28 7 ||| |||
, ||| - ||| 0.00239408 0.0030257 0.0412123 0.0044346 ||| 0-0 ||| 482 28 2 |||
|||
, |||   ||| 0.00754192 0.0020877 0.0118516 0.0022173 ||| 0-0 ||| 44 28 1 |||
|||
, |||    ||| 0.00188028 0.0021295 0.0699733 0.0110865 ||| 0-0 ||| 1042 28 3
||| |||
, |||      ||| 0.130617 0.0021295 0.0699733 8.18405e-06 ||| 0-0 ||| 15 28 3
||| |||
, |||        ||| 0.279893 0.0021295 0.0699733 1.34145e-07 ||| 0-0 ||| 7
```

28 3 ||| |||
, ||| ||| 0.00319081 0.0016502 0.0118516 0.0022173 ||| 0-0 ||| 104 28 1 |||
|||
, ||| ||| 0.000715578 0.0015903 0.0699733 0.0243902 ||| 0-0 ||| 2738 28 3
||| |||
, ||| ||| 0.0195203 0.211538 0.0118516 0.0243902 ||| 0-0 ||| 17 28 1 |||
|||
, ||| ||| 0.0678791 0.571429 0.0412123 0.0088692 ||| 0-0 ||| 17 28 2 |||
|||
, ||| ||| 0.000701574 0.0026093 0.0118516 0.0088692 ||| 0-0 ||| 473 28 1 |||
|||
, ||| ||| 0.0414805 0.0026093 0.0118516 9.73661e-06 ||| 0-0 ||| 8 28 1
||| |||
, က လေး ||| ||| 0.00904548 0.000115551 0.653084 0.0451709 ||| 1-0 2-0 |||
361 5 5 ||| |||
, ခင် ||| ||| 0.000280986 0.000141883 0.331844 0.0736693 ||| 1-0 ||| 1181 1
1 ||| |||
, ခု တော့ ||| ||| 9.72865e-05 5.742e-07 0.331844 0.117391 ||| 2-0 ||| 3411
1 1 ||| |||
, ခု တော့ ဒေါက် တာ ||| ||| 0.0132738 3.17003e-08 0.0829611 0.0480012
||| 2-0 3-1 4-1 ||| 25 4 1 ||| |||
, ခု တော့ ဒေါက် တာ ||| ||| 0.0829611 3.17003e-08 0.0829611
1.99877e-05 ||| 2-0 3-1 4-1 ||| 4 4 1 ||| |||
, ခု တော့ ဒေါက် တာ ||| ||| 0.0829611 3.17003e-08 0.0829611
8.70065e-09 ||| 2-0 3-1 4-1 ||| 4 4 1 ||| |||
, ခု တော့ ဒေါက် တာ ||| ||| 0.0829611 3.17003e-08 0.0829611
1.95638e-10 ||| 2-0 3-1 4-1 ||| 4 4 1 ||| |||
, ချက် နား ||| ||| 0.00078081 1.38085e-07 0.165922 0.0752754 ||| 2-0 ||| 425
2 1 ||| |||
, ချက် နား ||| ||| 0.00164279 1.38085e-07 0.165922 0.00075654 ||| 2-0
||| 202 2 1 ||| |||
, ချောင်း ဆိုး ရာ က ||| ||| 0.0288125 3.59054e-12 0.39185 0.0002329 ||| 4-0
||| 68 5 3 ||| |||
, ချောင်း ဆိုး ရာ က ||| ||| 0.0237032 3.59054e-12 0.0663689 2.37157e-06
||| 4-0 ||| 14 5 1 ||| |||
, ချောင်း ဆိုး ရာ က ||| ||| 0.0237032 3.59054e-12 0.0663689
3.88725e-08 ||| 4-0 ||| 14 5 1 ||| |||
, စ ||| ||| 0.00721401 0.00126866 0.110615 0.0008101 ||| 1-0 0-1 1-1
||| 46 3 1 ||| |||
, စ ||| ||| 0.0331844 0.00126866 0.110615 4.65355e-05 ||| 1-0 0-1
1-1 ||| 10 3 1 ||| |||
, စ ||| ||| 0.00222714 0.00155947 0.110615 0.000590288 ||| 0-0 1-0
1-1 ||| 149 3 1 ||| |||
, စ ခြင်း ||| ||| 0.0663689 9.68179e-05 0.165922 0.000137694 |||
0-0 1-0 1-1 2-2 ||| 5 2 1 ||| |||

, စ ခြင်း ||| ||| 0.0663689 9.68179e-05 0.165922
5.2186e-09 ||| 0-0 1-0 1-1 2-2 ||| 5 2 1 ||| |||
, စ ပြီး ||| ||| 0.00721401 1.0213e-05 0.165922 0.0008101 ||| 1-0 0-1
1-1 ||| 46 2 1 ||| |||
, စ ပြီး ||| ||| 0.0331844 1.0213e-05 0.165922 4.65355e-05 ||| 1-0
0-1 1-1 ||| 10 2 1 ||| |||
, စ အို , နဲ့ ||| ||| 0.0414805 2.56902e-06 0.331844
6.19381e-06 ||| 0-0 1-0 2-0 3-0 2-1 4-1 2-2 2-3 ||| 8 1 1 ||| |||
, တစ် မိ နစ် ||| ||| 0.00816355 7.35281e-07 0.653084 0.206612 ||| 2-0 3-0
||| 240 3 3 ||| |||
, တာ သို့ ||| ||| 0.0255265 3.309e-06 0.110615 0.000177367 ||| 2-1
||| 13 3 1 ||| |||
, တာ သို့ ||| ||| 0.0001212 3.309e-06 0.110615 0.585756 ||| 2-0 ||| 2738 3
1 ||| |||
, တာ သို့ ||| ||| 0.0255265 3.309e-06 0.110615 1.34267e-08 |||
2-2 ||| 13 3 1 ||| |||
, တာ သို့ မ ဟုတ် ||| ||| 0.0255265 1.47634e-07 0.110615 0.000102439
||| 2-1 3-1 4-1 ||| 13 3 1 ||| |||
, တာ သို့ မ ဟုတ် ||| ||| 0.0001212 1.47634e-07 0.110615 0.338307 ||| 2-0
3-0 4-0 ||| 2738 3 1 ||| |||
, တာ သို့ မ ဟုတ် ||| ||| 0.0255265 1.47634e-07 0.110615
7.75465e-09 ||| 2-2 3-2 4-2 ||| 13 3 1 ||| |||
, တာ သို့ သွား ပြီး ||| ||| 0.0207403 3.21226e-08
0.165922 5.01016e-12 ||| 3-1 4-1 2-4 ||| 16 2 1 ||| |||
, တာ သို့ သွား ပြီး ||| ||| 0.0174655 3.21226e-08 0.165922
1.61405e-09 ||| 3-0 4-0 2-3 ||| 19 2 1 ||| |||
, တူ လေး က ||| ||| 0.0829611 2.01684e-05 0.331844 6.20324e-05 ||| 3-0
0-1 1-1 2-1 3-1 ||| 4 1 1 ||| |||
, တော် ||| ||| 0.0114429 0.00233411 0.331844 0.0094034 ||| 0-0 1-0 ||| 29 1
1 ||| |||
, တဲ့ ||| ||| 0.153667 0.163265 0.435389 0.00457625 ||| 0-0 1-0 ||| 17 6 4
||| |||
, တဲ့ ||| ||| 0.0829611 0.163265 0.0553074 4.90244e-05 ||| 0-0 1-0 ||| 4
6 1 ||| |||
, တဲ့ ||| ||| 0.0829611 0.163265 0.0553074 5.02917e-07 ||| 0-0 1-0
||| 4 6 1 ||| |||
, တဲ့ အ ||| ||| 0.0174655 1.29096e-06 0.0255265 1.28011e-08 |||
2-3 ||| 19 13 1 ||| |||
, တဲ့ အ ||| ||| 0.230789 0.000798538 0.088765 1.13535e-10
||| 2-3 0-4 ||| 5 13 2 ||| |||
, တဲ့ အ ||| ||| 0.00118941 1.29096e-06 0.0255265 0.0047749 ||| 2-0 ||| 279
13 1 ||| |||
, တဲ့ အ ||| ||| 0.128216 0.000798538 0.088765 4.23495e-05 ||| 2-0 0-1
||| 9 13 2 ||| |||

, တဲ့ အ ||| ||| 5.06015e-05 4.50514e-07 0.0255265 0.0169496 ||| 2-0 ||| 6558 13 1 ||| |||
, တဲ့ အ ||| ||| 0.0174655 1.29096e-06 0.0255265 2.22844e-07 ||| 2-2 ||| 19 13 1 ||| |||
, တဲ့ အ ||| ||| 0.230789 0.000798538 0.088765 1.97645e-09 ||| 2-2 0-3 ||| 5 13 2 ||| |||
, တဲ့ အ ||| ||| 0.00253316 1.29096e-06 0.0255265 4.5459e-05 ||| 2-1 ||| 131 13 1 ||| |||
, တဲ့ အ ||| ||| 0.230789 0.000798538 0.088765 4.03185e-07 ||| 2-1 0-2 ||| 5 13 2 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.0174655 3.32215e-09 0.0195203 1.28011e-08 ||| 2-3 ||| 19 17 1 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.230789 2.05496e-06 0.0678791 1.13535e-10 ||| 2-3 0-4 ||| 5 17 2 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.00118941 3.32215e-09 0.0195203 0.0047749 ||| 2-0 ||| 279 17 1 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.290259 0.0015252 0.153667 8.0088e-05 ||| 2-0 3-0 0-1 1-1 ||| 9 17 4 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.165922 0.0015252 0.0195203 8.57967e-07 ||| 2-0 3-0 0-1 1-1 ||| 2 17 1 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.165922 0.0015252 0.0195203 8.80146e-09 ||| 2-0 3-0 0-1 1-1 ||| 2 17 1 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.00976013 1.89365e-08 0.0195203 0.000262265 ||| 2-0 3-1 ||| 34 17 1 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.0174655 3.32215e-09 0.0195203 2.22844e-07 ||| 2-2 ||| 19 17 1 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.230789 2.05496e-06 0.0678791 1.97645e-09 ||| 2-2 0-3 ||| 5 17 2 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.00253316 3.32215e-09 0.0195203 4.5459e-05 ||| 2-1 ||| 131 17 1 ||| |||
, တဲ့ အ ချိန် ||| ||| 0.230789 2.05496e-06 0.0678791 4.03185e-07 ||| 2-1 0-2 ||| 5 17 2 ||| |||
, တဲ့ အ ချိန် က ||| ||| 0.00976013 7.14199e-10 0.331844 0.000262265 ||| 2-0 3-1 ||| 34 1 1 ||| |||
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.0174655 4.73187e-11 0.0414805 1.28011e-08 ||| 2-3 ||| 19 8 1 ||| |||
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.0663689 2.92696e-08 0.0414805 1.13535e-10 ||| 2-3 0-4 ||| 5 8 1 ||| |||
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.00118941 4.73187e-11 0.0414805 0.0047749 ||| 2-0 ||| 279 8 1 ||| |||
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.0368716 2.92696e-08 0.0414805 4.23495e-05 ||| 2-0 0-1 ||| 9 8 1 ||| |||
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.0174655 4.73187e-11 0.0414805 2.22844e-07 ||| 2-2 ||| 19 8 1 ||| |||

, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.0663689 2.92696e-08 0.0414805 1.97645e-09 ||| 2-2 0-3 ||| 5 8 1 ||| |||
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.00253316 4.73187e-11 0.0414805 4.5459e-05 ||| 2-1 ||| 131 8 1 ||| |||
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.0663689 2.92696e-08 0.0414805 4.03185e-07 ||| 2-1 0-2 ||| 5 8 1 ||| |||
, တံ တွေး ||| ||| 0.0824246 3.97623e-05 0.230789 0.00687799 ||| 2-0 1-1 2-1 ||| 14 5 2 ||| |||
, တံ တွေး ||| ||| 0.00263369 7.86861e-05 0.0663689 0.30951 ||| 1-0 2-0 ||| 126 5 1 ||| |||
, တံ တွေး ||| ||| 0.110615 7.86861e-05 0.0663689 0.00363792 ||| 1-0 2-0 ||| 3 5 1 ||| |||
, တံ တွေး ||| ||| 0.110615 7.86861e-05 0.0663689 1.34956e-05 ||| 1-0 2-0 ||| 3 5 1 ||| |||
, တံ တွေး တွေ ||| ||| 0.0237032 3.90727e-06 0.331844 0.00461487 ||| 2-0 1-1 2-1 3-1 ||| 14 1 1 ||| |||
, တံ တွေး တွေ ရော ||| ||| 0.0184358 5.2926e-08 0.165922 9.44808e-05 ||| 2-0 1-1 2-1 3-1 4-2 ||| 18 2 1 ||| |||
, တံ တွေး တွေ ရော ||| ||| 0.0184358 5.2926e-08 0.165922 3.50495e-07 ||| 2-0 1-1 2-1 3-1 4-2 ||| 18 2 1 ||| |||
, တံ တွေး ရယ် ||| ||| 0.0237032 3.56527e-07 0.110615 0.0125817 ||| 3-0 1-1 2-1 ||| 14 3 1 ||| |||
, တံ တွေး ရယ် ||| ||| 0.0184358 3.56527e-07 0.110615 0.000147883 ||| 3-0 1-1 2-1 ||| 18 3 1 ||| |||
, တံ တွေး ရယ် ||| ||| 0.0184358 3.56527e-07 0.110615 5.48601e-07 ||| 3-0 1-1 2-1 ||| 18 3 1 ||| |||
, တံ တွေး ရယ် ကို ||| ||| 0.0237032 1.07714e-08 0.110615 0.0125817 ||| 3-0 1-1 2-1 ||| 14 3 1 ||| |||
, တံ တွေး ရယ် ကို ||| ||| 0.0184358 1.07714e-08 0.110615 0.000147883 ||| 3-0 1-1 2-1 ||| 18 3 1 ||| |||
, တံ တွေး ရယ် ကို ||| ||| 0.0184358 1.07714e-08 0.110615 5.48601e-07 ||| 3-0 1-1 2-1 ||| 18 3 1 ||| |||
, တံ တွေး ရော ||| ||| 0.0641081 5.386e-07 0.288486 0.000140814 ||| 2-0 1-1 2-1 3-2 ||| 18 4 2 ||| |||
, တံ တွေး ရော ||| ||| 0.0641081 5.386e-07 0.288486 5.22377e-07 ||| 2-0 1-1 2-1 3-2 ||| 18 4 2 ||| |||
, တံ တွေး ရော ကို ||| ||| 0.0641081 1.62721e-08 0.288486 0.000140814 ||| 2-0 1-1 2-1 3-2 ||| 18 4 2 ||| |||
, တံ တွေး ရော ကို ||| ||| 0.0641081 1.62721e-08 0.288486 5.22377e-07 ||| 2-0 1-1 2-1 3-2 ||| 18 4 2 ||| |||
, ဒီ ထက် ||| ||| 0.0114429 1.92016e-08 0.331844 1.82008e-05 ||| 2-0 0-1 1-1 2-1 ||| 29 1 1 ||| |||
, ဒေါ် ဒေါ် ||| ||| 0.00829611 2.9782e-05 0.110615 0.458599 ||| 1-0 2-0 ||| 40 3 1 ||| |||

, ဒေါ် ဒေါ် ||| ||| 0.0663689 2.9782e-05 0.110615 4.14735e-06 |||
1-2 2-2 ||| 5 3 1 ||| |||
, ဒေါ် ဒေါ် ||| ||| 0.0663689 2.9782e-05 0.110615 0.00486949 ||| 1-1 2-1
||| 5 3 1 ||| |||
, ဒေါ် ဒေါ် က ရ ||| ||| 0.0331844 4.3287e-07 0.165922
3.76661e-08 ||| 1-0 2-0 4-1 3-3 ||| 10 2 1 ||| |||
, ဒေါ် ဒေါ် က ရ ||| ||| 0.0368716 4.3287e-07 0.165922
3.99946e-10 ||| 1-1 2-1 4-2 3-4 ||| 9 2 1 ||| |||
, နက် ||| ||| 0.0165922 0.000711996 0.331844 0.0140397 ||| 0-0 1-0 ||| 20 1
1 ||| |||
, နက် ပါ ဘူး ||| ||| 0.165922 2.12619e-06 0.331844 0.001052 ||| 0-0 1-0
2-1 3-1 ||| 2 1 1 ||| |||
, နိုင် တော့ ||| ||| 0.0199415 0.000264939 0.653084 0.0135831 ||| 0-0 1-0
2-0 ||| 131 4 4 ||| |||
, နိုင် တော့ ရင် ||| ||| 0.0199415 3.18923e-06 0.653084 0.0135831 ||| 0-0
1-0 2-0 ||| 131 4 4 ||| |||
, နေ တဲ့ ||| ||| 0.000104981 1.08214e-06 0.165922 0.024791 ||| 2-0 ||| 3161
2 1 ||| |||
, နေ တဲ့ ||| ||| 0.00237032 1.08214e-06 0.165922 7.36689e-05 ||| 2-0 |||
140 2 1 ||| |||
, ပါ ||| ||| 0.0137011 0.000289146 0.39185 0.0021069 ||| 1-0 1-1 ||| 143
5 3 ||| |||
, ပါ ||| ||| 0.000163309 0.000358751 0.0663689 0.11327 ||| 1-0 ||| 2032 5 1
||| |||
, ပါ ||| ||| 0.000106874 0.000358751 0.0663689 0.00116198 ||| 1-0 ||| 3105
5 1 ||| |||
, ပါ နေ ||| ||| 0.000221821 1.64934e-05 0.331844 0.0628414 ||| 2-0 ||| 1496
1 1 ||| |||
, ပါ နေ ပါ ||| ||| 0.000221821 8.6553e-07 0.331844 0.0628414 ||| 2-0 |||
1496 1 1 ||| |||
, ပါ နေ ပါ တယ် ||| ||| 0.000221821 3.68538e-08 0.331844 0.0628414 ||| 2-0
||| 1496 1 1 ||| |||
, ပါ ဘူး ||| ||| 0.000163309 5.09631e-06 0.165922 0.11327 ||| 1-0 ||| 2032 2
1 ||| |||
, ပါ ဘူး ||| ||| 0.000106874 5.09631e-06 0.165922 0.00116198 ||| 1-0 |||
3105 2 1 ||| |||
, ပါ ရှင် ||| ||| 0.00734997 7.13717e-05 0.576972 0.000784465 ||| 1-0
1-1 2-2 ||| 157 2 2 ||| |||
, ပါ လား ရှင် ||| ||| 0.00211366 7.20612e-06 0.331844 0.000479803 |||
1-0 1-1 2-2 3-2 ||| 157 1 1 ||| |||
, ပေး ||| ||| 9.78892e-05 0.000164555 0.331844 0.215962 ||| 1-0 ||| 3390 1 1
||| |||
, ပေး ပါ ||| ||| 9.78892e-05 8.63543e-06 0.331844 0.215962 ||| 1-0 ||| 3390
1 1 ||| |||

```
, ပေး ပါ ရှင် |||   ||| 9.78892e-05 2.51533e-08 0.331844 0.215962 ||| 1-0 |||
3390 1 1 ||| |||
, ပြောင်း လဲ ရ မယ် |||              ||| 0.331844 3.31792e-05 0.331844
5.23214e-07 ||| 1-0 2-0 0-1 1-1 0-2 3-2 4-2 ||| 1 1 1 ||| |||
, ဖြစ် |||   ||| 0.00319081 2.35045e-05 0.331844 0.0022173 ||| 0-0 ||| 104 1
1 ||| |||
, ဖြစ် တာ |||     ||| 0.022123 3.08496e-06 0.331844 0.000116891 ||| 0-0 2-1
||| 15 1 1 ||| |||
, ဘယ် |||   ||| 0.00298959 0.000331767 0.331844 0.0328612 ||| 1-0 ||| 111 1 1
||| |||
, ဘယ် ဘက် |||   ||| 0.00298959 4.09699e-07 0.331844 0.0328612 ||| 1-0 ||| 111
1 1 ||| |||
, ဘယ် ဘက် ကို |||   ||| 0.00298959 1.23778e-08 0.331844 0.0328612 ||| 1-0 |||
111 1 1 ||| |||
, ဘယ် ဘက် ကို နှိပ် |||   ||| 8.25894e-05 1.39342e-13 0.165922 0.0451128 |||
4-0 ||| 4018 2 1 ||| |||
```

1.18 Reordering Table

ဒီတခါတော့ reordering table ကို လေ့လာကြည့်ရအောင်။

[29]:
```
%pwd
```

[29]: '/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model/tmp'

[30]:
```
!gunzip ./reordering-table.1.wbe-msd-bidirectional-fe.gz
```

[31]:
```
!head -n 300 ./reordering-table.1.wbe-msd-bidirectional-fe
```

```
! ! ! ||| !!! ||| 0.6 0.2 0.2 0.6 0.2 0.2
% ||| % ||| 0.69697 0.030303 0.272727 0.030303 0.030303 0.939394
% ||| %   ||| 0.777778 0.111111 0.111111 0.777778 0.111111 0.111111
% က |||   ||| 0.2 0.2 0.6 0.2 0.2 0.6
% က နည်း သည် |||      ||| 0.6 0.2 0.2 0.6 0.2 0.2
% ဆို ||| % ||| 0.6 0.2 0.2 0.2 0.2 0.6
% ဆို ||| %    ||| 0.6 0.2 0.2 0.2 0.2 0.6
% ဆို တော့ ||| %    ||| 0.777778 0.111111 0.111111 0.555556 0.111111
0.333333
% ဆို တော့ ||| %        ||| 0.6 0.2 0.2 0.2 0.2 0.6
% ဆို တော့ ပြင်း ထန် ||| %           ||| 0.714286 0.142857 0.142857
0.714286 0.142857 0.142857
% ဆို ရင် ||| % ||| 0.6 0.2 0.2 0.2 0.2 0.6
% ဆို ရင် အ ||| % ||| 0.6 0.2 0.2 0.2 0.2 0.6
% ဆို ရင် အ ဆ ||| % ||| 0.6 0.2 0.2 0.2 0.2 0.6
% တော့ ||| % ||| 0.714286 0.142857 0.142857 0.142857 0.142857 0.714286
```

```
% နည်း |||      ||| 0.6 0.2 0.2 0.2 0.2 0.6
% နည်း နေ |||      ||| 0.6 0.2 0.2 0.2 0.2 0.6
% နည်း နေ သည် |||      ||| 0.6 0.2 0.2 0.6 0.2 0.2
% နည်း လာ သည် |||      ||| 0.6 0.2 0.2 0.6 0.2 0.2
% နည်း သည် |||      ||| 0.714286 0.142857 0.142857 0.714286 0.142857
0.142857
% ပါ ||| % ||| 0.6 0.2 0.2 0.6 0.2 0.2
% ဘဲ ||| % ||| 0.714286 0.142857 0.142857 0.142857 0.142857 0.714286
% ဘဲ ရှိ ||| % ||| 0.714286 0.142857 0.142857 0.142857 0.142857 0.714286
% ဘဲ ရှိ နေ ||| % ||| 0.714286 0.142857 0.142857 0.142857 0.142857 0.714286
% ဘဲ ရှိ နေ တယ် ||| % ||| 0.714286 0.142857 0.142857 0.714286 0.142857 0.142857
% လောက် ||| % ||| 0.714286 0.142857 0.142857 0.142857 0.142857 0.714286
% လောက် ရှိ တယ် ||| % ||| 0.6 0.2 0.2 0.6 0.2 0.2
% အ ||| % ||| 0.6 0.2 0.2 0.2 0.2 0.6
% အ သေ ||| % ||| 0.6 0.2 0.2 0.2 0.2 0.6
% အ သေ အ ||| % ||| 0.6 0.2 0.2 0.2 0.2 0.6
&#91; ||| &#91; ||| 0.2 0.2 0.6 0.2 0.2 0.6
&#91; ||| &#91;        &#93; ||| 0.0909091 0.0909091 0.818182
0.0909091 0.0909091 0.818182
&#91; ||| &#93; ||| 0.0909091 0.0909091 0.818182 0.0909091 0.0909091 0.818182
&#91; ||| &#93;    ||| 0.0909091 0.0909091 0.818182 0.0909091 0.818182
0.0909091
&#91; ||| &#93;    ||| 0.111111 0.777778 0.111111 0.111111 0.111111 0.777778
&#91; ||| , ||| 0.2 0.6 0.2 0.2 0.2 0.6
&#91; |||        &#93; ||| 0.0909091 0.0909091 0.818182 0.0909091
0.0909091 0.818182
&#91; |||        &#93;    ||| 0.0909091 0.0909091 0.818182 0.0909091
0.818182 0.0909091
&#91; |||      &#93; ||| 0.0909091 0.0909091 0.818182 0.0909091
0.0909091 0.818182
&#91; |||      &#93;    ||| 0.0909091 0.0909091 0.818182 0.0909091
0.818182 0.0909091
&#91; |||    &#93; ||| 0.0909091 0.0909091 0.818182 0.0909091 0.0909091
0.818182
&#91; |||    &#93;    ||| 0.0909091 0.0909091 0.818182 0.0909091 0.818182
0.0909091
&#91; စစ် ဆေး |||    &#93;    ||| 0.111111 0.777778 0.111111 0.111111
0.111111 0.777778
&#91; အ ||| &#91;         ||| 0.555556 0.111111 0.333333 0.111111
0.111111 0.777778
&#91; အ မည် ||| &#91;   , ||| 0.111111 0.333333 0.555556 0.111111 0.111111
0.777778
&#91; အ မည် ||| &#91;   ,   ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
&#91; အ မည် ||| &#91;   ,    ||| 0.142857 0.142857 0.714286 0.142857
```

0.142857 0.714286
[အ မည် ||| , ||| 0.2 0.2 0.6 0.6 0.2 0.2
[အ မည် တည် ||| [, ||| 0.2 0.2 0.6 0.6 0.2 0.2
[အ မည် တည် နေ ||| [, ||| 0.2 0.2 0.6 0.2 0.2 0.6
[အ မည် နေ ||| [, ||| 0.142857 0.142857 0.714286 0.142857 0.142857
0.714286
[အ မည် နေ ||| [, ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
[အ မည် နေ ||| [, ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
[အ မည် နေ ||| , ||| 0.2 0.6 0.2 0.2 0.2 0.6
[အ မည် နေ ||| , , ||| 0.2 0.6 0.2 0.2 0.2 0.6
[အ မည် နေ ||| , ||| 0.2 0.6 0.2 0.2 0.2 0.6
[အ မည် နေ ရပ် ||| [, ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
[အ မည် နေ ရပ် ||| [, ||| 0.111111 0.111111 0.777778 0.111111
0.111111 0.777778
[အ မည် နေ ရပ် ||| [, ||| 0.111111 0.111111 0.777778
0.111111 0.111111 0.777778
[အ မည် နေ ရပ် ||| [, ||| 0.2 0.2 0.6 0.2 0.2 0.6
] ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2
] |||] " ||| 0.0909091 0.0909091 0.818182 0.818182 0.0909091
0.0909091
] |||] ||| 0.0769231 0.0769231 0.846154 0.0769231 0.0769231 0.846154
] |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] |||] ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
] |||] " ||| 0.0909091 0.0909091 0.818182 0.818182 0.0909091
0.0909091
] |||] ||| 0.0909091 0.0909091 0.818182 0.0909091 0.0909091
0.818182
] ||| [||| 0.2 0.6 0.2 0.2 0.6 0.2
] |||] " ||| 0.818182 0.0909091 0.0909091 0.818182
0.0909091 0.0909091
] |||] ||| 0.818182 0.0909091 0.0909091 0.0909091 0.0909091
0.818182
] ကို ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2
] ကို ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2
] ကို သွား ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2
] ကို သွား ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2
] ကို သွား ခဲ့ ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2
] ကို သွား ခဲ့ ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2
] ကို သွား ခဲ့ တာ ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2
] ကို သွား ခဲ့ တာ ||| [||| 0.2 0.2 0.6 0.2 0.6 0.2

] တို့ |||] ||| 0.111111 0.111111 0.777778 0.111111 0.111111 0.777778
] တွေ |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ |||] ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
] တွေ ရော |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ ရော ရှိ |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ ရော ရှိ သ |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ သ |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ သ |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ သ င့် |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ သ င့် |||] ||| 0.2 0.2 0.6 0.2 0.2 0.6
] တွေ သ င့် မှာ |||] ||| 0.2 0.2 0.6 0.6 0.2 0.2
] တွေ သ င့် မှာ |||] ||| 0.2 0.2 0.6 0.6 0.2 0.2
" B " ||| " D" ||| 0.2 0.6 0.2 0.2 0.2 0.6
" B " နဲ့ " ||| B " D" ||| 0.6 0.2 0.2 0.2
0.2 0.6
" B " နဲ့ " ||| " D" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" B " နဲ့ " ||| " D" ||| 0.2 0.2 0.6 0.2 0.2
0.6
" B " နဲ့ ||| B " D" ||| 0.6 0.2 0.2 0.2 0.2 0.6
" B " နဲ့ ||| " D" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" B " နဲ့ ||| " D" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" C " က ||| O ||| 0.2 0.2 0.6 0.2 0.2 0.6
" C " က ||| O ||| 0.2 0.2 0.6 0.2 0.6 0.2
" C " က တူ ||| O ||| 0.2 0.2 0.6 0.2 0.2 0.6
" C " က တူ ||| O ||| 0.2 0.2 0.6 0.2 0.6 0.2
" F " ||| " I " ||| 0.2 0.6 0.2 0.2 0.2 0.6
" F " နှ ||| " I " ||| 0.2 0.2 0.6 0.2 0.2 0.6
" F " နှ င့် ||| " I " ||| 0.2 0.2 0.6 0.2 0.2 0.6
" I " ကို ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" I " ကို ||| F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" I " ကို ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" I " ကို လဲ ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" I " ကို လဲ ||| F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" I " ကို လဲ ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" O " ||| " C" ||| 0.2 0.6 0.2 0.2 0.2 0.6
" O " နဲ့ " ||| " C" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" O " နဲ့ " ||| " C" ||| 0.2 0.2 0.6 0.2 0.2
0.6
" O " နဲ့ ||| " C" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" O " နဲ့ ||| " C" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" က ||| O ||| 0.2 0.2 0.6 0.2 0.2 0.6

" က ||| 0 ||| 0.2 0.2 0.6 0.2 0.2 0.6
" က တူ ||| 0 ||| 0.2 0.2 0.6 0.2 0.2 0.6
" က တူ ||| 0 ||| 0.2 0.2 0.6 0.2 0.2 0.6
" က တူ နေ ||| 0 ||| 0.2 0.2 0.6 0.2 0.2 0.6
" က တူ နေ ||| 0 ||| 0.2 0.2 0.6 0.2 0.2 0.6
" က တူ နေ တယ် ||| 0 ||| 0.2 0.2 0.6 0.2 0.2 0.6
" က တူ နေ တယ် ||| 0 ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို ||| F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ||| F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ခွဲ ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ခွဲ ||| F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ခွဲ ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ခွဲ ခြား ||| " F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ခွဲ ခြား ||| F" ||| 0.2 0.2 0.6 0.2 0.2 0.6
" ကို လဲ ခွဲ ခြား ||| " F" ||| 0.2 0.6 0.2 0.2 0.2 0.6
" ကျွန် တော် ||| ||| 0.6 0.2 0.2 0.6 0.2 0.2
" ကျွန် တော် လမ်း လျှောက် ||| ||| 0.6 0.2 0.2 0.2 0.2 0.6
(||| ||| 0.0909091 0.0909091 0.818182 0.0909091 0.818182 0.0909091
* ||| * ||| 0.967213 0.0163934 0.0163934 0.213115 0.0163934 0.770492
* ထုံ ကျဉ် ||| * ||| 0.2 0.6 0.2 0.2 0.2 0.6
* ထုံ ကျဉ် ဆေး ထိုး ||| * ||| 0.2 0.6 0.2 0.2 0.2 0.6
* ထုံ ကျဉ် သော ||| * ||| 0.2 0.6 0.2 0.2 0.2 0.6
* ထုံ ဆေး ||| * ||| 0.142857 0.714286 0.142857 0.142857 0.142857 0.714286
* ထုံ ဆေး ထိုး ||| * ||| 0.142857 0.428571 0.428571 0.142857 0.142857
0.714286
* ထုံ ဆေး ထိုး နေ ||| * ||| 0.2 0.6 0.2 0.2 0.2 0.6
* ထုံ ဆေး ထိုး နှံ ||| * ||| 0.2 0.6 0.2 0.2 0.2 0.6
* ထုံ အောင် ||| * ||| 0.2 0.6 0.2 0.2 0.2 0.6
* ထုံ အောင် ဆေး ထိုး ||| * ||| 0.2 0.6 0.2 0.2 0.2 0.6
* ပြီး ခဲ့ ပါ ပြီ ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* ပြီး သွား ပါ ပြီ ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* ပြီး သွား ပြီ ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* ပြီး သွား ပြီ ရှင် ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* သွား ကို ခွဲ ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* သွား ခွဲ ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* သွား ခွဲ ထုတ် ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* သွား ခွဲ သည် * ||| * * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* အ ||| * ||| 0.777778 0.111111 0.111111 0.111111 0.111111 0.777778

* အ နာ ||| * ||| 0.777778 0.111111 0.111111 0.111111 0.111111 0.777778
* အ နာ ကြ ည့် ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* အ နာ ဆေး ||| * ||| 0.777778 0.111111 0.111111 0.111111 0.111111 0.777778
* အ နာ ဆေး ချုပ် ||| * ||| 0.6 0.2 0.2 0.2 0.2 0.6
* အ နာ ပြန် ချုပ် ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
* အံ ဆုံး သွား ခွဲ ||| * ||| 0.6 0.2 0.2 0.6 0.2 0.2
, 0 0 0 - ||| , 000 - ||| 0.6 0.2 0.2 0.2 0.2 0.6
, 0 0 0 ||| < 100 , 000 ||| 0.142857 0.142857 0.714286 0.714286 0.142857 0.142857
, 0 0 0 ||| , 000 ||| 0.6 0.2 0.2 0.6 0.2 0.2
, 0 0 0 ||| 10 , 000 ||| 0.2 0.2 0.6 0.6 0.2 0.2
, 0 0 0 ||| 100 , 000 ||| 0.142857 0.142857 0.714286 0.714286 0.142857 0.142857
, 0 0 0 ||| < 100 , 000 ||| 0.142857 0.142857 0.714286 0.714286 0.142857 0.142857
, 0 0 0 ဆဲလ် ||| < 100 , 000 ||| 0.142857 0.142857 0.714286 0.714286 0.142857 0.142857
, 0 0 0 ဆဲလ် ||| 100 , 000 ||| 0.142857 0.142857 0.714286 0.714286 0.142857 0.142857
, 0 0 0 ဘတ် ||| 10 , 000 ||| 0.2 0.2 0.6 0.2 0.2 0.6
, 0 ||| < 100 ||| 0.142857 0.142857 0.714286 0.714286 0.142857 0.142857
, 0 ||| 10 ||| 0.2 0.2 0.6 0.6 0.2 0.2
, 0 ||| 100 ||| 0.142857 0.142857 0.714286 0.714286 0.142857 0.142857
, 0 ||| < 100 ||| 0.142857 0.142857 0.714286 0.714286 0.142857 0.142857
, ||| , ||| 0.411765 0.0588235 0.529412 0.411765 0.0588235 0.529412
, ||| - ||| 0.714286 0.142857 0.142857 0.714286 0.142857 0.142857
, ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ||| ||| 0.111111 0.111111 0.777778 0.111111 0.111111 0.777778
, ||| ||| 0.111111 0.111111 0.777778 0.111111 0.111111 0.777778
, ||| ||| 0.111111 0.111111 0.777778 0.111111 0.111111 0.777778
, ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ||| ||| 0.111111 0.111111 0.777778 0.777778 0.111111 0.111111
, ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ||| ||| 0.142857 0.142857 0.714286 0.142857 0.142857 0.714286
, ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ||| ||| 0.2 0.2 0.6 0.2 0.6 0.2
, က လေး ||| ||| 0.0769231 0.0769231 0.846154 0.0769231 0.0769231 0.846154
, ခင် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ခု တော့ ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, ခု တော့ ဒေါက် တာ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ခု တော့ ဒေါက် တာ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ခု တော့ ဒေါက် တာ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ခု တော့ ဒေါက် တာ ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, ချက် နား ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ချက် နား ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, ချောင်း ဆိုး ရာ က ||| ||| 0.777778 0.111111 0.111111 0.111111 0.111111

0.777778
, ချောင်း ဆိုး ရာ က ||| ||| 0.6 0.2 0.2 0.2 0.2 0.6
, ချောင်း ဆိုး ရာ က ||| ||| 0.6 0.2 0.2 0.2 0.2 0.6
, စ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, စ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, စ ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, စ ခြင်း ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, စ ခြင်း ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, စ ပြီး ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, စ ပြီး ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, စ အို , နဲ့ ||| ||| 0.6 0.2 0.2 0.2 0.2 0.6
, တစ် မိ နစ် ||| ||| 0.111111 0.777778 0.111111 0.111111 0.111111 0.777778
, တာ သို့ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တာ သို့ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တာ သို့ ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, တာ သို့ မ ဟုတ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တာ သို့ မ ဟုတ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တာ သို့ မ ဟုတ် ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, တာ သို့ သွား ပြီး ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တာ သို့ သွား ပြီး ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တူ လေး က ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တော် ||| ||| 0.2 0.2 0.6 0.2 0.6 0.2
, တဲ့ ||| ||| 0.0909091 0.818182 0.0909091 0.0909091 0.0909091 0.818182
, တဲ့ ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, တဲ့ ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, တဲ့ အ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ||| ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
, တဲ့ အ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ||| ||| 0.142857 0.142857 0.714286 0.142857 0.142857 0.714286
, တဲ့ အ ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, တဲ့ အ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ||| ||| 0.142857 0.142857 0.714286 0.142857 0.142857
0.714286
, တဲ့ အ ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ||| ||| 0.142857 0.142857 0.714286 0.142857 0.142857
0.714286
, တဲ့ အ ချိန် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ||| ||| 0.142857 0.428571 0.428571 0.142857
0.142857 0.714286
, တဲ့ အ ချိန် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ||| ||| 0.0909091 0.0909091 0.818182 0.0909091 0.0909091
0.818182
, တဲ့ အ ချိန် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6

, တဲ့ အ ချိန် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ||| ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
, တဲ့ အ ချိန် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ||| ||| 0.142857 0.142857 0.714286 0.142857 0.142857
0.714286
, တဲ့ အ ချိန် က ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တဲ့ အ ချိန် ဖြစ် ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, တံ တွေး ||| ||| 0.142857 0.142857 0.714286 0.714286 0.142857
0.142857
, တံ တွေး ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, တံ တွေး ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, တံ တွေး ||| ||| 0.2 0.6 0.2 0.2 0.2 0.6
, တံ တွေး တွေ ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, တံ တွေး တွေ ရော ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တံ တွေး တွေ ရော ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တံ တွေး ရယ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တံ တွေး ရယ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တံ တွေး ရယ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တံ တွေး ရယ် ကို ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တံ တွေး ရယ် ကို ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, တံ တွေး ရယ် ကို ||| ||| 0.2 0.2 0.6 0.6 0.2 0.2
, တံ တွေး ရော ||| ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
, တံ တွေး ရော ||| ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
, တံ တွေး ရော ကို ||| ||| 0.142857 0.142857 0.714286 0.142857
0.142857 0.714286
, တံ တွေး ရော ကို ||| ||| 0.142857 0.142857 0.714286
0.714286 0.142857 0.142857
, ဒီ ထက် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ဒေါ် ဒေါ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ဒေါ် ဒေါ် ||| ||| 0.2 0.2 0.6 0.2 0.2 0.6

```
, ဒေါ် ဒေါ် |||      ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ဒေါ် ဒေါ် က ရ |||            ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ဒေါ် ဒေါ် က ရ |||               ||| 0.2 0.2 0.6 0.2 0.2 0.6
, နက် |||    ||| 0.6 0.2 0.2 0.6 0.2 0.2
, နက် ပါ ဘူး |||      ||| 0.6 0.2 0.2 0.6 0.2 0.2
, နိုင် တော့ |||    ||| 0.818182 0.0909091 0.0909091 0.0909091 0.0909091
0.818182
, နိုင် တော့ ရင် |||    ||| 0.818182 0.0909091 0.0909091 0.818182 0.0909091
0.0909091
, နေ တဲ့ |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, နေ တဲ့ |||      ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပါ |||      ||| 0.111111 0.111111 0.777778 0.777778 0.111111 0.111111
, ပါ |||   ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပါ |||      ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပါ နေ |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပါ နေ ပါ |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပါ နေ ပါ တယ် |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပါ ဘူး |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပါ ဘူး |||      ||| 0.2 0.2 0.6 0.6 0.2 0.2
, ပါ ရှင် |||         ||| 0.142857 0.142857 0.714286 0.714286 0.142857
0.142857
, ပါ လား ရှင် |||          ||| 0.2 0.2 0.6 0.6 0.2 0.2
, ပေး |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပေး ပါ |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ပေး ပါ ရှင် |||    ||| 0.2 0.2 0.6 0.6 0.2 0.2
, ပြောင်း လဲ ရ မယ် |||                ||| 0.6 0.2 0.2 0.6 0.2 0.2
, ဖြစ် |||    ||| 0.2 0.2 0.6 0.6 0.2 0.2
, ဖြစ် တာ |||      ||| 0.2 0.2 0.6 0.6 0.2 0.2
, ဘယ် |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ဘယ် ဘက် |||    ||| 0.2 0.2 0.6 0.2 0.2 0.6
, ဘယ် ဘက် ကို |||    ||| 0.2 0.6 0.2 0.2 0.2 0.6
, ဘယ် ဘက် ကို နှိပ် |||    ||| 0.2 0.2 0.6 0.2 0.6 0.2
```

1.19 moses.ini

Training လုပ်ပြီးထွက်လာတဲ့ moses.ini ဖိုင်ကိုလေ့လာကြည့်ရအောင်။

[32]:
```
%cd ..
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model
```

[33]:
```
!ls --color=auto
```

```
aligned.1.grow-diag-final-and  lex.1.f2e
extract.1.inv.sorted.gz        moses.ini.1
```

```
extract.1.o.sorted.gz          phrase-table.1.gz
extract.1.sorted.gz            reordering-table.1.wbe-msd-
bidirectional-fe.gz
lex.1.e2f                      tmp
```

[34]:
```
!cat moses.ini.1
```

```
#########################
### MOSES CONFIG FILE ###
#########################

# input factors
[input-factors]
0

# mapping steps
[mapping]
0 T 0

[distortion-limit]
6

# feature functions
[feature]
UnknownWordPenalty
WordPenalty
PhrasePenalty
PhraseDictionaryMemory name=TranslationModel0 num-features=4
path=/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model/phrase-table.1
input-factor=0 output-factor=0
LexicalReordering name=LexicalReordering0 num-features=6 type=wbe-msd-
bidirectional-fe-allff input-factor=0 output-factor=0 path=/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/reordering-table.1.wbe-msd-
bidirectional-fe.gz
Distortion
KENLM name=LM0 factor=0 path=/home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/lm/thmy.binlm.1 order=5

# dense weights for feature functions
[weight]
# The default weights are NOT optimized for translation quality. You MUST tune
the weights.
# Documentation for tuning is here:
http://www.statmt.org/moses/?n=FactoredTraining.Tuning
UnknownWordPenalty0= 1
WordPenalty0= -1
PhrasePenalty0= 0.2
```

```
TranslationModel0= 0.2 0.2 0.2 0.2
LexicalReordering0= 0.3 0.3 0.3 0.3 0.3 0.3
Distortion0= 0.3
LM0= 0.5
```

============

moses.ini ဖိုင်က moses decoder အတွက် blueprint ပါပဲ။ ဒီဖိုင်ထဲကနေ decoder က ဘယ် path အောက်မှာရှိတဲ့ မော်ဒယ်ကို load လုပ်ယူရမယ်။ အကောင်းဆုံး translation ရလဒ် ရလာနိုင်ဖို့အတွက် ဘယ်မော်ဒယ်တွေကို (i.e. language model, phrase table etc.) ဘယ်လို အလေးပေး (i.e. weight values) ပြီးတွက်ရမယ်ဆိုတာကို ညွှန်ကြားထားတာပါ။ Training လုပ်ပြီးသွားတဲ့အချိန်မှာ training data pair တွေကနေ သူလေ့လာထားသလောက် weight တွေကို သတ်မှတ်ပေးထားတာပါ။

1.19.1 1. Structural Settings

[input-factors] - `0`: This indicates a single-factor model (usually just the surface word form). In factored models, you might see 0,1,2 representing words, POS tags, and lemmas.

[mapping] - `0 T 0`: This defines the decoding steps. T 0 means "apply Translation Model 0." If you had a generation model (e.g., to create a lemma from a word), it would be listed here.

[distortion-limit] - `6`: This controls reordering. It limits how far a source phrase can "jump" to find its target translation. A limit of 6 means the decoder can move 6 positions away from the current word. For language pairs with very different structures (like Myanmar and English), this is sometimes increased.

1.19.2 2. Feature Functions

This section lists the "ingredients" of the log-linear translation model.

Feature	Description
`UnknownWordPenalty`	Penalizes the model for every word it cannot translate (OOV).
`WordPenalty`	Controls the length of the translation. Negative weights encourage longer sentences; positive weights encourage shorter ones.
`PhrasePenalty`	Controls the number of phrases used. It balances between using many short phrases versus a few long phrases.
`PhraseDictionaryMemory`	Loads the Phrase Table into RAM. num-features=4 refers to the four probabilities we discussed earlier (inverse/direct phrase and lexical weights).
`LexicalReordering`	Loads the reordering table. wbe-msd-bidirectional-fe-allff indicates it uses word-based extraction and tracks swaps in both directions.
`Distortion`	A basic reordering feature based on the distance words move.

Feature	Description
`KENLM`	The Language Model (LM). Your order=5 means it checks sequences of up to 5 words to ensure the Thai output sounds natural.

1.19.3 3. Weights ([weight])

The weights determine the "influence" of each feature. In SMT, the final score for a translation is the weighted sum of all features:

$$Score = \sum_i w_i \cdot f_i(x, y)$$

- `TranslationModel0= 0.2 0.2 0.2 0.2`: These are the weights for the four scores in your phrase table.
- `LexicalReordering0= 0.3 0.3 0.3 0.3 0.3 0.3`: These are the weights for the six re-ordering scores (Monotonic/Swap/Discontinuous for both directions).
- `LM0= 0.5`: Notice the Language Model often has a high weight. This is because the LM is the primary judge of whether the output is grammatically correct.

1.20 moses.ini (After Tuning)

အတန်းထဲမှာ ရှင်းပြခဲ့တဲ့အတိုင်းပါပဲ။ Tuning step ပြီးသွားတဲ့အခါမှာတော့ moses.ini ဖိုင်ရဲ့ တချို့ weight value တွေက အသေးစိတ်သတ်မှတ်လာတာကို အောက်ပါအတိုင်း တွေ့ရပါလိမ့်မယ်။

[35]:
```
%cd /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/tuning
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/tuning
```

[36]:
```
!ls --color=auto
```

```
filtered.1  moses.filtered.ini.1  moses.ini.1  moses.tuned.ini.1
tmp.1
```

Tuning သို့မဟုတ် optimization အဆင့်ကတော့ PBSMT, HPBSMT, OSM နည်းလမ်းတွေနဲ့ မော်ဒယ်ဆောက်တဲ့အခါမှာ အကြာဆုံး အပိုင်းဖြစ်ပါတယ်။ ဘာကြောင့်လဲ ဆိုတော့ moses က ဖြစ်နိုင်ခြေ phrase အတွဲတွေနဲ့ တခြား မော်ဒယ်တွေနဲ့ ပေါင်းစပ်မှုတွေလုပ်ပြီး development data ထဲက စာကြောင်းတစ်ကြောင်းချင်းစီအတွက် အကောင်းဆုံး ဘာသာပြန်ပေးနိုင်ခြေ (i.e. best weight တန်ဖိုး) တွေကို ရှာဖွေတာမို့လို့ပါ။ အဆင်ပြေပြေနဲ့ ဘာ error မှမပေးပဲ ပြီးသွားရင်တော့ အထက်မှာ မြင်ရတဲ့အတိုင်း moses.ini ဖိုင်အသစ်နှစ်မျိုး (i.e. filtered, tuned) ထပ်ထွက်လာပါလိမ့်မယ်။ အဲဒါကြောင့် စုစုပေါင်း ini ဖိုင် သုံးမျိုး ရလာပါလိမ့်မယ်။ အသေးစိတ်က အောက်ပါ ဇယားမှာ ရှင်းပြထားသလိုပါပဲ။

1.20.1 1. The Three .ini Files

File	What it is
moses.ini.1	**The Original.** This is a copy of your untuned configuration from the model folder. It uses default weights and points to your full, massive phrase table.
moses.filtered.ini.1	**The MERT Engine.** This is the config Moses uses during the tuning process. It points to the **filtered** tables in the filtered.1 folder. This makes tuning run much faster because it doesn't have to search the entire phrase table-only the parts relevant to your tuning set.
moses.tuned.ini.1	**The Winner.** This is your **final product**. It contains the optimized weights (e.g., TranslationModel0= 0.12 0.05 0.21...). You should use this file for your final translation/testing.

filtered.ini ဖိုင်ကို လေ့လာကြည့်ရအောင်။

[37]:
```
!cat moses.filtered.ini.1
```

```
#########################
### MOSES CONFIG FILE ###
#########################

# input factors
[input-factors]
0

# mapping steps
[mapping]
0 T 0

[distortion-limit]
6

# feature functions
[feature]
UnknownWordPenalty
WordPenalty
PhrasePenalty
PhraseDictionaryMemory name=TranslationModel0 num-features=4
path=/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-
th/tuning/filtered.1/phrase-table.0-0.1.1.gz input-factor=0 output-factor=0
LexicalReordering name=LexicalReordering0 num-features=6 type=wbe-msd-
bidirectional-fe-allff input-factor=0 output-factor=0 path=/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/tuning/filtered.1/reordering-table.1.wbe-
```

```
msd-bidirectional-fe.0-0.1
Distortion
KENLM name=LM0 factor=0 path=/home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/lm/thmy.binlm.1 order=5

# dense weights for feature functions
[weight]
# The default weights are NOT optimized for translation quality. You MUST tune
the weights.
# Documentation for tuning is here:
http://www.statmt.org/moses/?n=FactoredTraining.Tuning
UnknownWordPenalty0= 1
WordPenalty0= -1
PhrasePenalty0= 0.2
TranslationModel0= 0.2 0.2 0.2 0.2
LexicalReordering0= 0.3 0.3 0.3 0.3 0.3 0.3
Distortion0= 0.3
LM0= 0.5
```

Tuning လုပ်ပြီးတော့ ရလာတဲ့ final ini ဖိုင်ကို လေ့လာကြည့်ရအောင်။
ဒီမို ဘာသာလုပ်ဖို့ အသုံးပြုတဲ့အခါမှာလည်း ဒီ moses.tuned.ini.1 ကိုပဲ သုံးကြတာပါ။

[38]:
```
!cat moses.tuned.ini.1
```

```
#########################
### MOSES CONFIG FILE ###
#########################

# input factors
[input-factors]
0

# mapping steps
[mapping]
0 T 0

[distortion-limit]
6

# feature functions
[feature]
UnknownWordPenalty
WordPenalty
PhrasePenalty
PhraseDictionaryMemory name=TranslationModel0 num-features=4
path=/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/model/phrase-table.1
input-factor=0 output-factor=0
LexicalReordering name=LexicalReordering0 num-features=6 type=wbe-msd-
```

```
bidirectional-fe-allff input-factor=0 output-factor=0 path=/home/ye/exp/SMT-
NMT_tutorial/pbsmt-big/baseline/my-th/model/reordering-table.1.wbe-msd-
bidirectional-fe.gz
Distortion
KENLM name=LM0 factor=0 path=/home/ye/exp/SMT-NMT_tutorial/pbsmt-
big/baseline/my-th/lm/thmy.binlm.1 order=5

# core weights
[weight]
LexicalReordering0= 0.0461365 0.0337038 0.0660466 0.0478805 0.00992487 0.126209
Distortion0= 0.0200328
LM0= 0.260847
WordPenalty0= 0.0706379
PhrasePenalty0= -0.0961515
TranslationModel0= 0.0973455 0.0601279 0.038421 -0.0265353
UnknownWordPenalty0= 1
```

1.21 BLEU Scores

ခုလောက်ဆိုရင်တော့ PBSMT နဲ့ ပတ်သက်ပြီးအတွင်းပိုင်းကို တော်တော်လေး သိသွားပြီလို့ ယုံကြည်ပါတယ်။ နောက်ဆုံးအနေနဲ့ translation performance ကို BLEU score နဲ့ အကဲဖြတ်ကြရအောင်။

ပထမဆုံး မြန်မာကနေ ထိုင်းကို ဘာသာပြန်တာကို test data စာကြောင်းရေ ရှစ်ထောင်နဲ့ BLEU score ဘယ်လောက် ရသလဲ ဆိုတာ ရှာဖွေကြည့်ရအောင်။

[39]:
```
%cd /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/evaluation
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/my-th/evaluation
```

[40]:
```
!ls
```

```
test.analysis.1          test.filtered.ini.1  test.nist-bleu-c.1
test.cleaned.1           test.input.txt.1     test.output.1
test.detokenized.sgm.1   test.multi-bleu.1    test.output.1.wa
test.filtered.1          test.nist-bleu.1     test.reference.txt.1
```

[41]:
```
!cat test.multi-bleu.1
```

```
BLEU = 36.80, 57.3/39.7/31.1/25.9 (BP=1.000, ratio=1.004, hyp_len=56314,
ref_len=56113)
```

အထက်မှာ မြင်ရတဲ့အတိုင်း 36.80 ရပါတယ်။

ထိုင်းကနေ မြန်မာဘက် ကို ဘာသာပြန်တဲ့ ရလဒ်ကိုလည်း လေ့လာကြည့်ရအောင်။

[42]:
```
%cd /home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/th-my/evaluation
```

```
/home/ye/exp/SMT-NMT_tutorial/pbsmt-big/baseline/th-my/evaluation
```

[44]:
```
!cat test.multi-bleu.1
```

```
BLEU = 32.76, 59.7/39.4/27.4/19.9 (BP=0.973, ratio=0.974, hyp_len=91434,
ref_len=93908)
```

1.22 Summary

ခုချိန်ထိ အစအဆုံး လေ့လာခဲ့ရင် SMT system overview ကို တော်တော်လေး တီးမိခေါက်မိရှိသွားပြီလို့ ပြောလို့ ရပါလိမ့်မယ်။ ဒါပေမဲ့ လက်တွေ့ ဘာသာစကားအတွဲတွေ အမျိုးမျိုးကို ဘာသာပြန်ကြည့်တာမျိုး လုပ်ရင်းနဲ့မှပဲ သိလာမယ့် ကိစ္စတွေ အများကြီး ကျန်ရှိပါသေးတယ်။

လောလောဆယ်တော့ လက်ရှိ BLEU score တွေကို ဘယ်လို တက်အောင် လုပ်ကြမလဲ ဆိုတာကိုပဲ နောက်ဆက်တွဲ experiment လုပ်ရင်း လေ့လာသွားကြရအောင်။ ပြောရရင်တော့ translation performance တက်အောင်လုပ်လို့ရတဲ့ နည်းလမ်းက အမျိုးမျိုး ရှိပါတယ်။

- Parallel corpus ကို လက်ရှိ ဖြသောင်းကျော် ကနေ တစ်သိန်းလောက် တိုးလိုက်တာမျိုး
- ထိုင်း နဲ့ မြန်မာစာကြောင်းတွေကို word segmentation အမျိုးမျိုးနဲ့ ဘာသာပြန်ခိုင်းကြည့်တာမျိုး။ ဥပမာ word-word, word-syl, char-char
- လက်ရှိ ရှိနေတဲ့ ဒေတာကို cleaning သေချာပြန်လုပ်တာမျိုး။ ဥပမာ normalization, spelling checking, masking
- Language model ကို လက်ရှိမှာက training data ကပဲ ယူထားတာပါ။ အဲဒါကို တခြား ပိုသန့်တဲ့ ပိုကြီးတဲ့ monolingual corpus နဲ့ ဆောက်ထားတဲ့ language model နဲ့ အစားထိုးကြည့်တာမျိုး စသည်ဖြင့် အမျိုးမျိုး လုပ်လို့ ရပါတယ်။

`SMT-Turotiral-for-AI-Class-3.ipynb` Notebook မှာတော့ မြန်မာစာဘက် တစ်ဖက်တည်းကို syl_normalizer.py (version 0.6) ကို သုံးပြီး normalization လုပ်လိုက်ပြီး ဘယ်လောက် ရလဒ်တက်လာသလဲ ဆိုတာကို experiment လုပ်ကြည့်ပါမယ်။

[ ]: